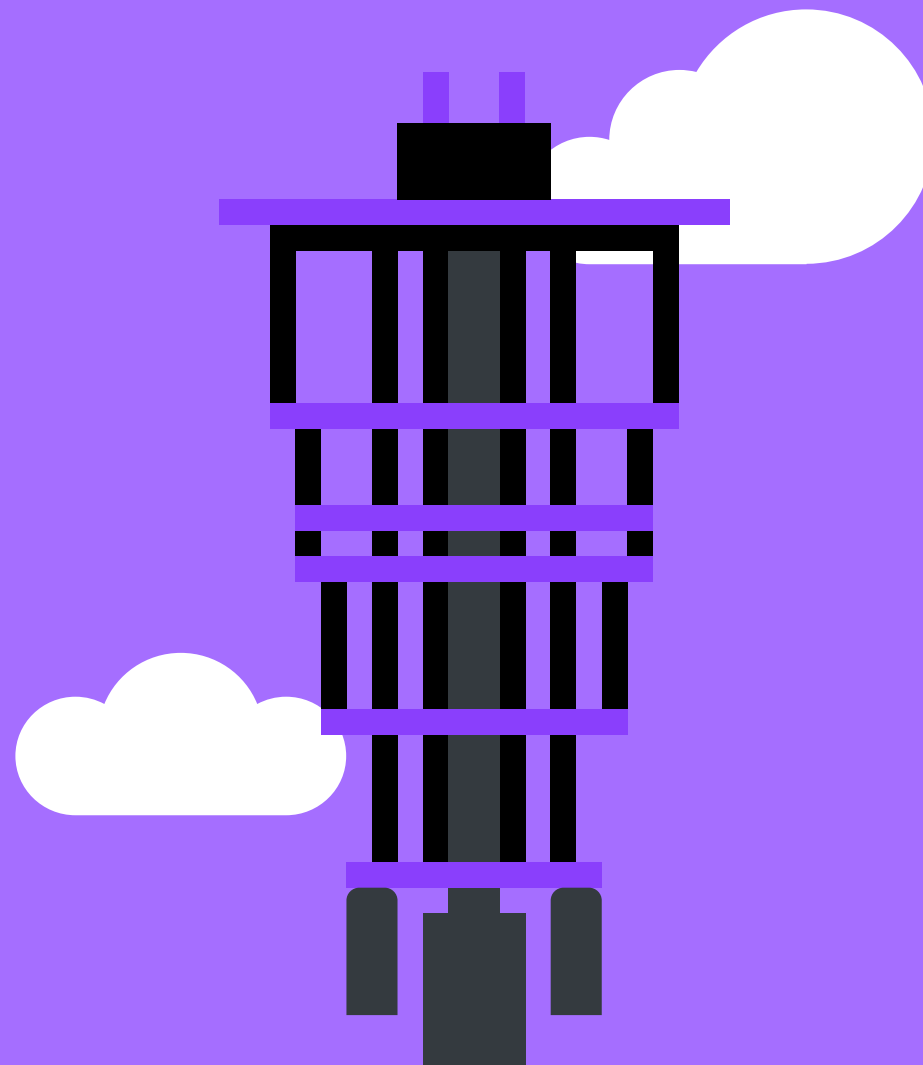
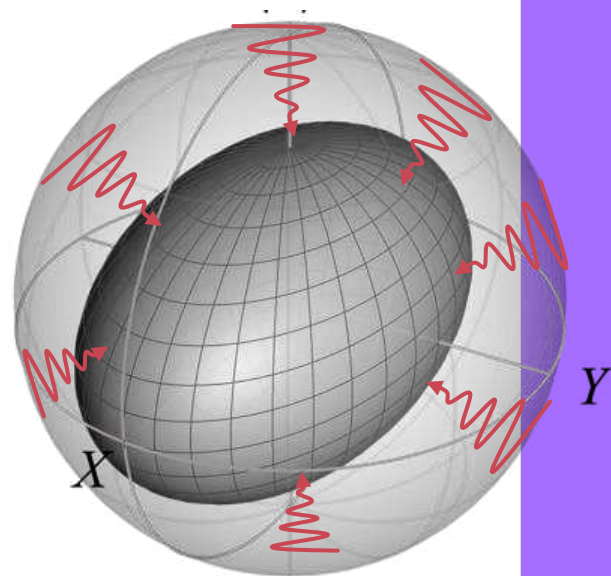


Introduction to real quantum computers (and noise)

Zlatko K. Minev
IBM Quantum



@zlatko_minev



zlatko-minev.com/blog

Have you used
a quantum computer?



Quantum Noise and Error Mitigation

Lecture 1

Big picture

Quantum computers status

Why error mitigation?

Noise in quantum computers
Overview of error mitigation

Mitigation fundamentals

Probabilistic error cancelation (PEC)

Introduction
One qubit example

Next lecture

Learning noise
State-of-art mitigation experiments
Hardware
Outlook



Where?

Slides from lecture

Posted here zlatko-minev.com/education
Tutorials [/blog](https://zlatko-minev.com/blog)

See

IBM Learning
learning.quantum.ibm.com

Weekly seminars

[qiskit](https://qiskit.org) [YouTube](https://www.youtube.com/qiskit)

7. Digital quantum circuits (pictorial)
7A. Basic elements

Quantum wire
Quantum wire bundle (n qubits)
Quantum wire bundle (alternating)

Classical wire
Entangled bit (ebit: Bell pair)

Quantum gate U

Control gate U (control on $|0\rangle$)

Control gate U (control on $|1\rangle$)

(Circuit: K. Minev)

Cheat sheet: Digital quantum circuits - pictorial 101

Making a qubit cheatsheet. On this festive Christmas day.

Primer on Pauli Twirling

Zlatko Minev
2022-04-26, 07-11

A tutorial on tailoring quantum noise - Twirling 101 (Parts I-IV)

Nutshell introduction to tailoring quantum noise by twirling into stochastic Pauli or

learn and cancel quantum noise

error cancellation with sparse Pauli-Lindblad models on quantum processors

Zlatko K. Minev

van den Berg, Zlatko K. Minev, Abhinav Kandala, Kristan Temme
arXiv:2201.11629 (2022)

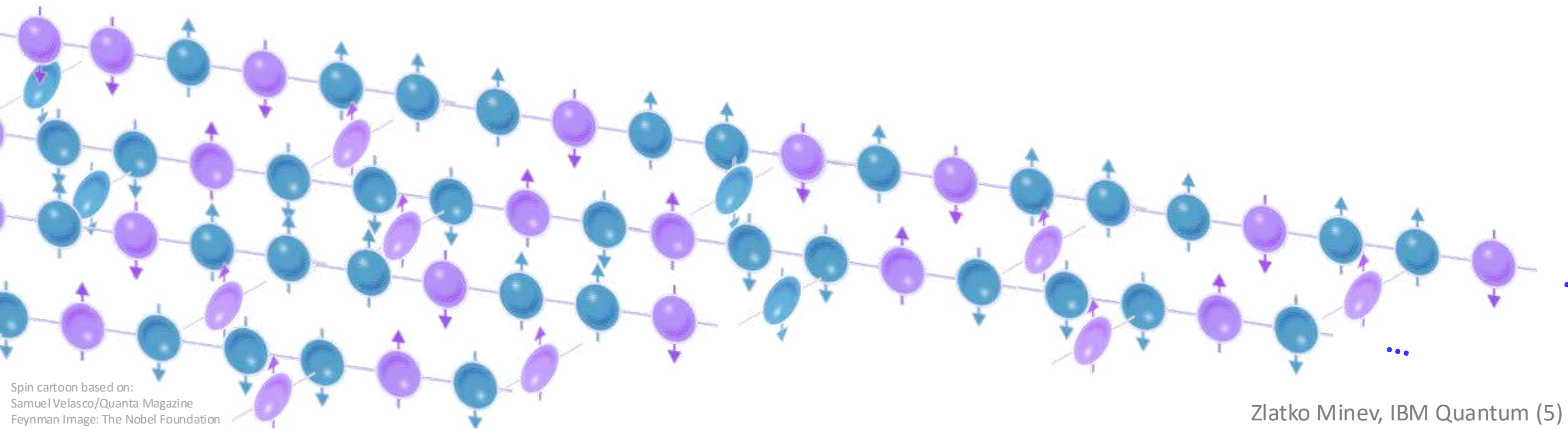
with: Seminar IBM Quantum team

Talk: To Learn and Cancel Noise: Probabilistic Error Cancellation with Sparse Pauli-Lindblad Models on Noisy Quantum Processors

#Quantum Seminar

Qiskit

How is it going for quantum computers?



This year 2023

A 127-qubit quantum computer installed in the the lobby cafeteria of a research building dutifully executing jobs almost all the time.



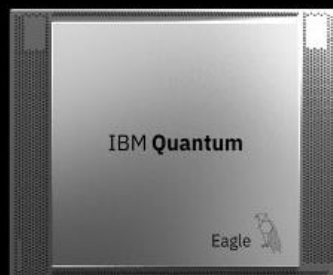




2019
Falcon
27 Qubits



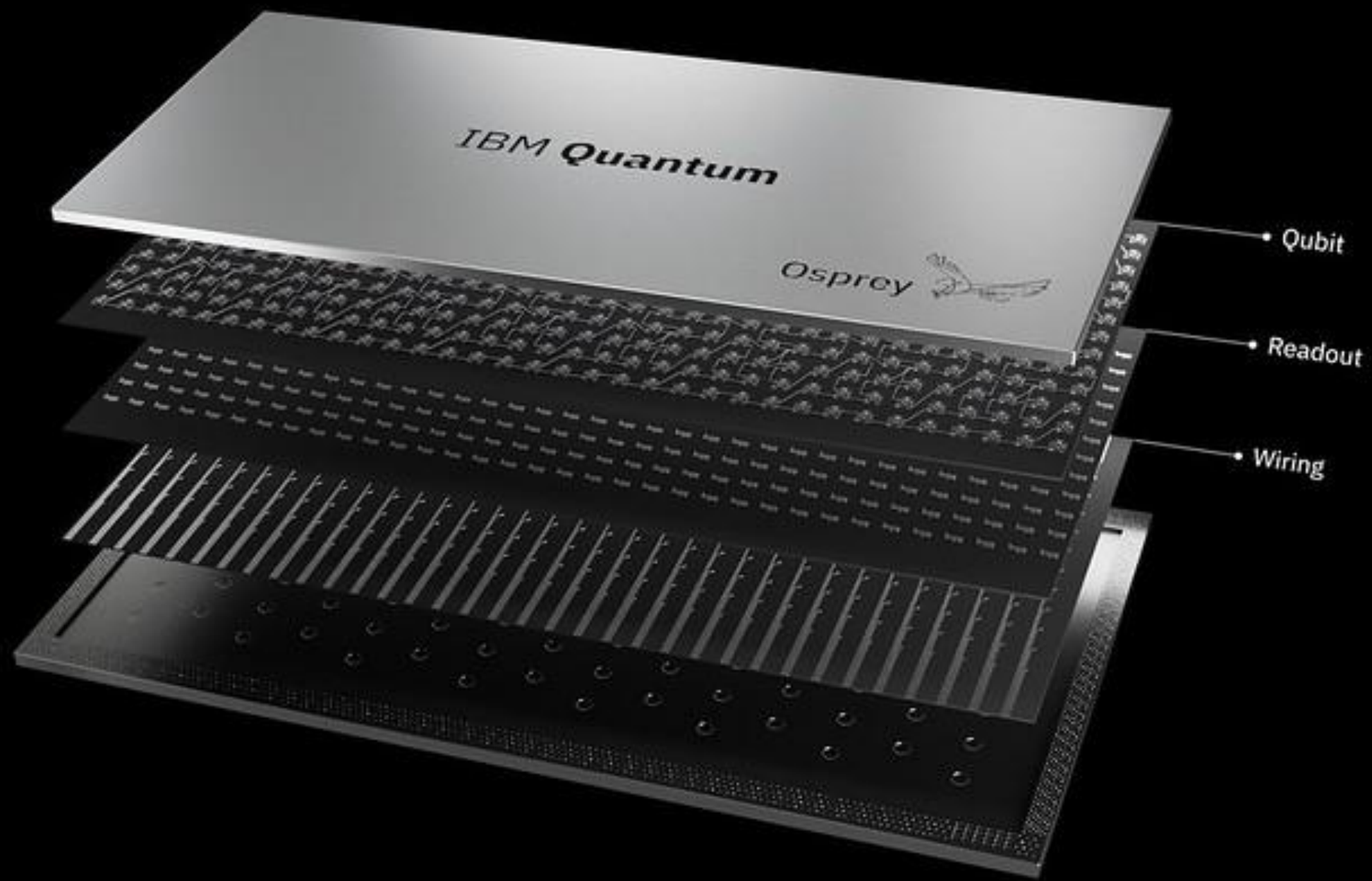
2020
Hummingbird
65 Qubits



2021
Eagle
127 Qubits

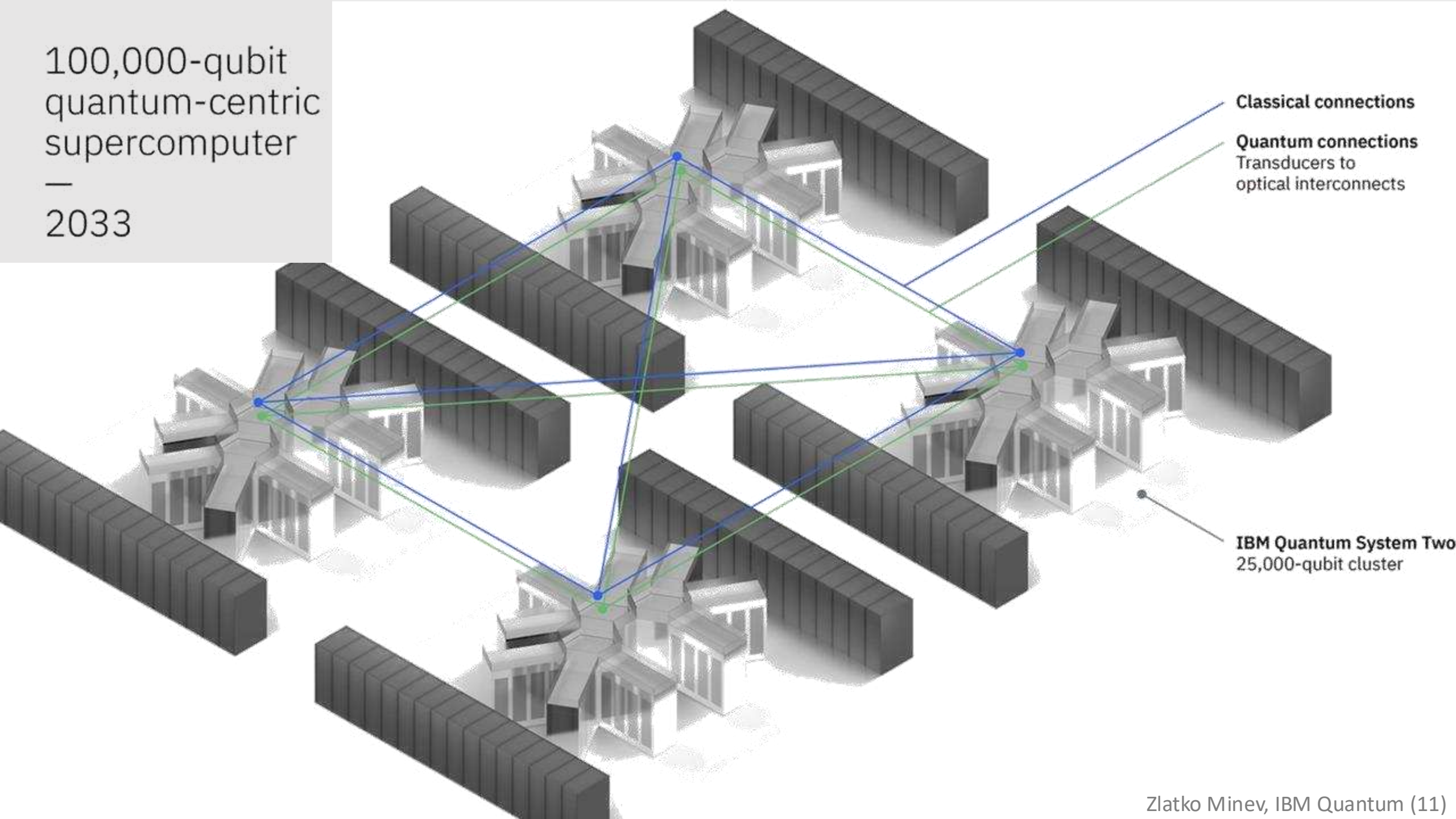


2022
Osprey
433 Qubits





100,000-qubit
quantum-centric
supercomputer
—
2033



Development Roadmap

	2016–2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2033+
	Run quantum circuits on the IBM Quantum Platform	Release multi-dimensional roadmap publicly with initial aim focused on scaling	Enhancing quantum execution speed by 100x with Qiskit Runtime	Bring dynamic circuits to unlock more computations	Enhancing quantum execution speed by 5x with quantum serverless and Execution modes	Improving quantum circuit quality and speed to allow 5K gates with parametric circuits	Enhancing quantum execution speed and parallelization with partitioning and quantum modularity	Improving quantum circuit quality to allow 7.5K gates	Improving quantum circuit quality to allow 10K gates	Improving quantum circuit quality to allow 15K gates	Improving quantum circuit quality to allow 100M gates	Beyond 2033, quantum-centric supercomputers will include 1000's of logical qubits unlocking the full power of quantum computing
Data Scientist						Platform						
						Code assistant	Functions	Mapping Collection	Specific Libraries			General purpose QC libraries
Researchers					Middleware							
					Quantum Serverless	Transpiler Service	Resource Management	Circuit Knitting x P	Intelligent Orchestration			Circuit libraries
Quantum Physicist			Qiskit Runtime									
	IBM Quantum Experience		QASM3	Dynamic circuits	Execution Modes	Heron (5K)	Flamingo (5K)	Flamingo (7.5K)	Flamingo (10K)	Flamingo (15K)	Starling (100M)	Blue Jay (1B)
	Early	Falcon				Error Mitigation	Error Mitigation	Error Mitigation	Error Mitigation	Error Mitigation	Error correction	Error correction
	Canary 5 qubits Albatross 16 qubits Penguin 20 qubits Prototype 53 qubits	Benchmarking 27 qubits				5k gates 133 qubits Classical modular 133x3 = 399 qubits	5k gates 156 qubits Quantum modular 156x7 = 1092 qubits	7.5k gates 156 qubits Quantum modular 156x7 = 1092 qubits	10k gates 156 qubits Quantum modular 156x7 = 1092 qubits	15k gates 156 qubits Quantum modular 156x7 = 1092 qubits	100M gates 200 qubits Error corrected modularity	1B gates 2000 qubits Error corrected modularity

Innovation Roadmap

Software Innovation	IBM Quantum Experience	Qiskit	Application modules	Qiskit Runtime	Serverless	AI enhanced quantum	Resource management	Scalable circuit knitting	Error correction decoder			
		Circuit and operator API with compilation to multiple targets	Modules for domain specific application and algorithm workflows	Performance and abstract through Primitives	Demonstrate concepts of quantum centric-supercomputing	Prototype demonstrations of AI enhanced circuit transpilation	System partitioning to enable parallel execution	Circuit partitioning with classical reconstruction at HPC scale	Demonstration of a quantum system with real-time error correction decoder			
Hardware Innovation	Early	Falcon	Hummingbird	Eagle	Osprey	Condor	Flamingo	Kookaburra	Cockatoo	Starling		
	Canary 5 qubits Albatross 16 qubits Penguin 20 qubits Prototype 53 qubits	Demonstrate scaling with I/O routing with Bump bonds	Demonstrate scaling with multiplexing readout	Demonstrate scaling with MLW and TSV	Enabling scaling with high density signal delivery	Single system scaling and fridge capacity	Demonstrate scaling with modular connectors	Demonstrate scaling with nonlocal c-coupler	Demonstrate path to improved quality with logical memory	Demonstrate path to improved quality with logical communication	Demonstrate path to improved quality with logical gates	
						Heron	Crossbill					
						Architecture based on tunable-couplers	m- coupler					

Executed by IBM

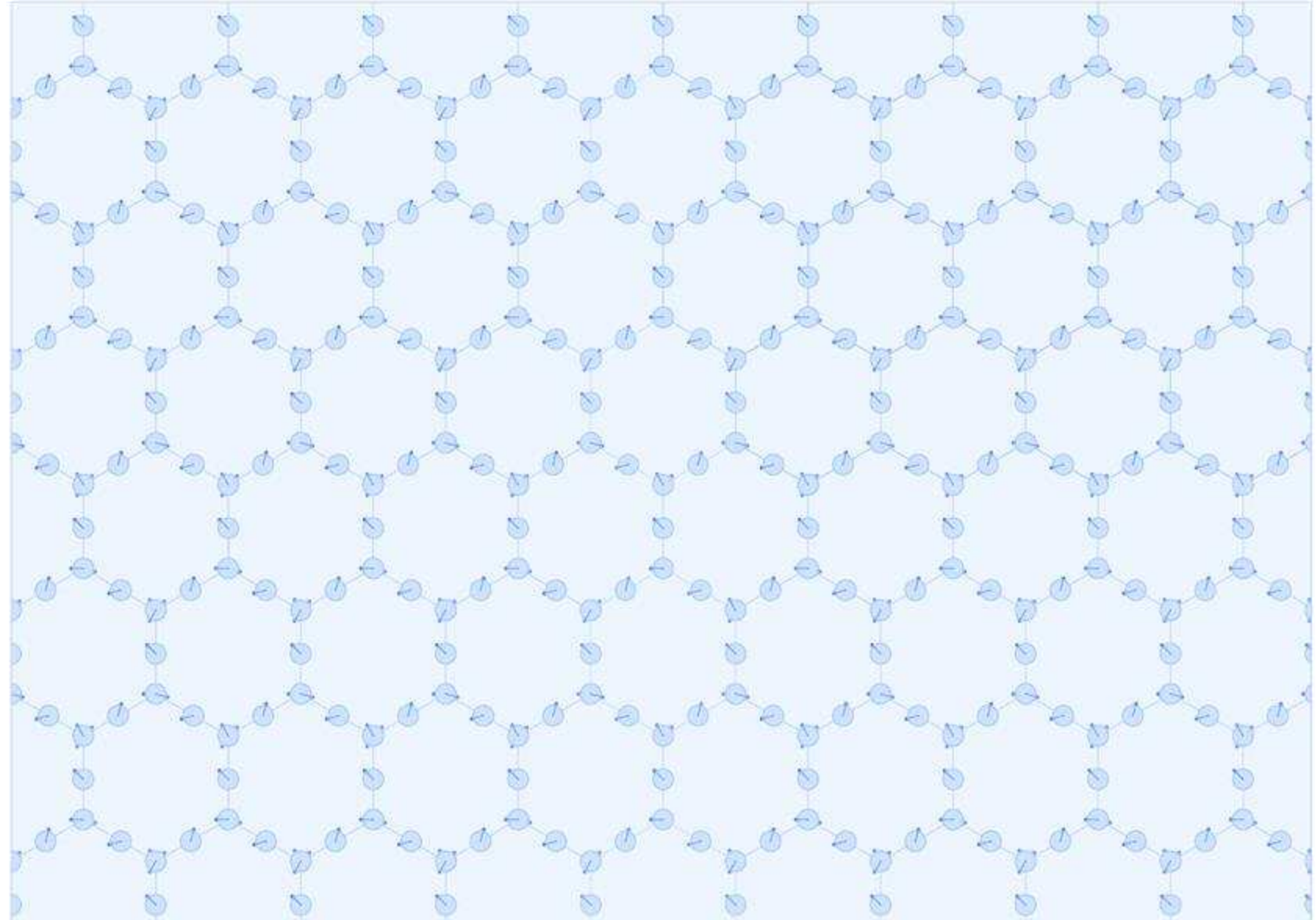
On target

Back to today ...

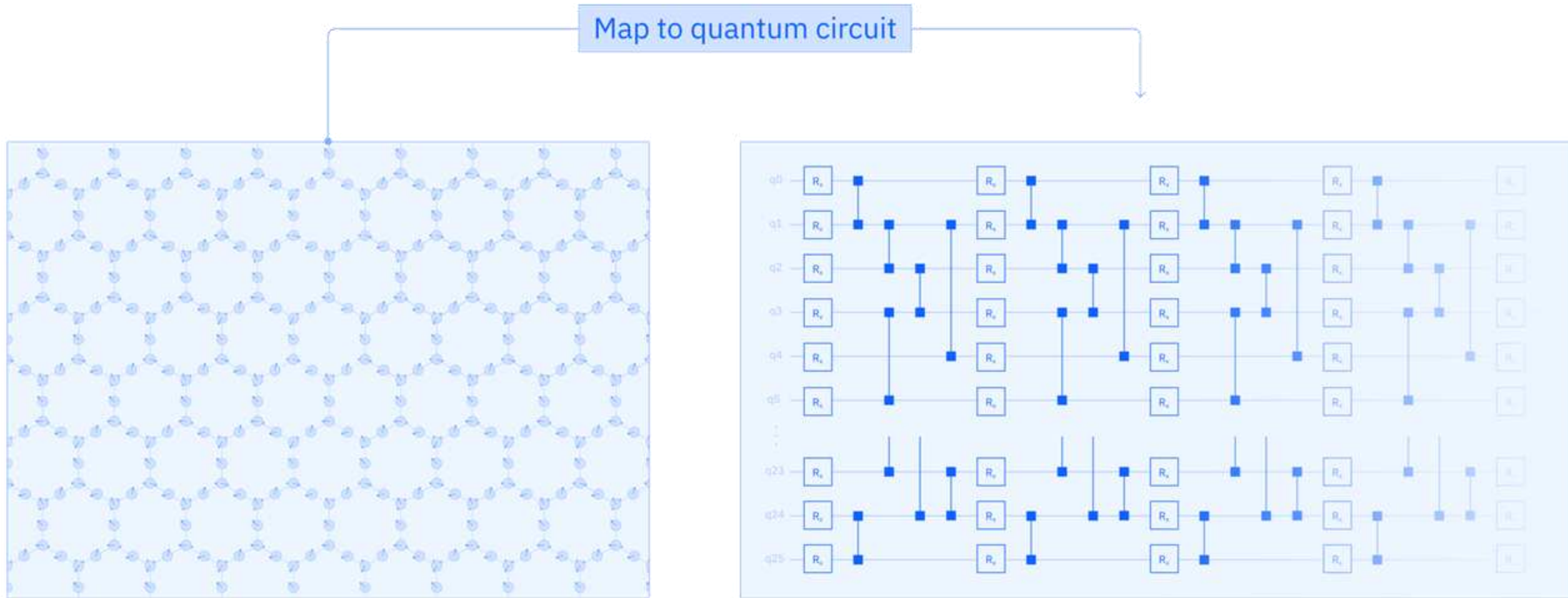
Example: Quantum simulation

Example task:

Simulate the out-of-equilibrium quantum dynamics of a 2D spin chain lattice to find the evolution of the global and local magnetization.

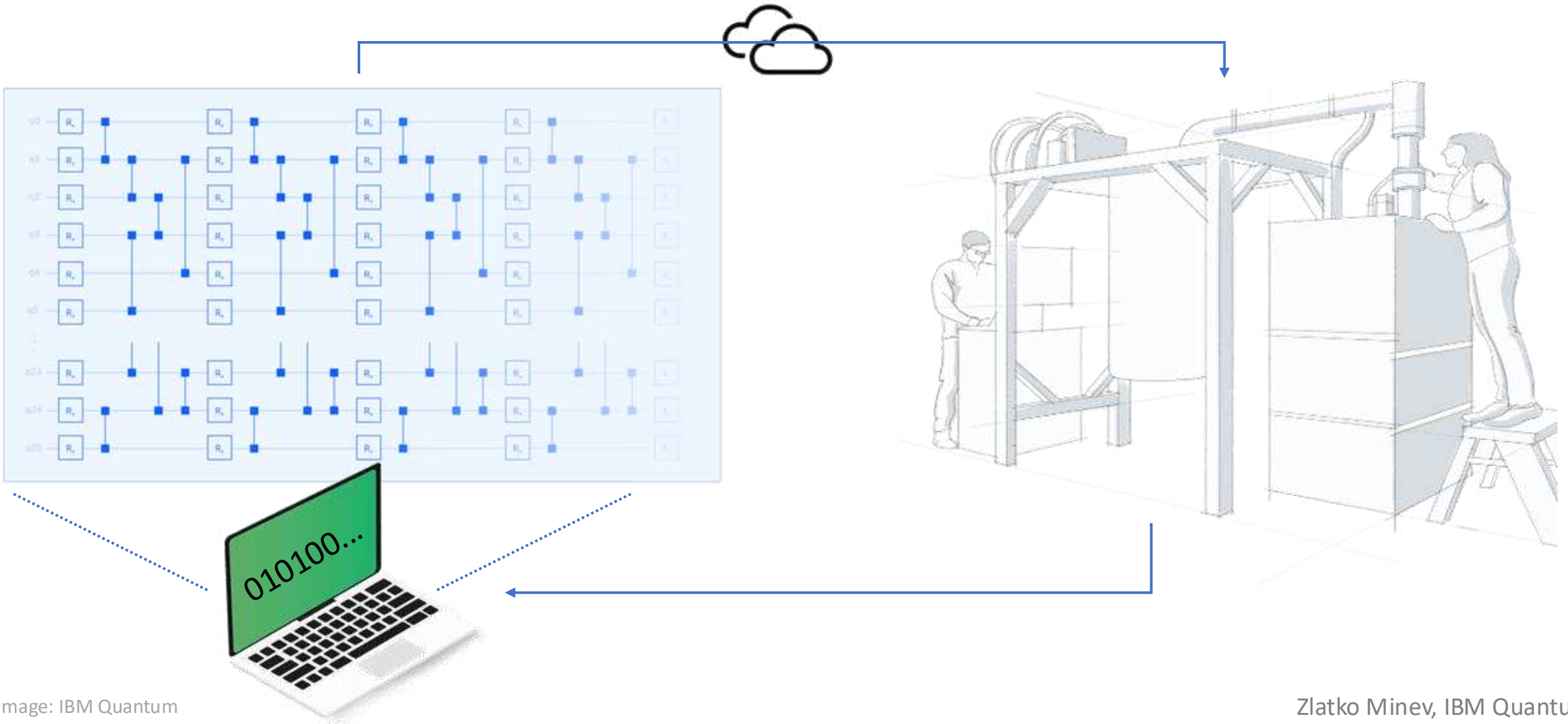


Quantum simulation on a quantum computer



Quantum simulation on a quantum computer

Execute on a real quantum computer device and obtain results as classical data



Biggest challenge?

Please do share

hardware
development

decoherence

loss

stability

heat

algo
development

error correction
overheads

high error rates

scalability

Noise
(Errors)

Biggest challenge?

engineering

modularization

importance of N
in NISQ

hype

need CS/EE
talent

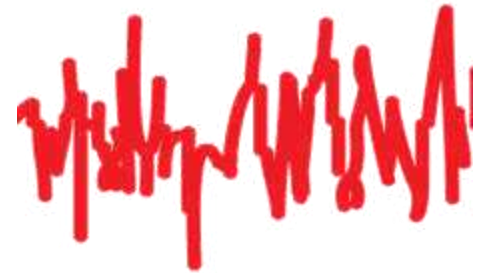
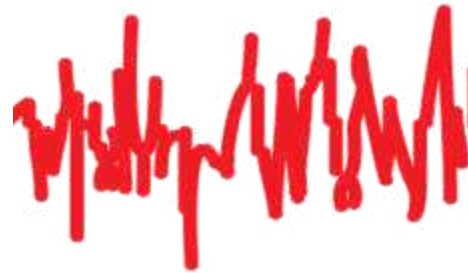
material
quality

gravity

expectations

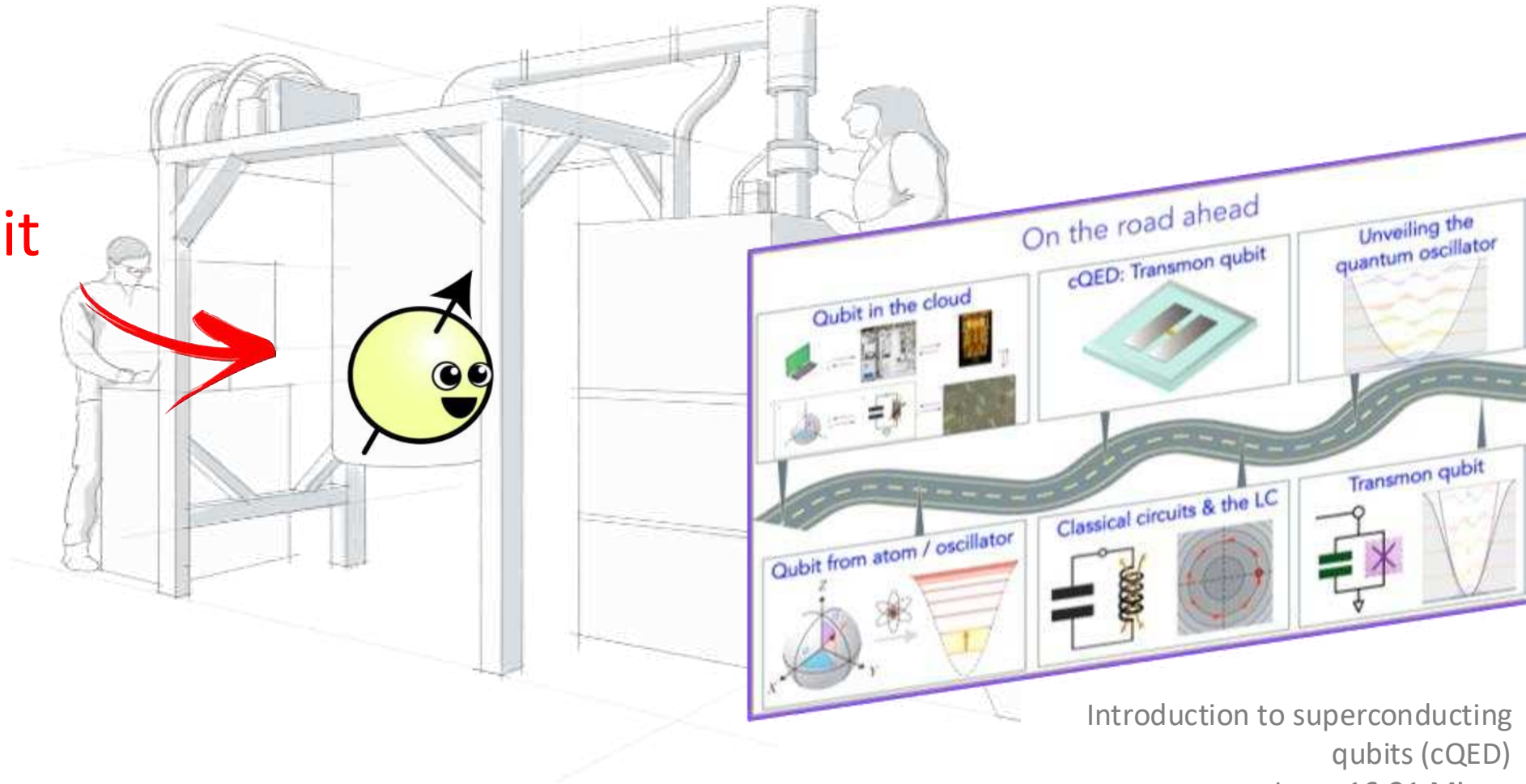
Biggest challenge

Noise
(Errors)



Hello World with a real experiment!

A qubit

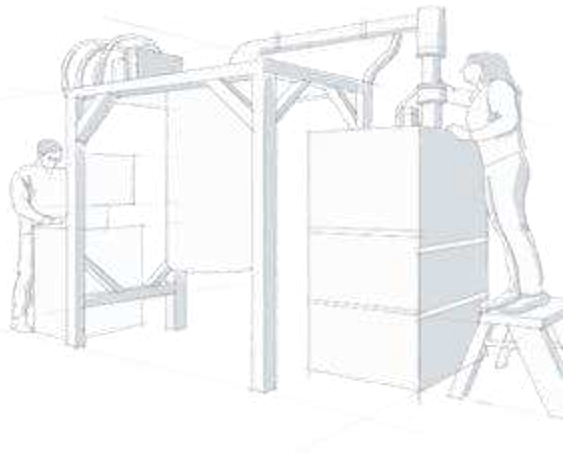


Introduction to superconducting
qubits (cQED)

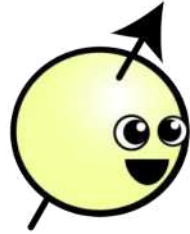
Lecs. 16-21 Minev

QGSS 2020 at qiskit.org/learn

Hello World! building blocks

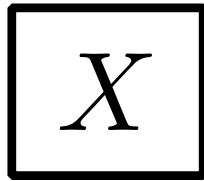


A qubit

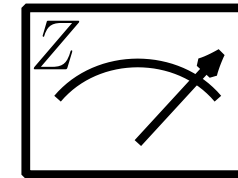
 $|1\rangle$ $|0\rangle$

Computational
basis states

Operations: qubit gate



Measurements: qubit observable



refresher:

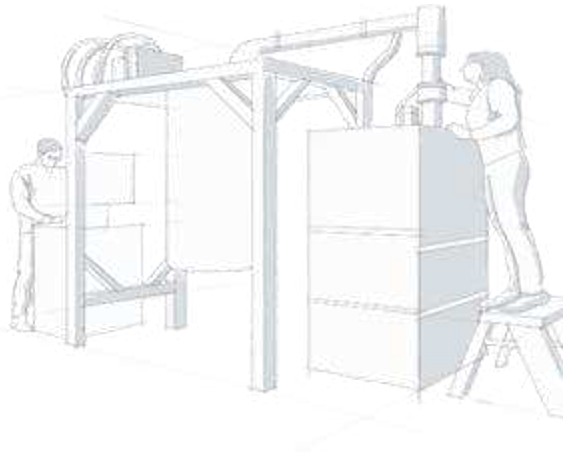
$$X |0\rangle = |1\rangle$$

$$X |1\rangle = |0\rangle$$

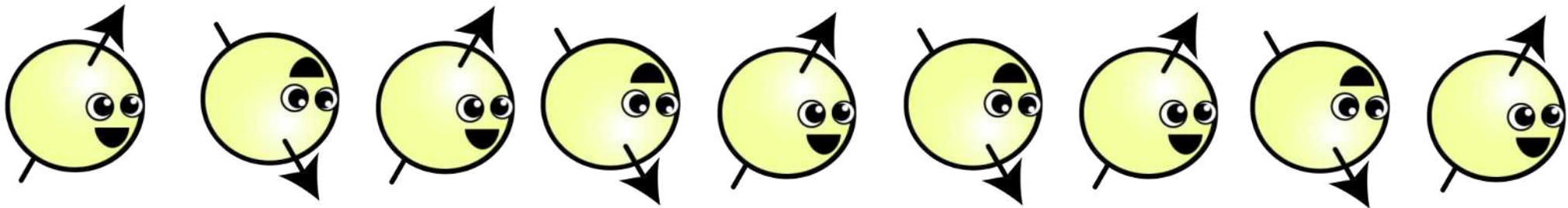
$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! Even-odd algo: qubit flipper



Task: Classify or report if a classical positive integer d is even or odd.



flip spin d times, measure polarization

refresher:

$$X |0\rangle = |1\rangle$$

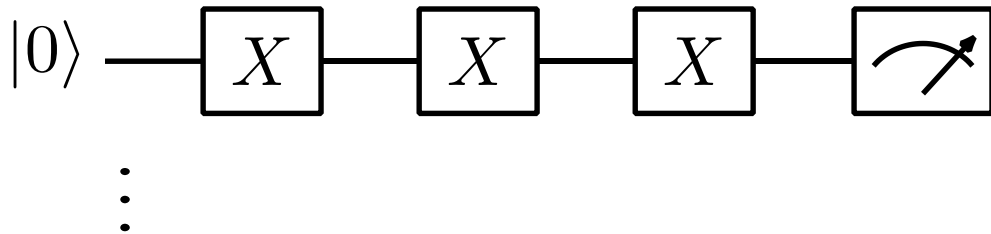
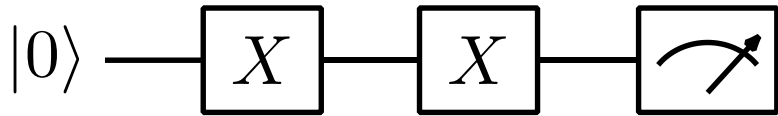
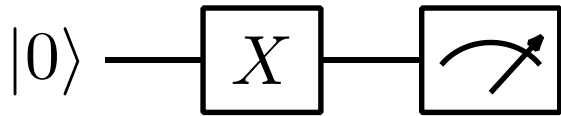
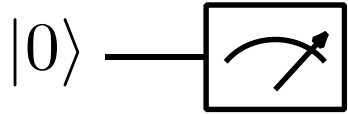
$$X |1\rangle = |0\rangle$$

$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! qubit flipper quantum circuits

depth



refresher:

$$X |0\rangle = |1\rangle$$

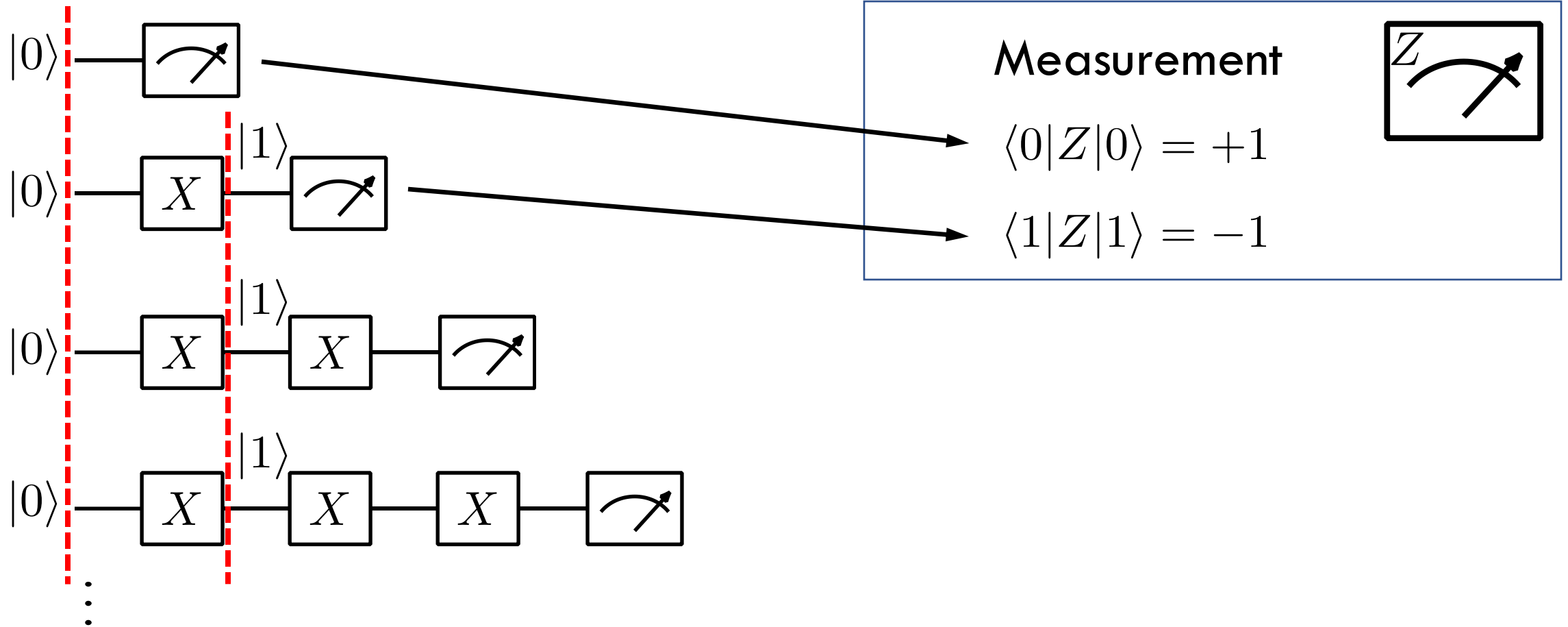
$$X |1\rangle = |0\rangle$$

$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! “debugger” step through

depth



refresher:

$$X |0\rangle = |1\rangle$$

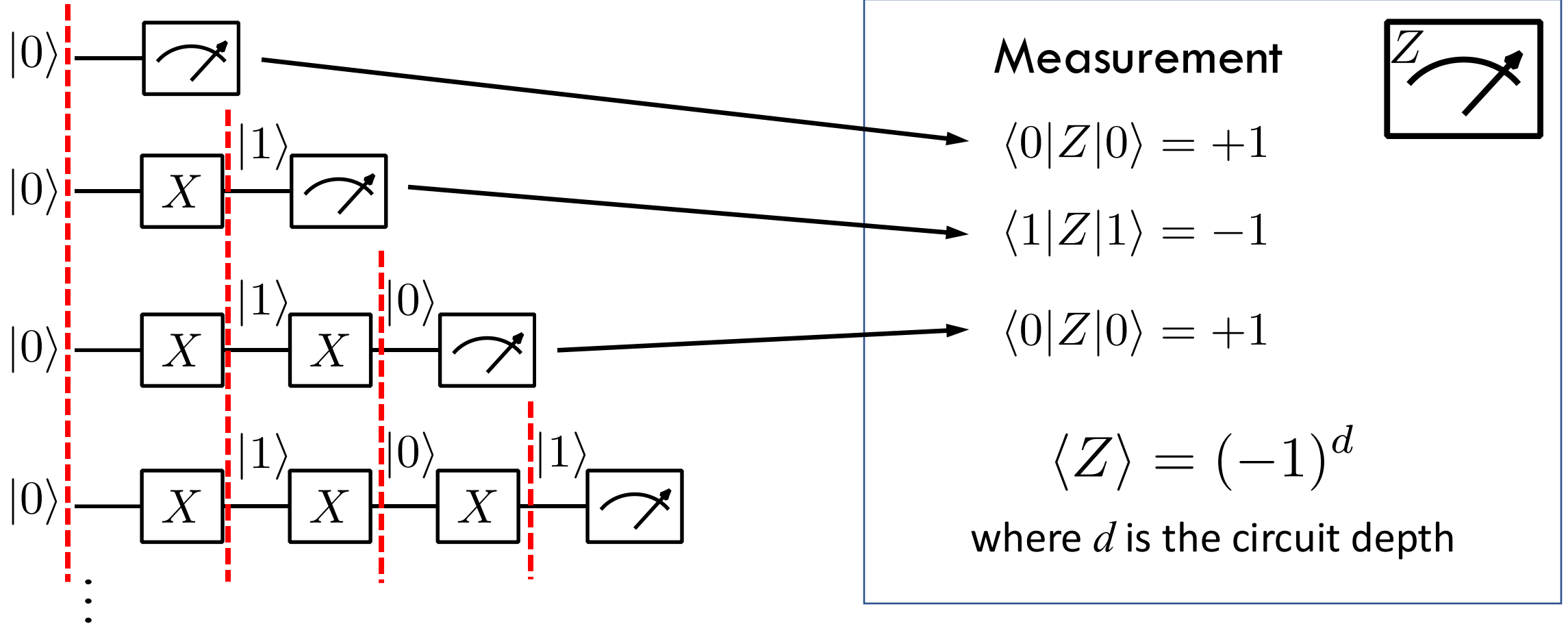
$$X |1\rangle = |0\rangle$$

$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! “debugger” step through

depth



refresher:

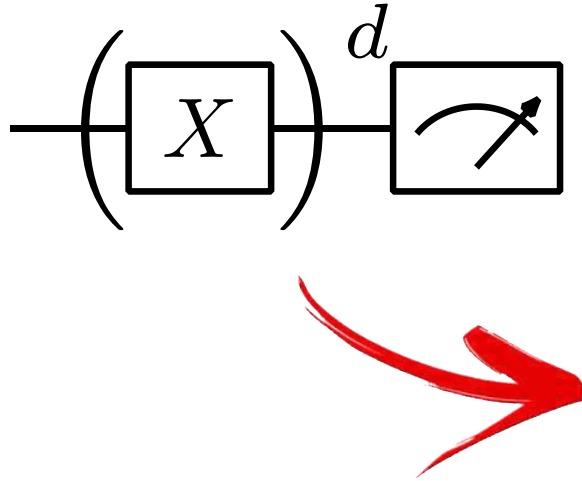
$$X |0\rangle = |1\rangle$$

$$X |1\rangle = |0\rangle$$

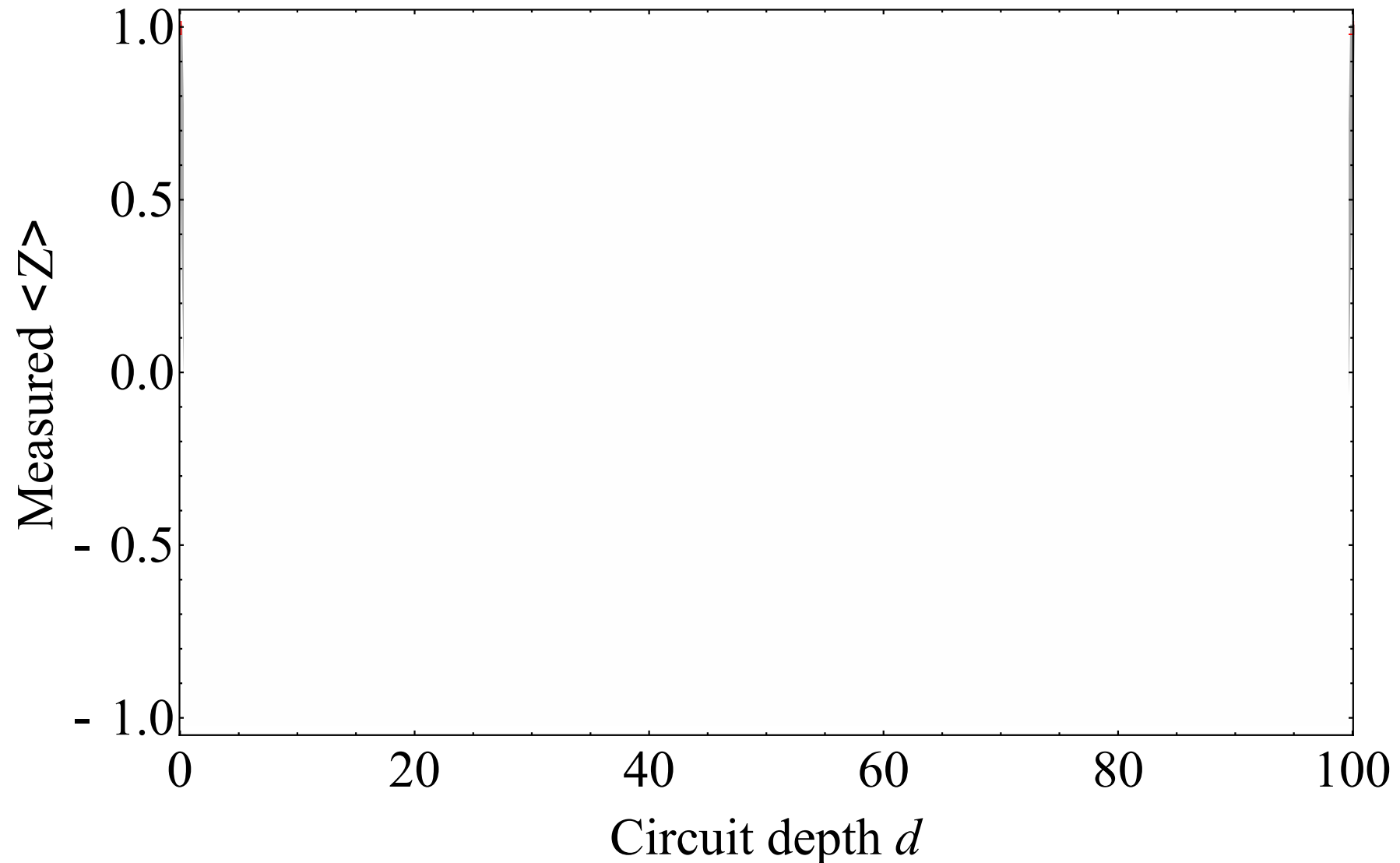
$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! Ideal expectation results

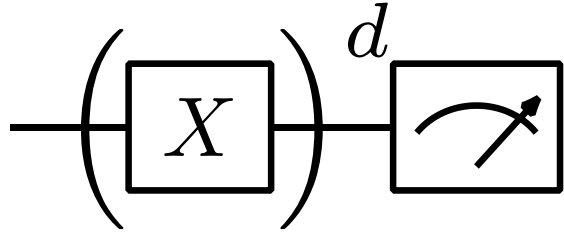


$$\langle Z \rangle = (-1)^d$$

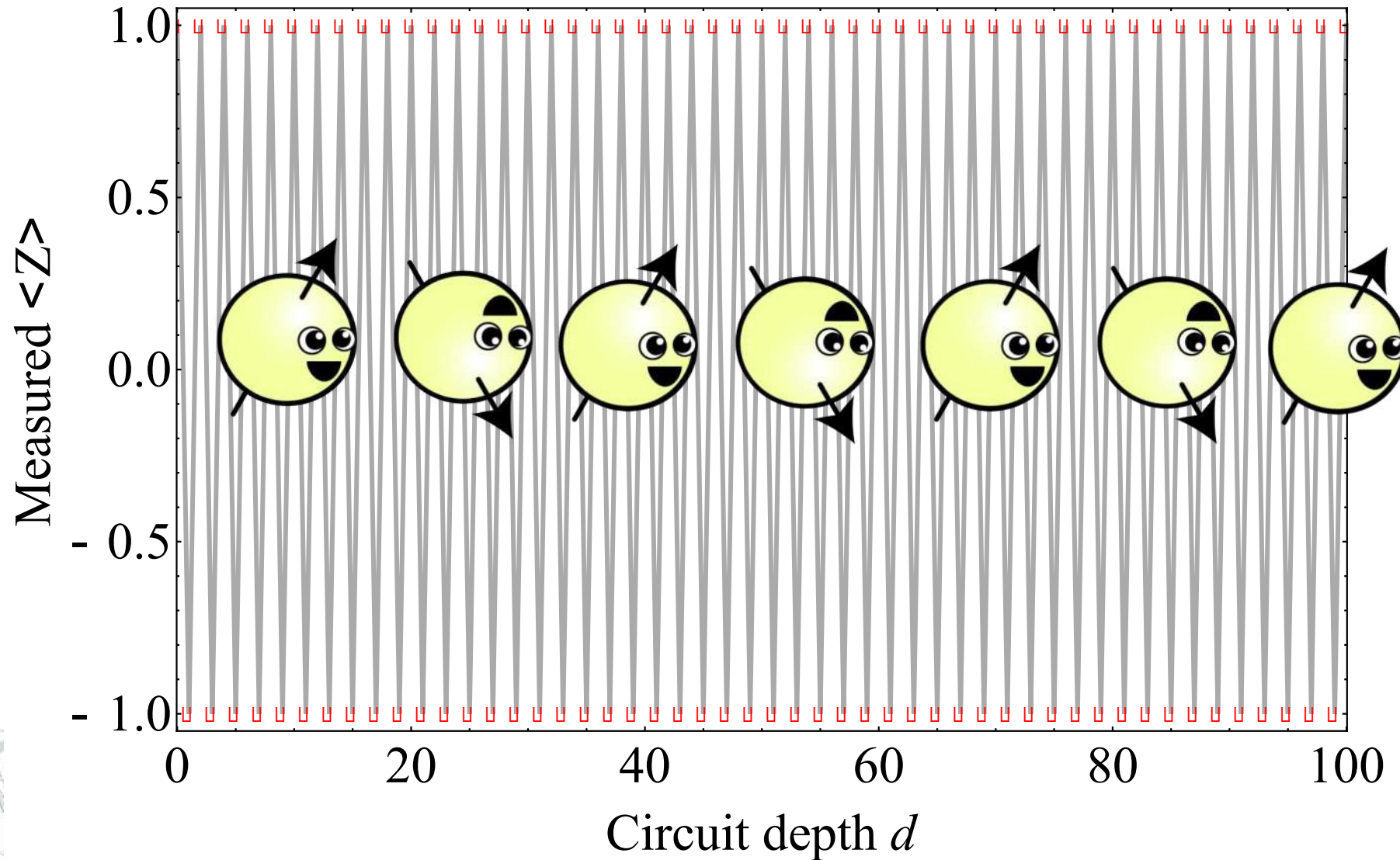
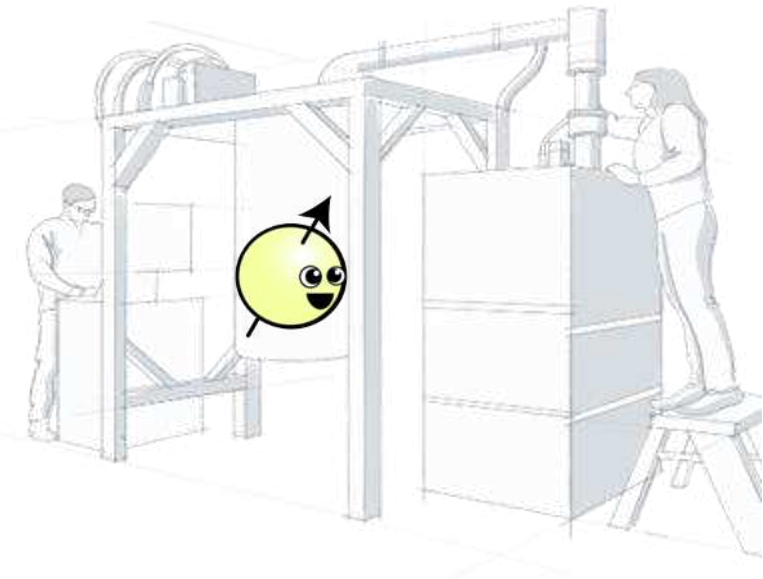


Hello World! Ideal expectation results

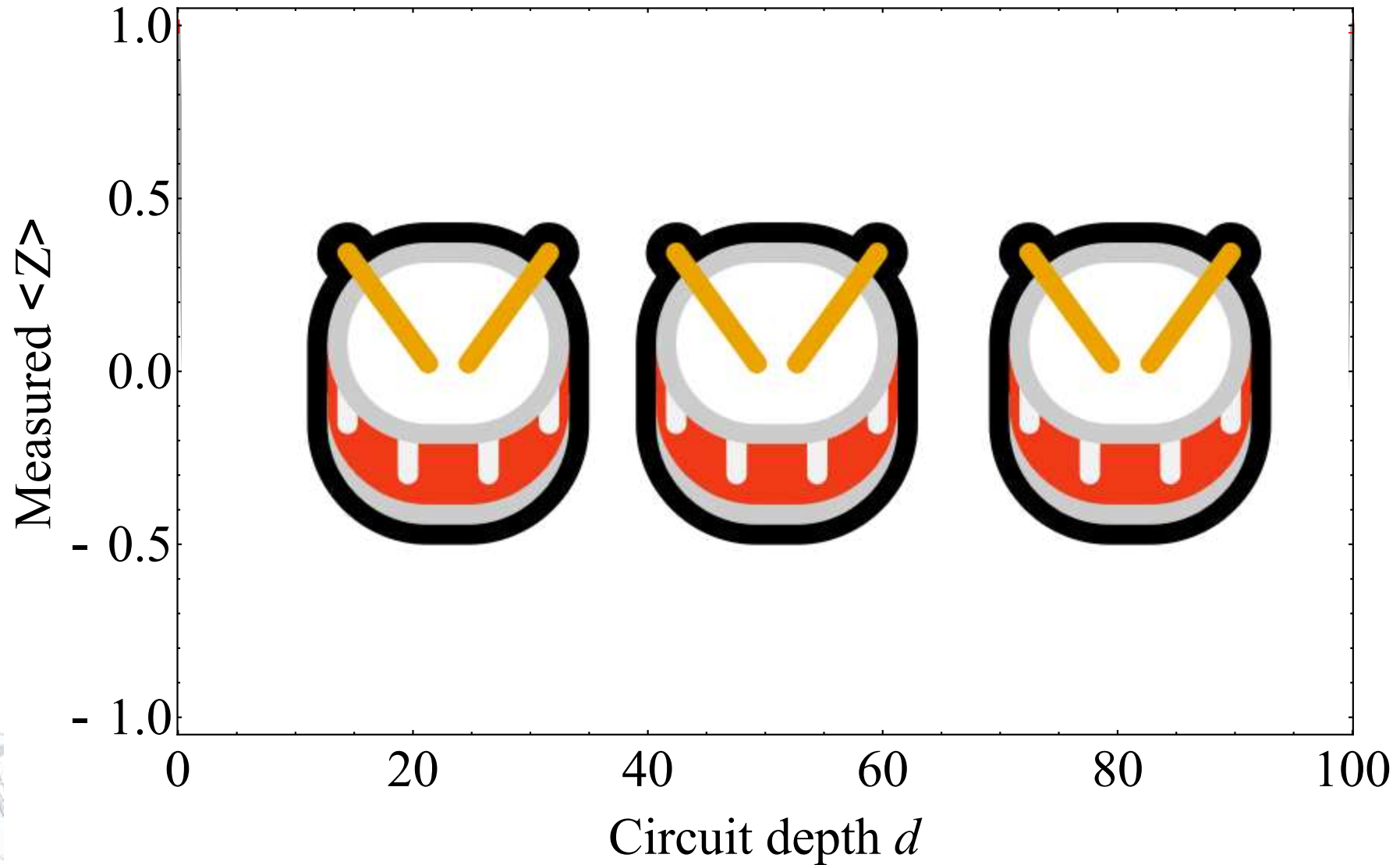
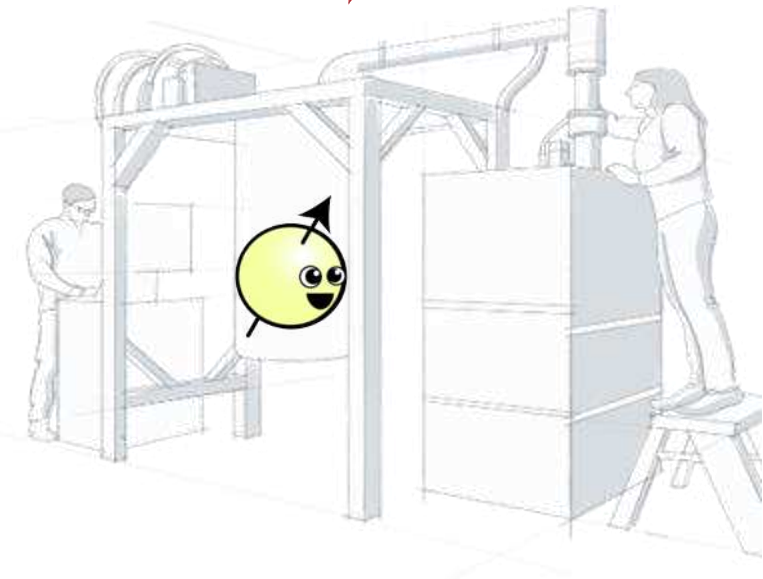
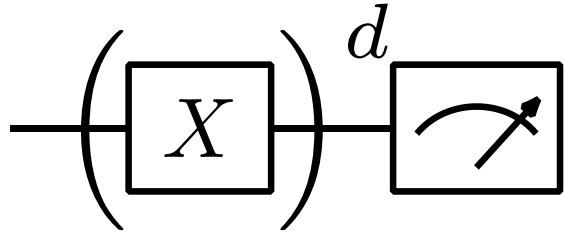
$$\langle Z \rangle = (-1)^d$$



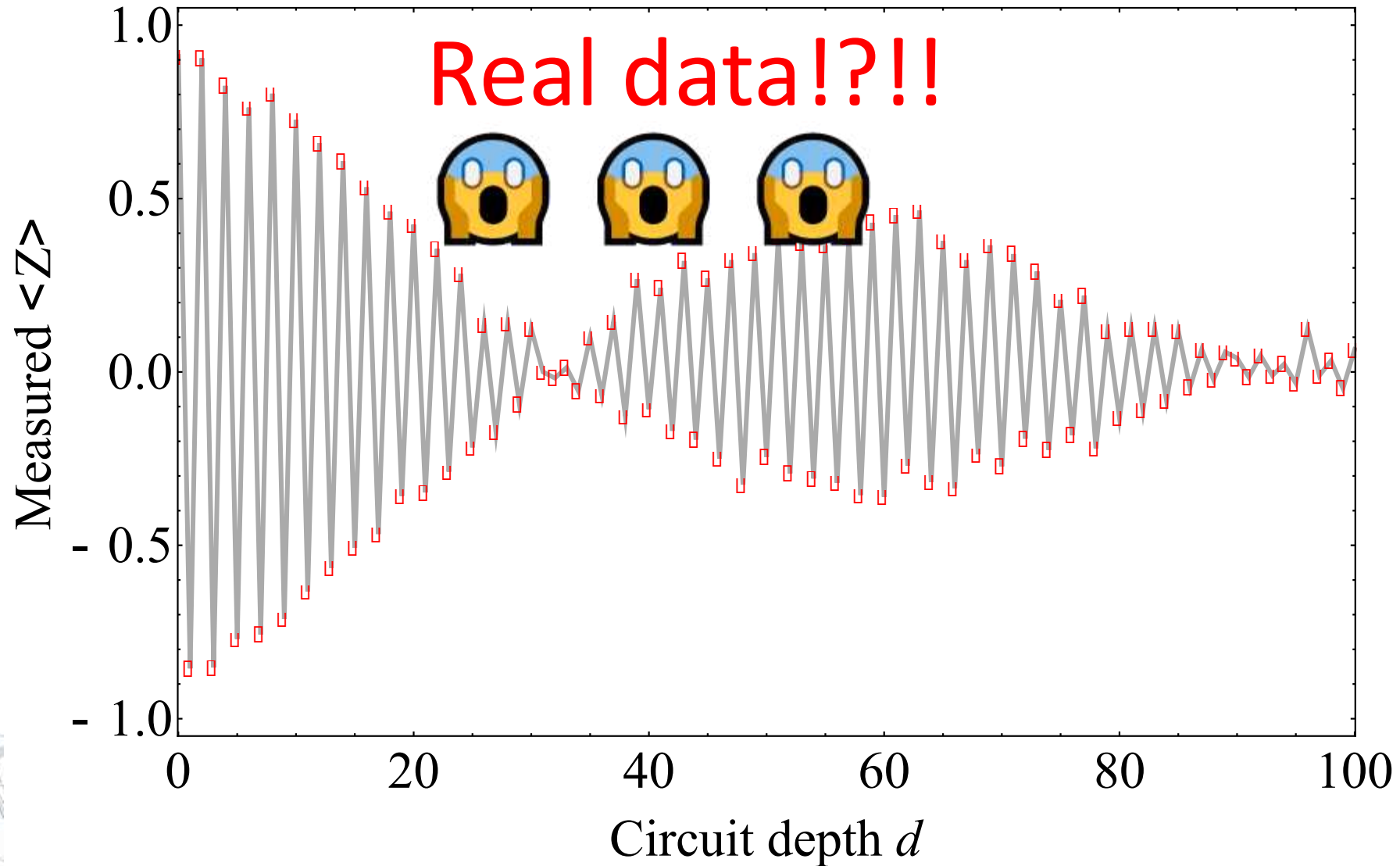
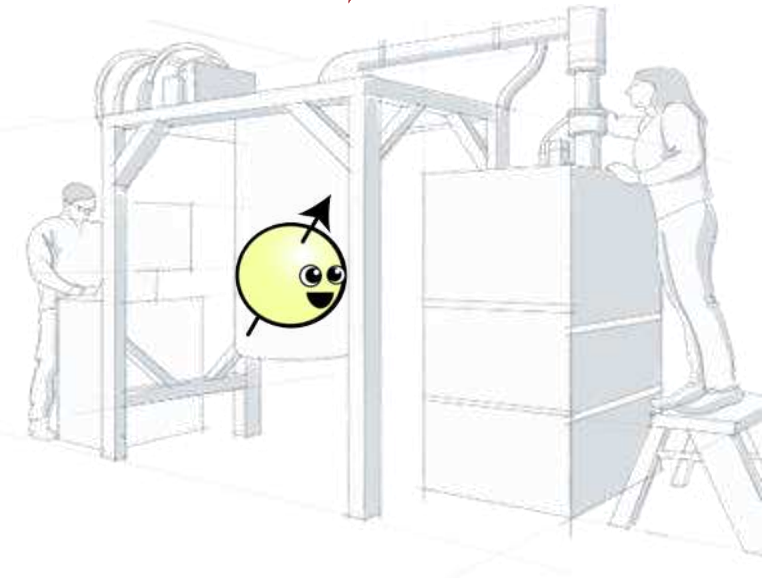
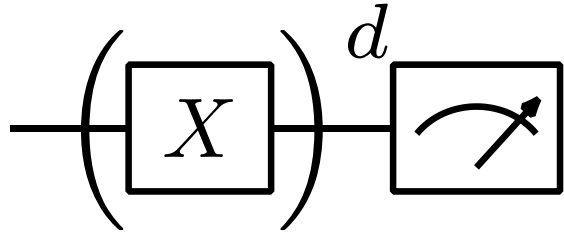
Let's run on a real device!



Hello World! Running



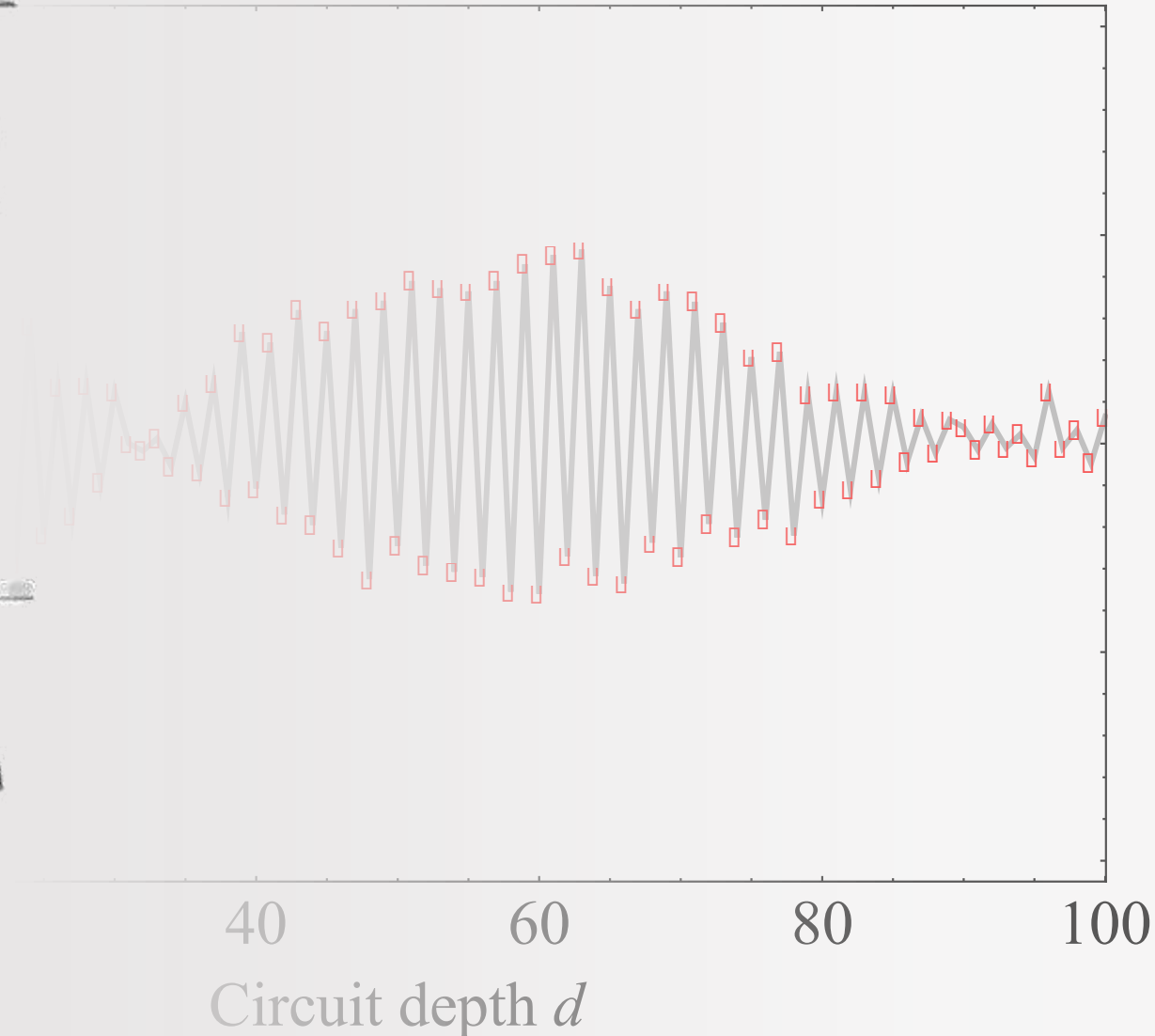
Hello World! Real expectation results

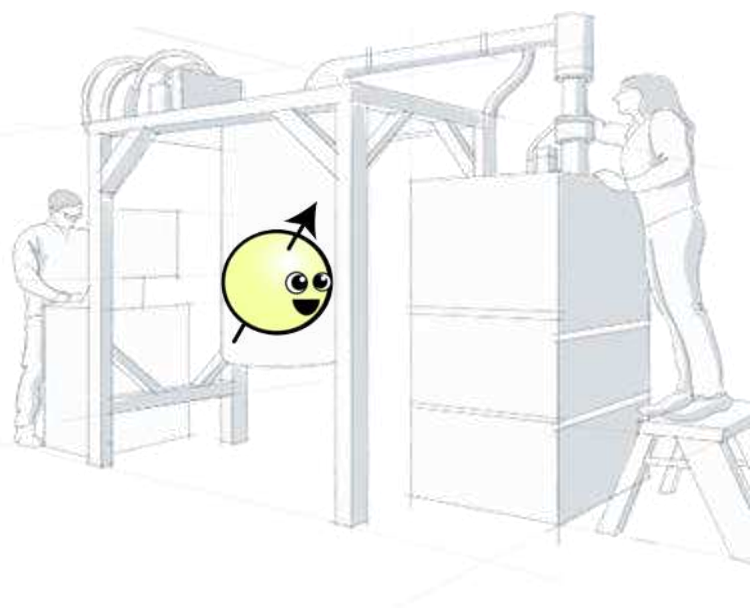


Real & noisy quantum processors: Why study noise?



“Well, your quantum computer is broken in every way possible simultaneously.”

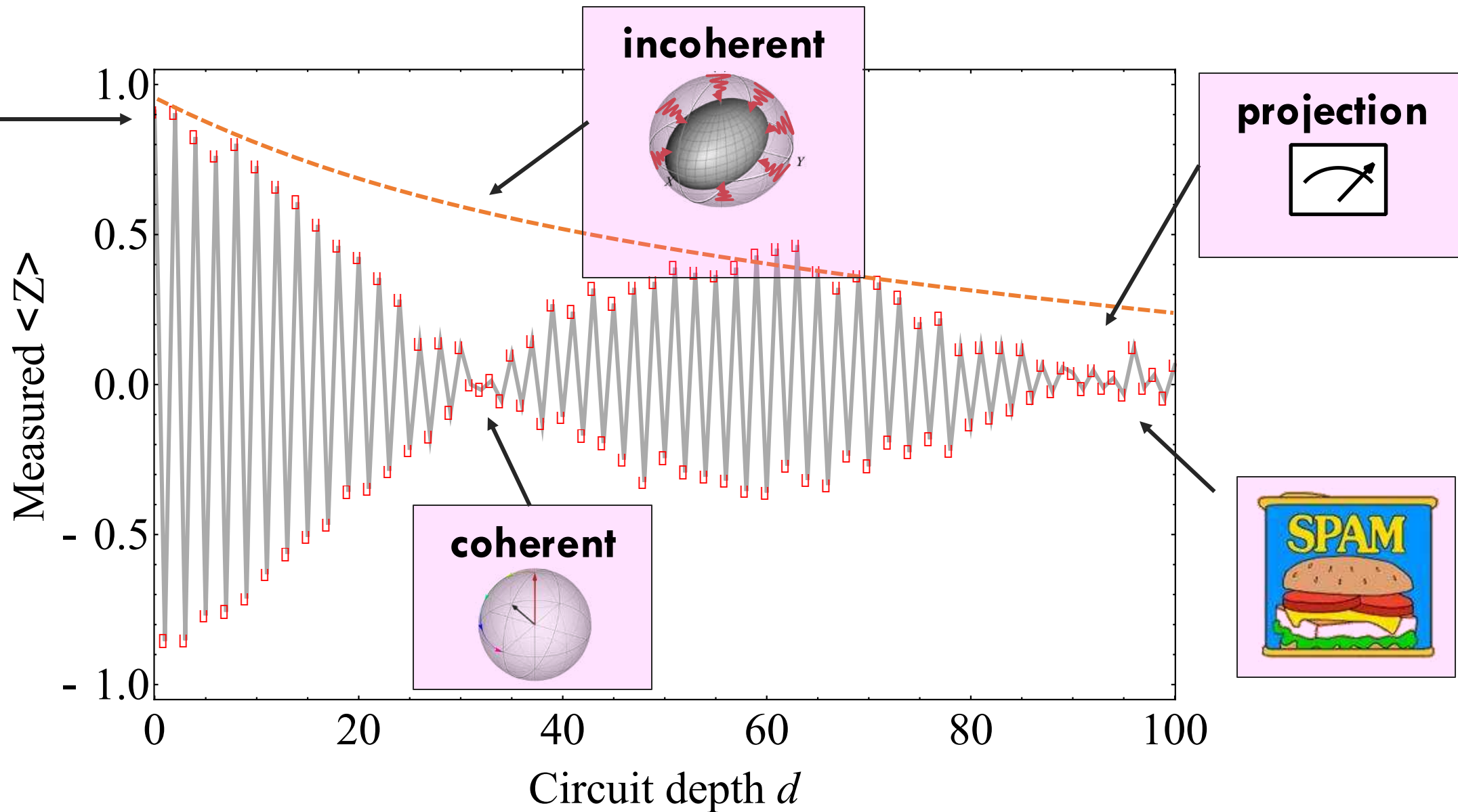




“Quantum phenomena
do *not* occur in a Hilbert space,
they occur in a laboratory.”

Asher Peres

Elements of 🤖 noise



How to deal with errors due to noise?

Monitor

Error occurs
Error detect

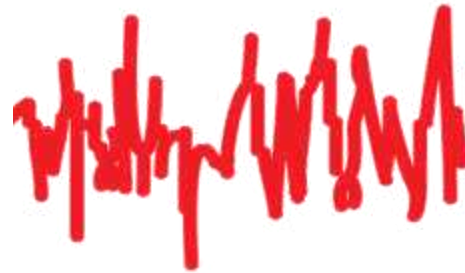


Quantum error correction

Shor, PRA (1995), ...

Monitor

Error anticipated
Tell signal detected

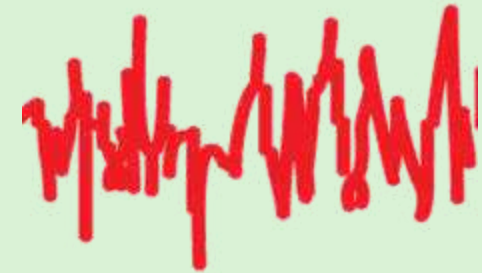


Catch and reverse

Minev, Nature (2019), ...

No monitor

Error occurs
Error undetected

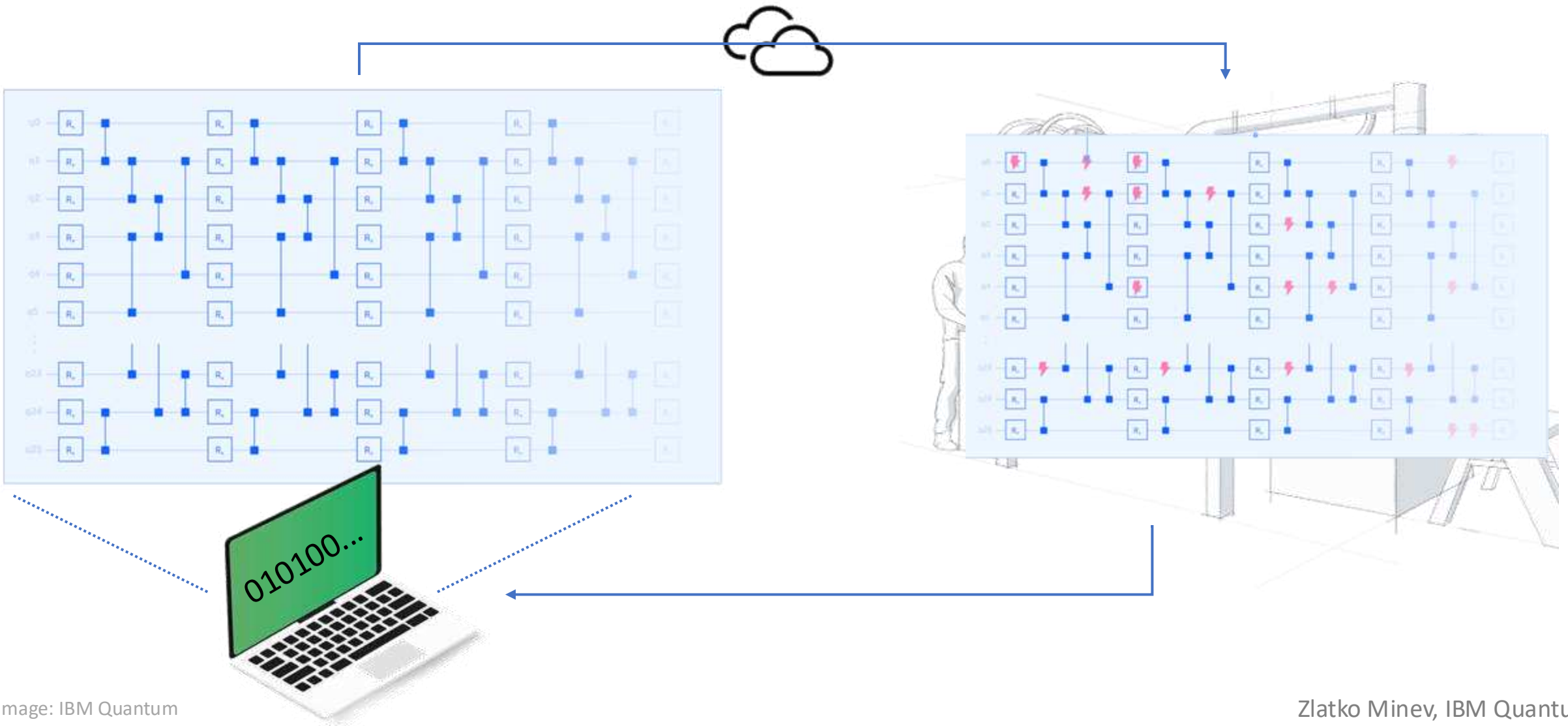


Error mitigation

... subject of today

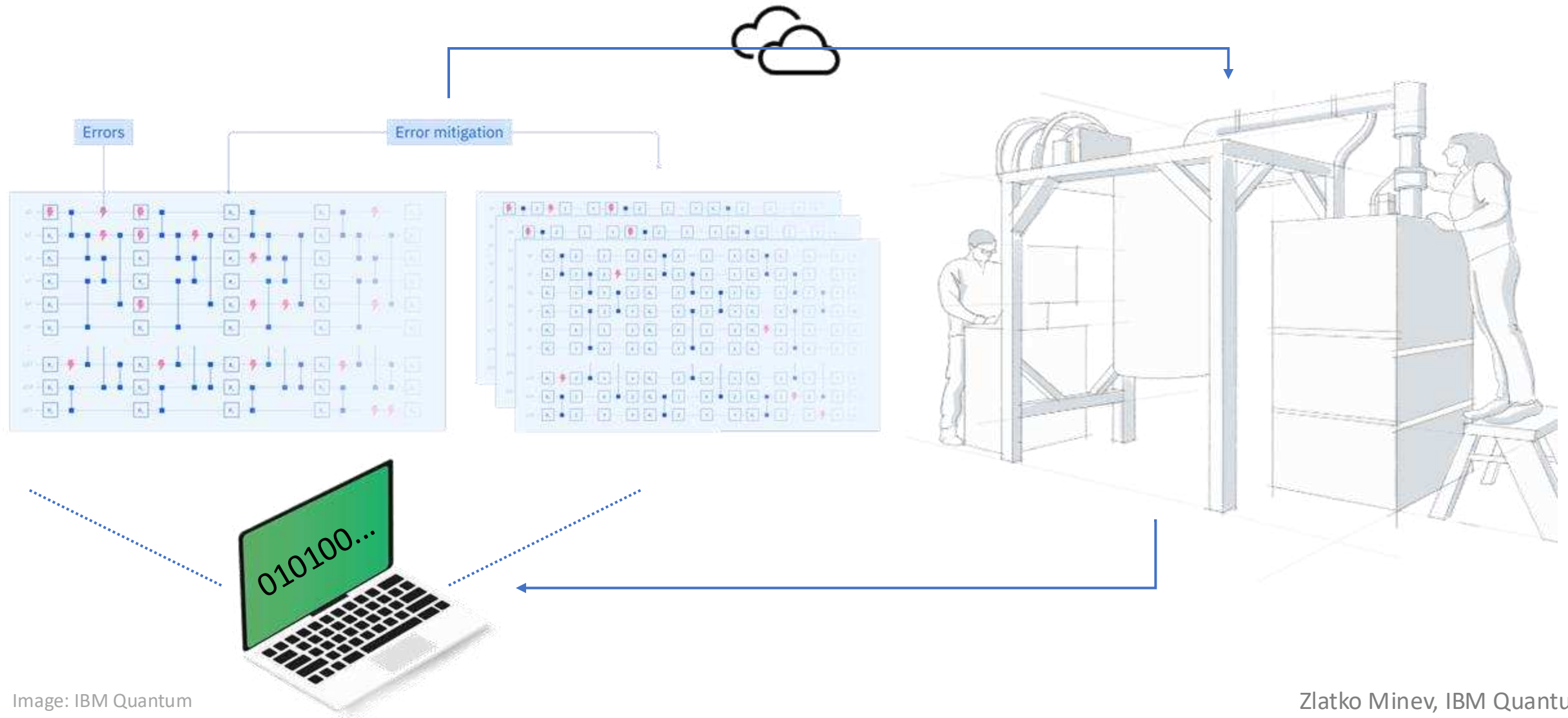
Quantum simulation on a noisy quantum computer

Execute on a real quantum computer device and obtain results as classical data

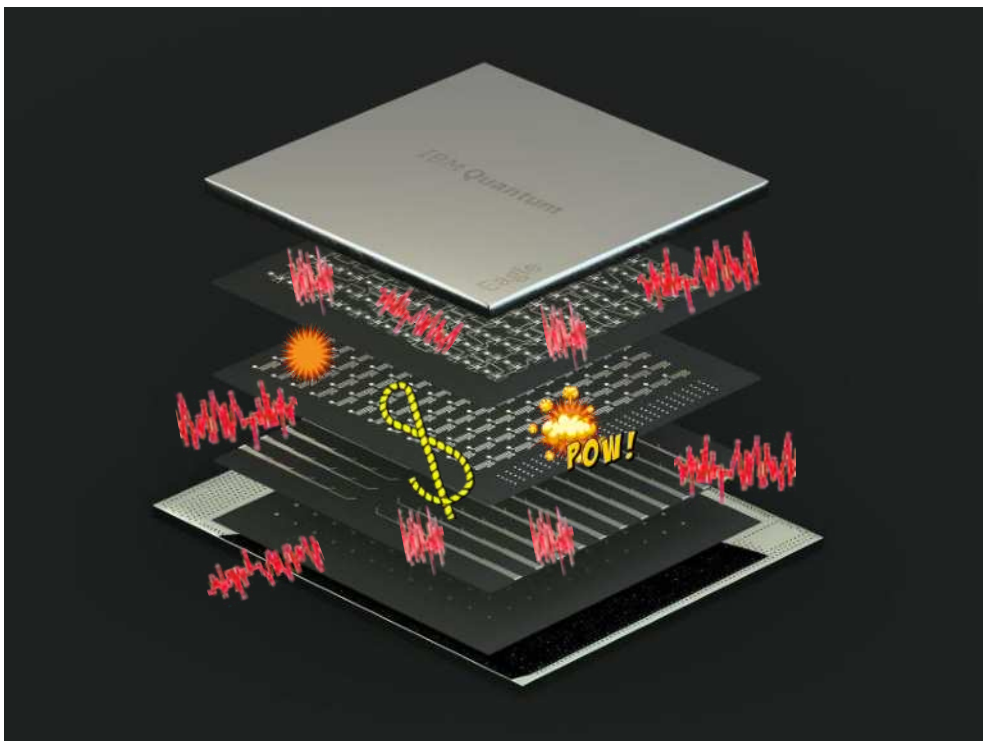


Quantum error mitigation overview

Execute on a real quantum computer device and obtain results as classical data



Error mitigation and error correction



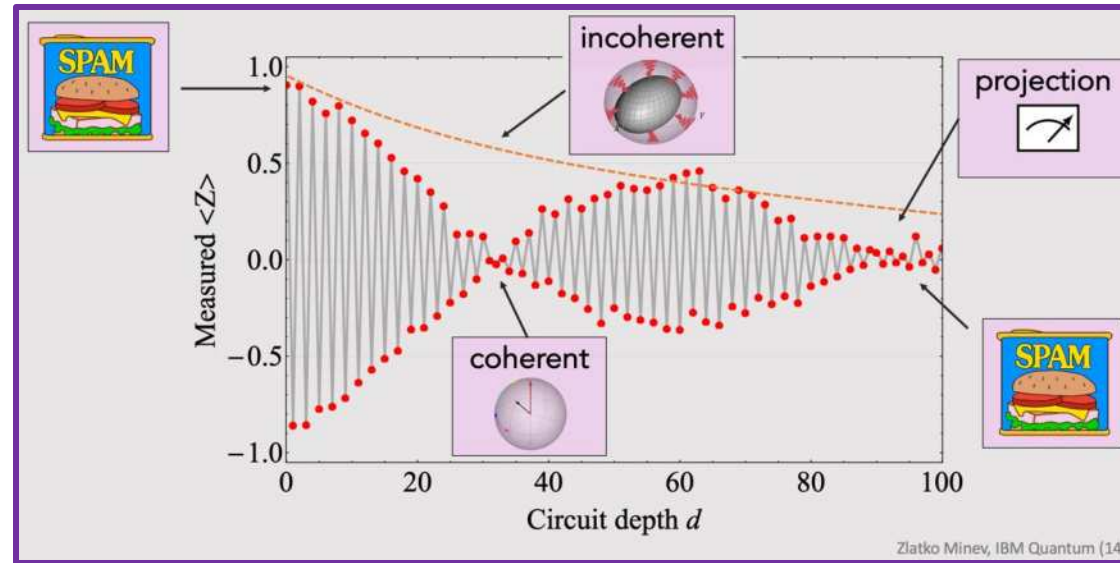
Error mitigation: working with what you have

- **benefit** suppress errors on classical results (expectation values)
- **q-cost** no extra qubits or hardware resources needed
- **c-cost** trades classical resources (post-processing) for lower error
- **limitation** bad asymptotic scaling: high number of samples & circuits

Error correction: protecting quantum information

- **benefit** suppress & correct errors to arbitrarily small level
- **q-cost** very large qubit and hardware overhead
- **c-cost** decoding and encoding can be classically costly
- **challenge** requires fault-tolerant operations and readout

Error mitigation landscape



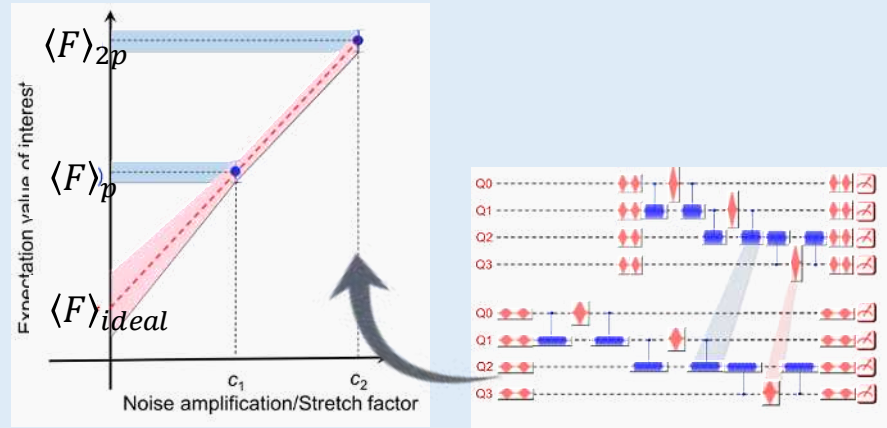
more
speed

more information,
accuracy



Error mitigation landscape

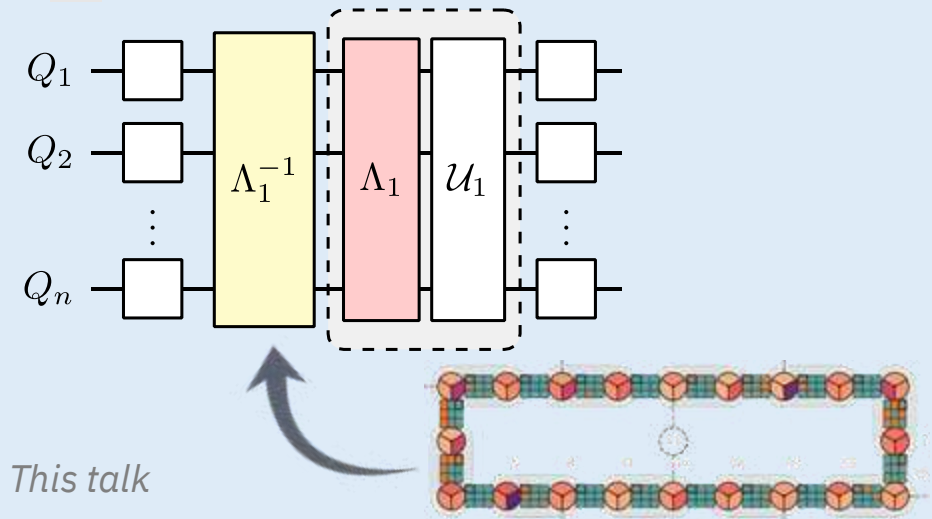
Zero-noise extrapolation (ZNE)



Nature 567, 491 (2019)

38

Probabilistic error cancelation (PEC)



more
speed



more information,
accuracy

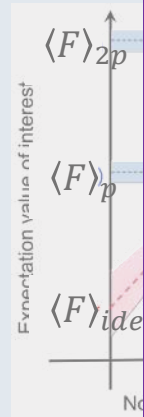


Zlatko Minev, IBM Quantum (38)

Error mitigation landscape

Zero-noise extrapolation (ZNE)

Probabilistic error cancelation (PEC)



 **Qiskit** IBM Qiskit

Overview Learn Community

English

Get started

- Overview
- Getting Started
- backend.run vs. Qiskit Runtime
- Introduction to primitives

Tutorials

- Get started with Estimator
- Get started with error suppression and error mitigation
- VQE with Estimator
- CHSH with Estimator
- Get started with Sampler

Qiskit Runtime IBM Client documentation > Error suppression and error mitigation with Qiskit Runtime

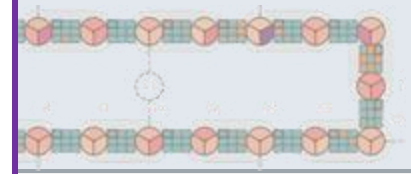
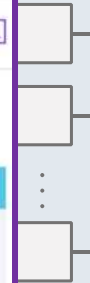
NOTE

This page was generated from [docs/tutorials/Error-Suppression-and-Error-Mitigation.ipynb](#).

Error suppression and error mitigation with Qiskit Runtime

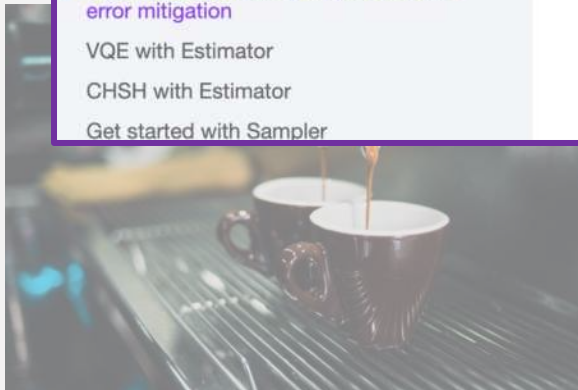
```
[1]: import datetime
import numpy as np
import matplotlib as mpl
import matplotlib.pyplot as plt
plt.rcParams.update({"text.usetex": True})
plt.rcParams["figure.figsize"] = (6,4)
mpl.rcParams["figure.dpi"] = 200

from qiskit_ibm_runtime import Estimator, Session, QiskitRuntimeService,
Options
from qiskit.quantum_info import SparsePauliOp
from qiskit import QuantumCircuit
```



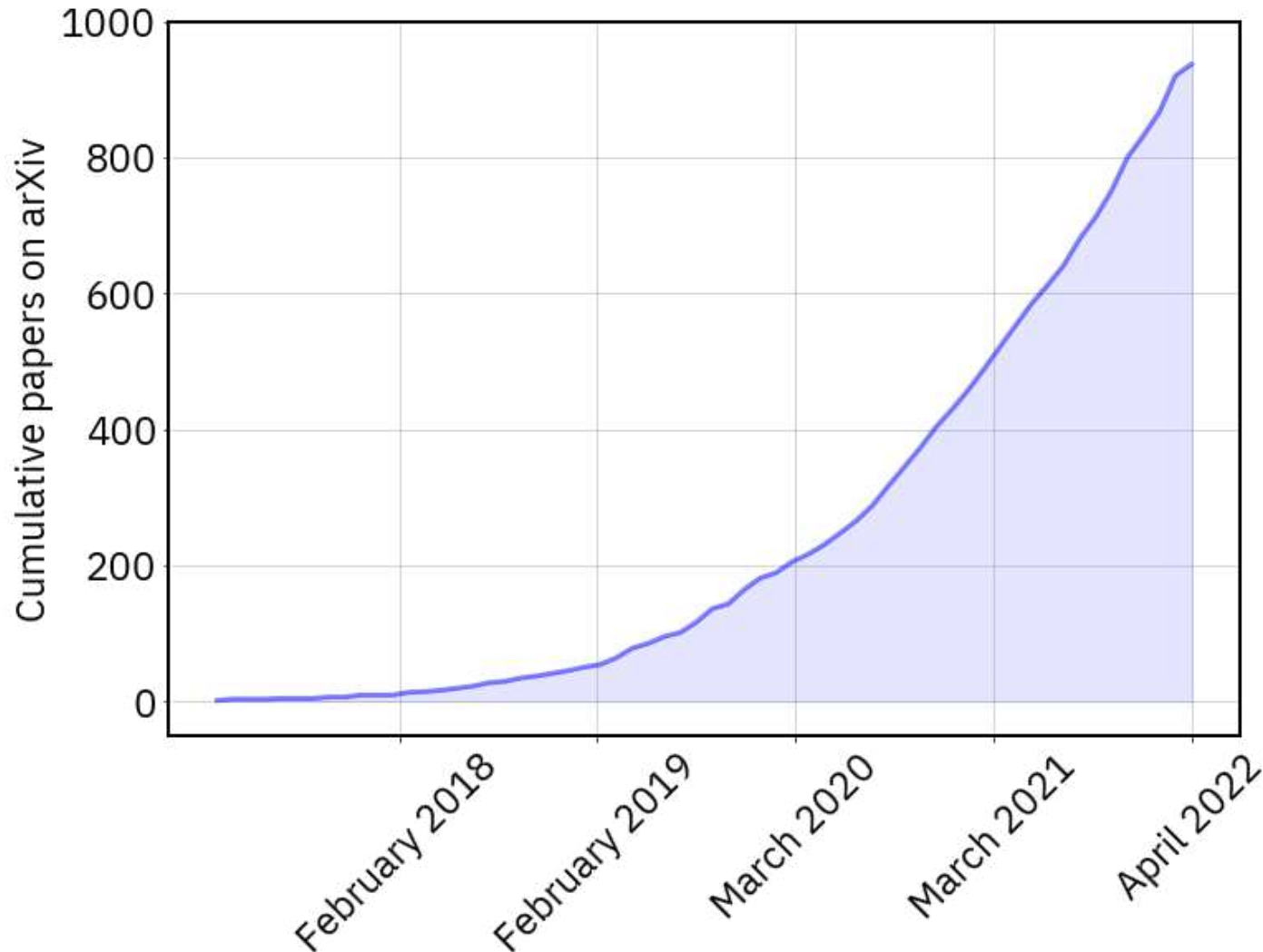
more
speed

more information,
accuracy



Adoption of error mitigation

Papers involving error mitigation over time



Examples

ARTICLE
<https://doi.org/10.1038/s41467-020-14374-z> OPEN
Error-mitigated quantum gates exceeding physical fidelities in a trapped-ion system
Shuaining Zhang¹, Yao Lu¹, Kuan Zhang^{1,2}, Wentao Chen¹, Ying Li^{3*}, Jing-Ning Zhang^{1,4*} & Kihwan Kim^{1*}

Matrix product channel: Variation to mitigate noise and reduce errors

Sergey Filippov,^{*} Boris Sokolov, Ma Borrelli, Daniel Cavalcanti, Sabrina Algorithmiq Ltd, Kanavakat

Article | Published: 08 May 2023

Probabilistic error cancellation with sparse Pauli-Lindblad models on noisy quantum processors

Ewout van den Berg, Zlatko K. Minev, Abhinav Kandala & Kristan Temme

Nature Physics (2023) | Cite this article

Quantum Error Mitigation

Zhenyu Cai,^{1,2,*} Ryan Babbush,³ McClean,³ and Thomas E. O'Brien,¹ Department of Materials, University of Cambridge, 9 Sterling Way, Google Quantum AI, Venice, California, NTT Computer and Data Science Graduate School of China Academy (Dated: July 3, 2023)

npi | quantum information

ARTICLE OPEN

Fundamental limits of quantum error mitigation

Ryuji Takagi^{1,2*}, Suguru Endo^{2,3*}, Shintaro Minagawa^{3,4*} and Mile Gu^{1,4,5*}

Single-shot error mitigation

Ewout van den Berg, Sergey B. P. Maslov, Dmitri Maslov, Quantum, IBM T.J. Watson Research Center, Yorktown Heights, NY 10598, USA (December 2022)

PHYSICAL REVIEW LETTERS **127**, 200505 (2021)

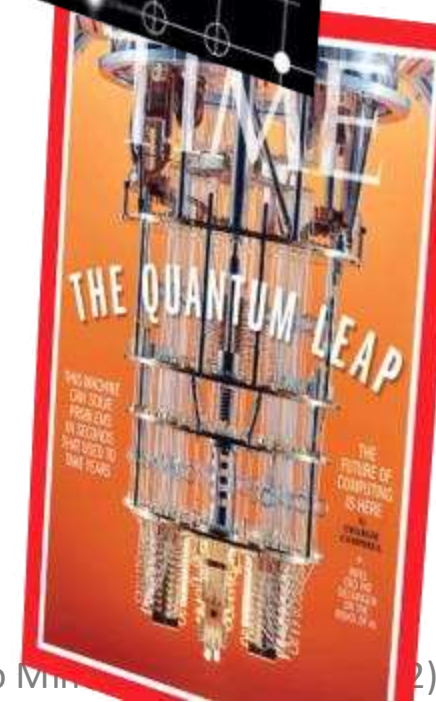
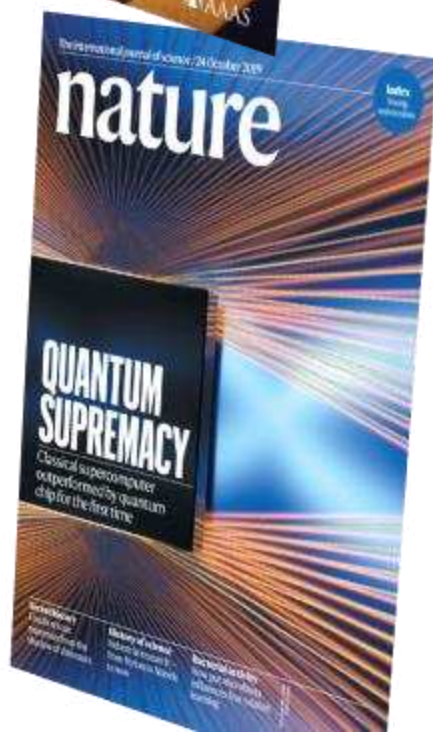
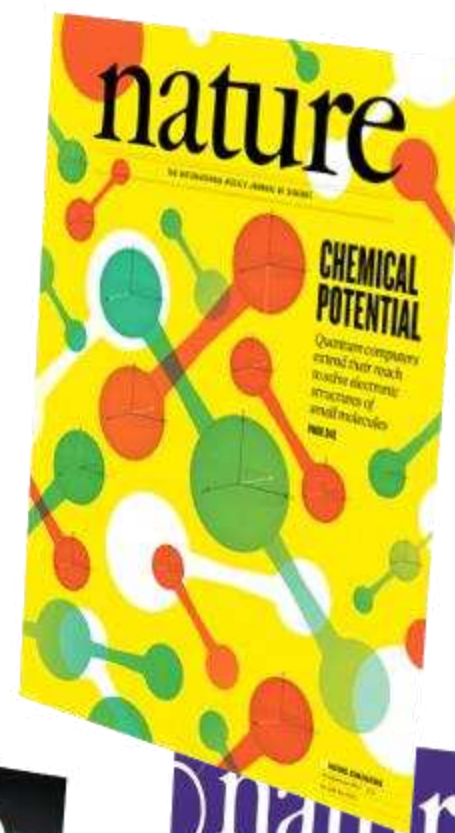
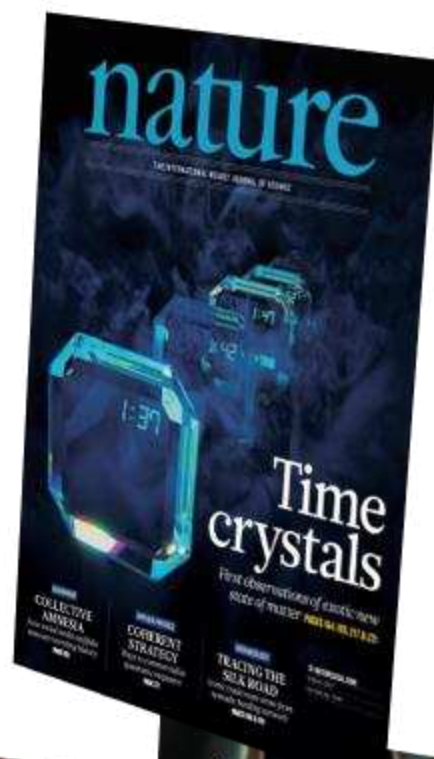
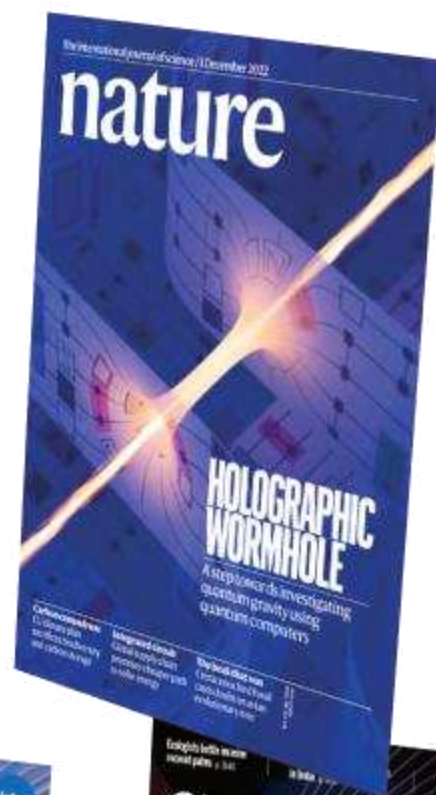
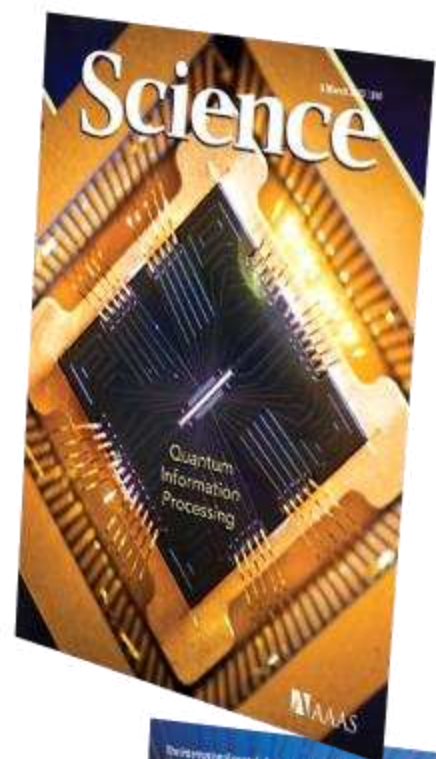
Error Mitigation for Universal Gates on Encoded Qubits

Christophe Piveteau
IBM Quantum, IBM Research—Zurich, 8803 Rüschlikon, Switzerland

Model-free readout-error mitigation for quantum expectation values

Ewout van den Berg, Zlatko K. Minev, and Kristan Temme
Phys. Rev. A **105**, 032620 – Published 30 March 2022

Is science with noisy devices of
broad interest today?



Some of these ideas covered
in other lectures at the
school

Deep dive:

Probabilistic error cancellation (PEC)

To learn and cancel quantum noise



Got Slides?



Ewout van den Berg, Zlatko K. Mineev, Abhinav Kandala, Kristan Temme
Nature Physics (2023)

Cancel quantum noise



High-level message

Learn

accurate, efficient, scalable



Cancel

noise with noise,
practical



Cost

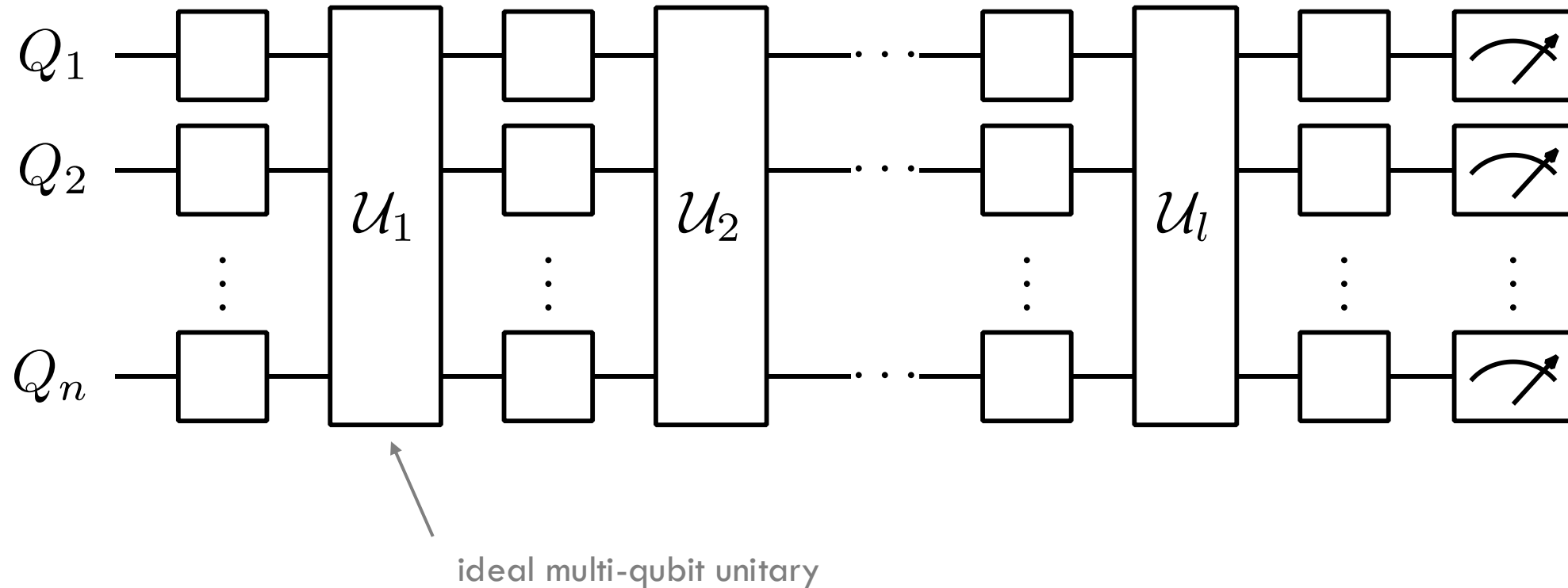
more noise more cost





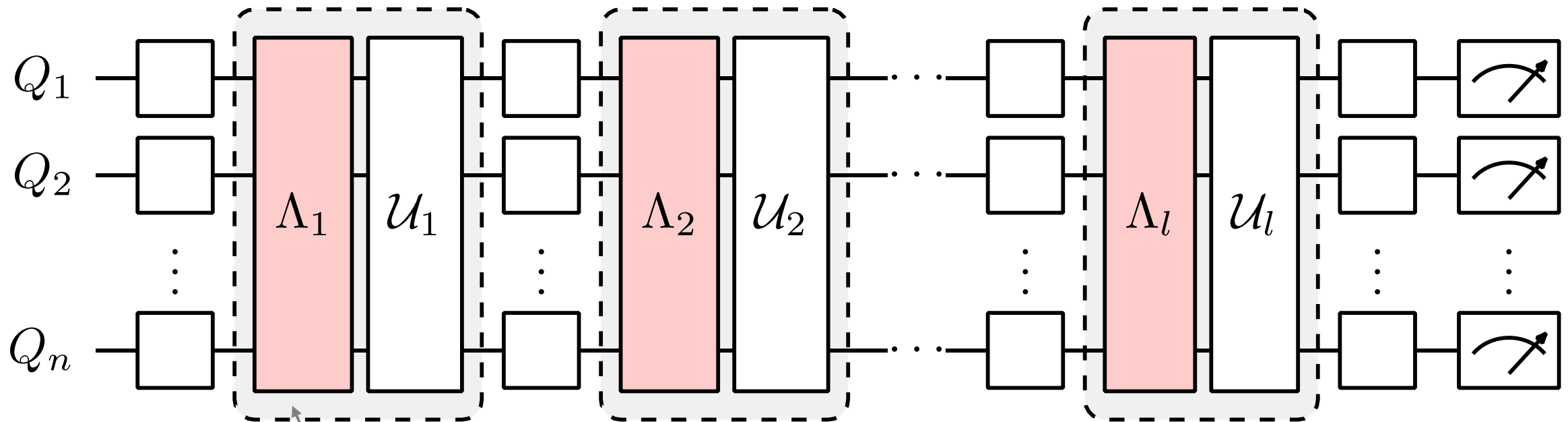
Idea

Ideal (noise-free) quantum circuit



A circuit can be decomposed into a layer construction
Example: Trotterization of Ising model simulation

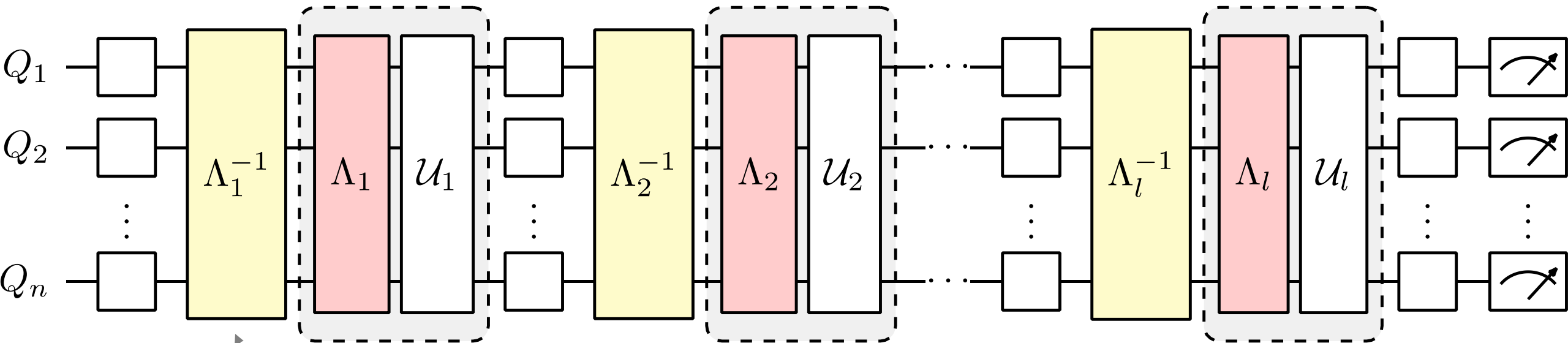
Real (noisy) quantum circuit



multi-qubit noise channel
inseparable from gate

completely positive and trace preserving (CPTP)
representable by a $4^n \times 4^n$ matrix

Why not invert noise?



inverse operation of noise channel

unphysical

would need to know lost information due to noise

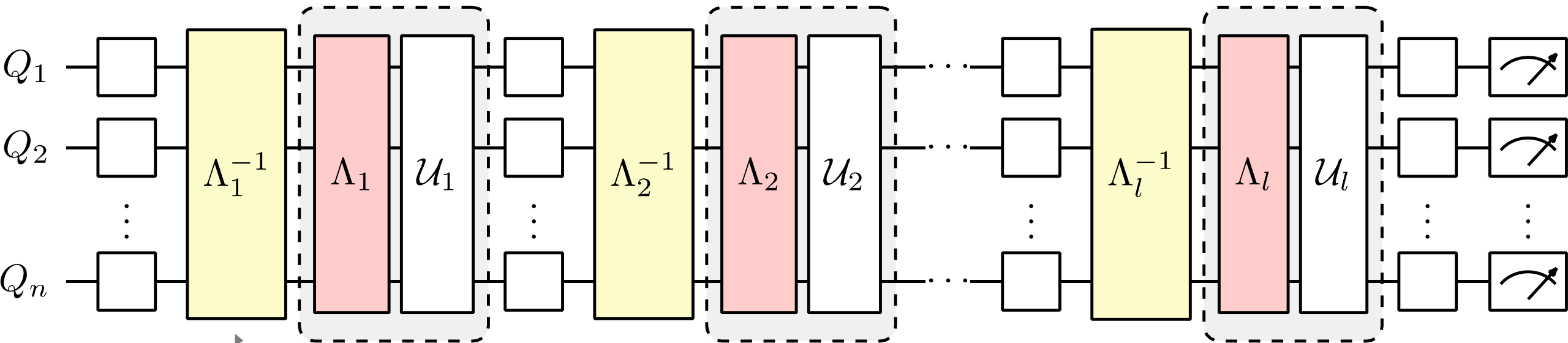
non CPTP map

has negative eigenvalues

...

Not possible?

Probabilistic error cancellation



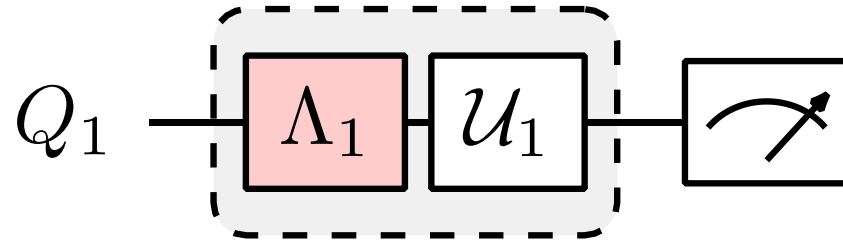
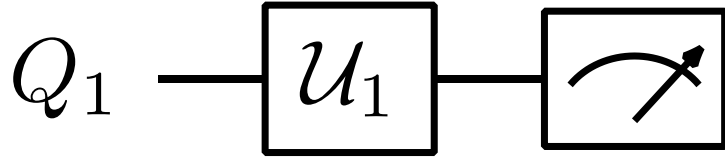
inverse operation of noise channel
implement on average



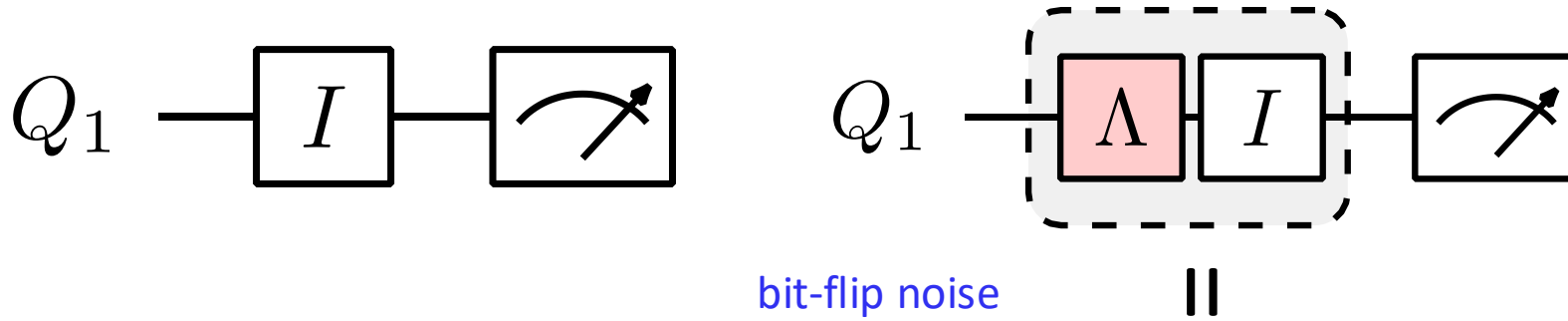
K. Temme, S. Bravyi, and J. M. Gambetta
PRL 119, 180509 (2017)

See also S. Endo, S. Benjamin, and Y. Li
Phys. Rev. X 8, 031027 (2018)

Toy model

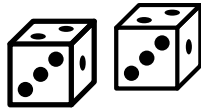


Toy model: noise unraveling into quantum trajectories

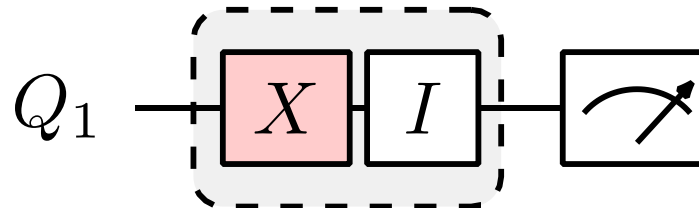
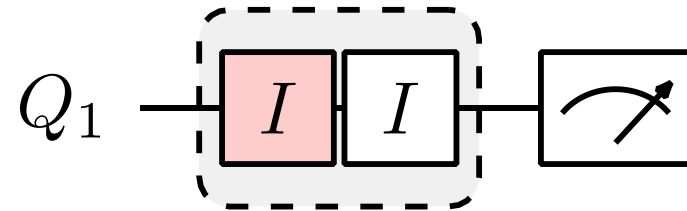


unraveling
(quantum trajectories)

probability $1-p$

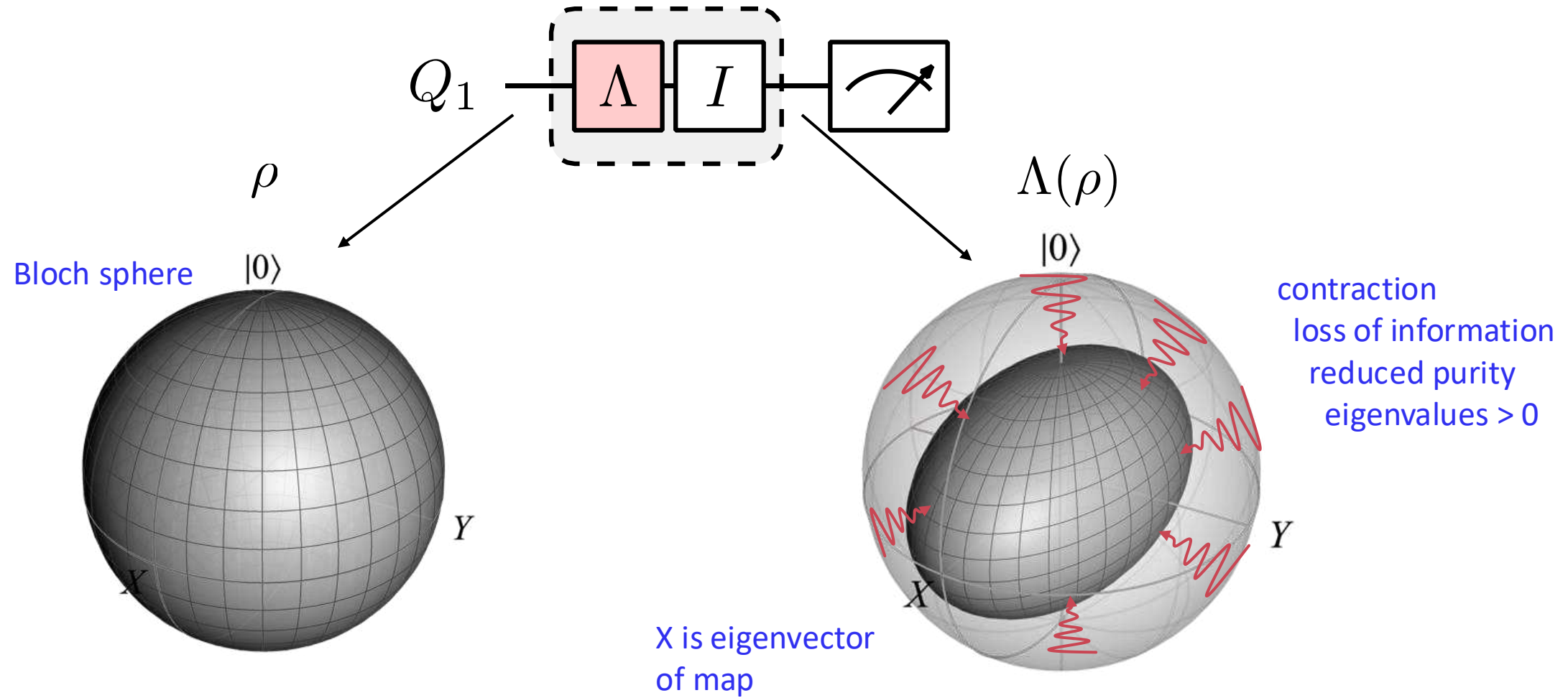


probability p

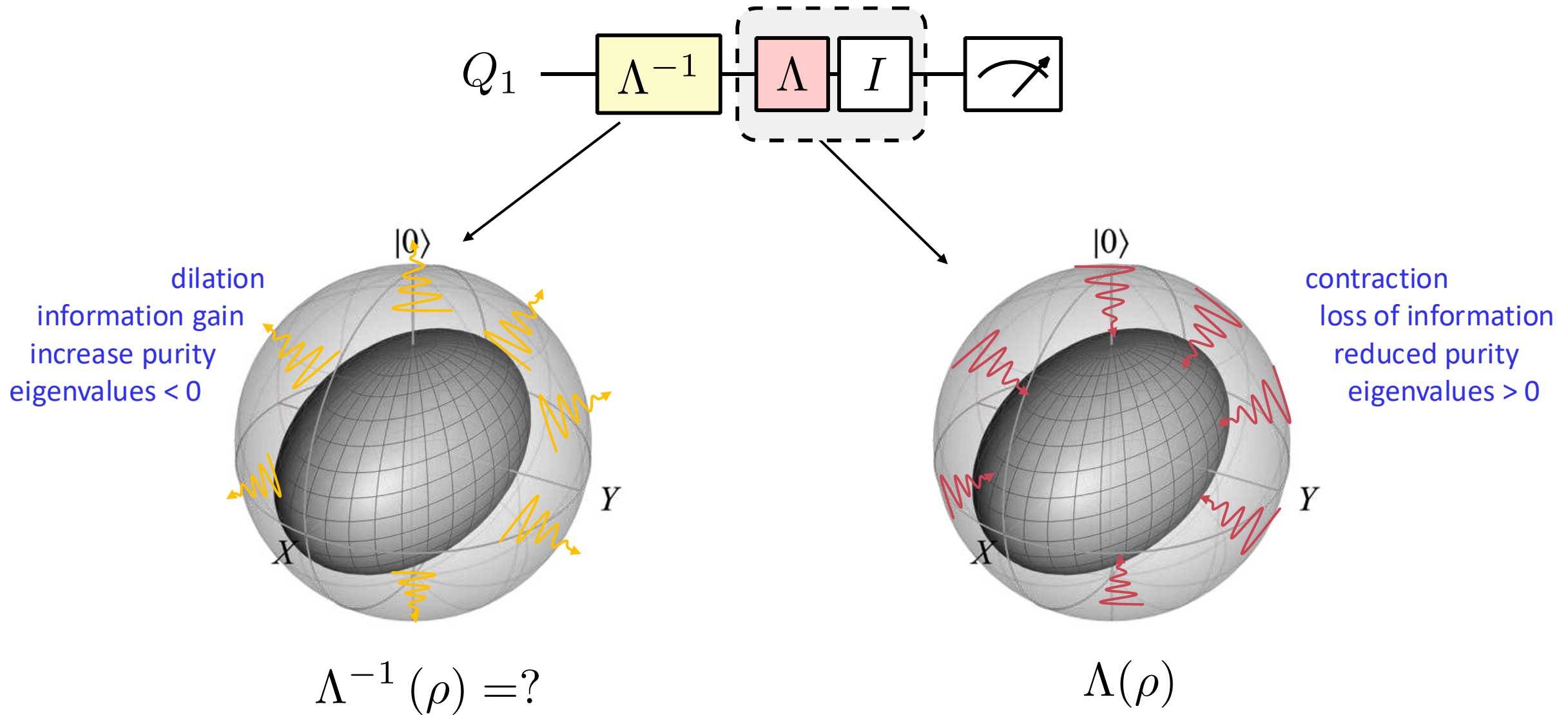


$$\Lambda(\rho) = (1-p)I\rho I + pX\rho X$$

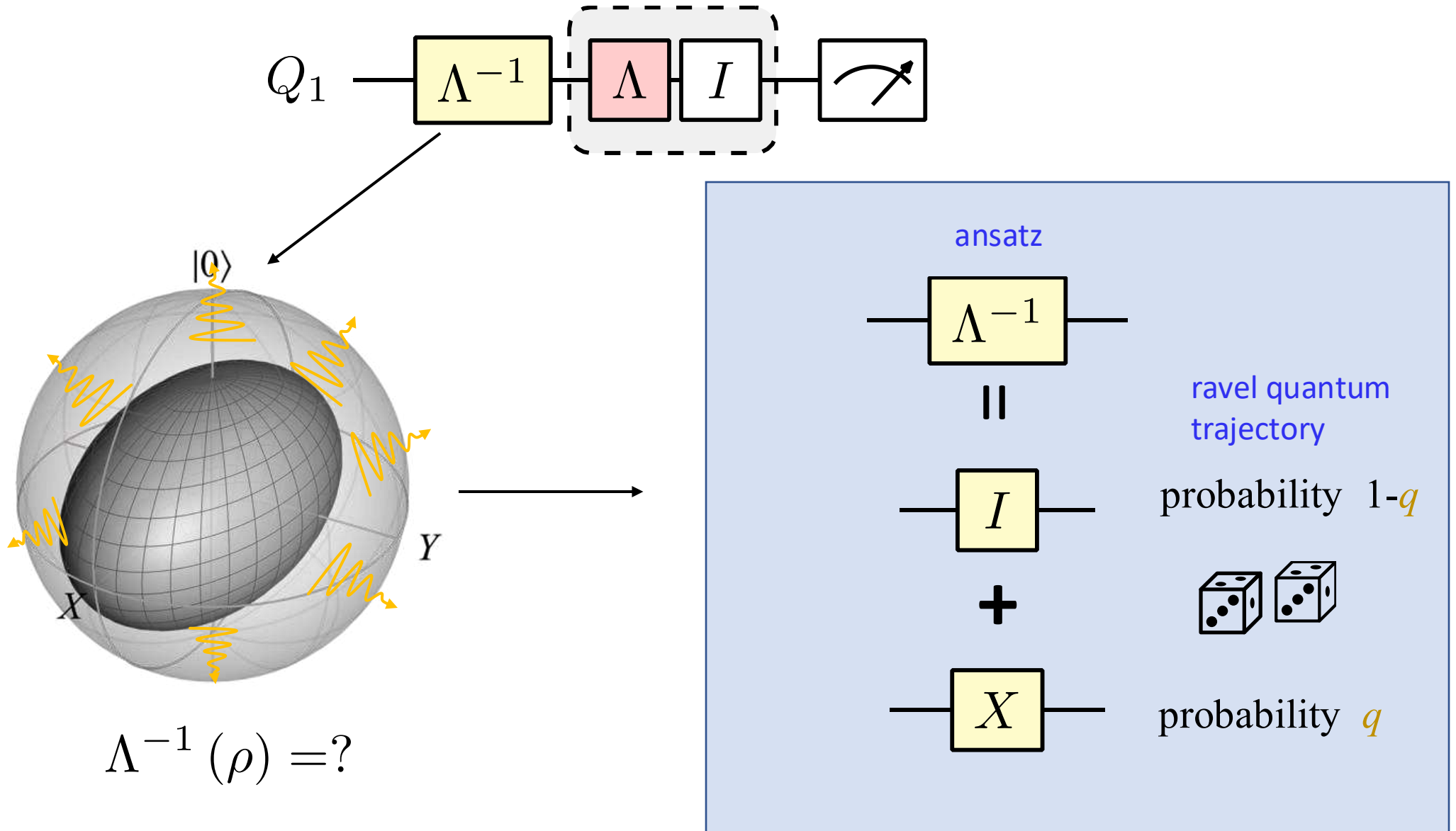
Toy model: noise unraveling into quantum trajectories



Inverse of noise map is **not physical**

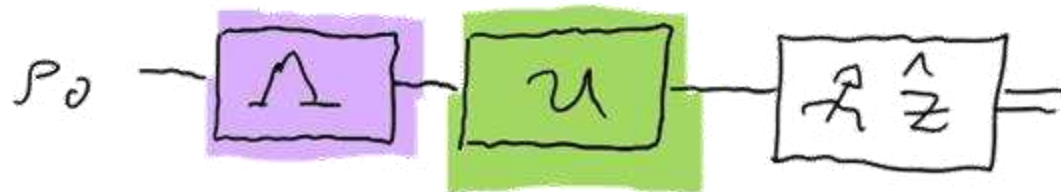
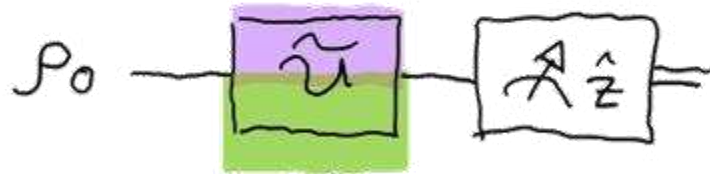


Inverse of noise map is not physical



Blackboard derivation

Setup



Details on notation:

Quantum register alphabet $\mathcal{E} = \{0, 1\}$
 Hilbert space $\mathcal{H} = \mathbb{C}^{\mathcal{E}}$
 Initial state $\rho_0 \in D(\mathcal{H}) \subset L(\mathcal{H})$
 Ideal unitary $U \in U(\mathcal{H}) \subset L(\mathcal{H})$
 Ideal u-channel $\mathcal{U}(\rho) = U \rho U^\dagger$
 $\mathcal{U} \in C(\mathcal{H}) \subset L(L(\mathcal{H}))$

Noisy gate / circuit $\tilde{\mathcal{U}} \in L(L(\mathcal{H}))$

Decompose noisy gate $\tilde{\mathcal{U}} = \mathcal{U} \Lambda$

Blackboard derivation

Simple Example

Keeping it simple and illustrative, let's do a simple case

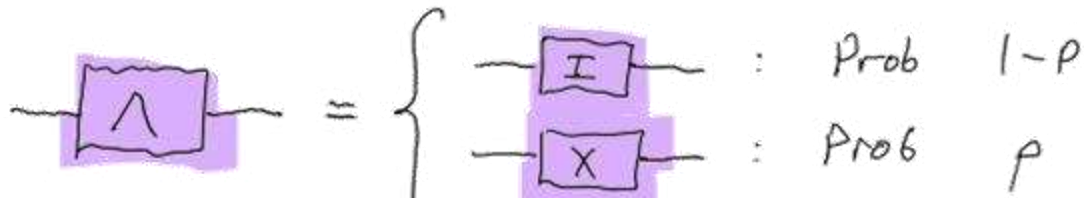
$$\text{Let } U = I \\ u = I \cdot I$$

For the noise, let's play with the simplest bit-flip channel

$$\Lambda_p = \underbrace{(1-p)I_p I}_{\text{prob of no error}} + \underbrace{pX_p X}_{\text{prob of a bit flip error}}$$

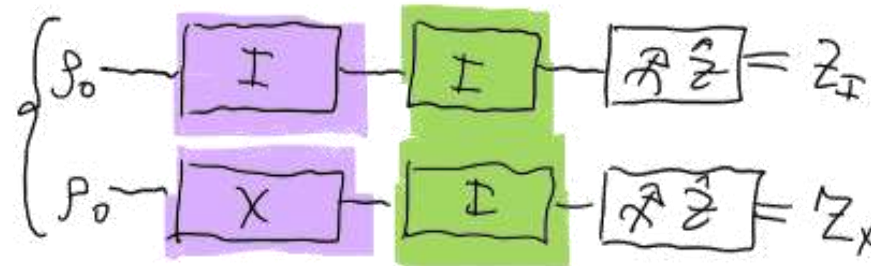
$$\left(\Lambda_p = (1-p)\mathcal{I}_p + p\mathcal{X}_p \right. \quad \left. \begin{array}{l} \text{Equivalent superoperator} \\ \text{channel representation} \\ \mathcal{X}_p = X_p X \\ \mathcal{I}_p = I_p I = p \end{array} \right)$$

Equivalent trajectory unraveling



Blackboard derivation

Our circuit then is equivalent to either



with prob $1-p$
with prob p

Simple Example

Keeping it simple and illustrative, let's do a simple case

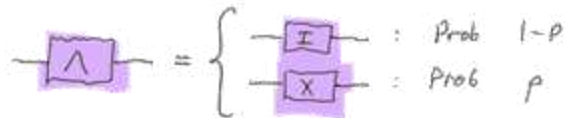
$$\text{let } U = I \\ \mathcal{U} = I \cdot I$$

For the noise, let's play with the simplest bit-flip channel

$$\mathcal{N}(\rho) = \underbrace{(1-p)I\rho I}_{\text{prob of no error}} + \underbrace{pX\rho X}_{\text{prob of a bit flip error}}$$

$$\left(\begin{aligned} \mathcal{N}_Z &= (1-p)\mathcal{Z}_\rho + p\mathcal{Z}_\rho \\ \mathcal{X}_\rho &= X\rho X \\ \mathcal{X}_\rho &= I\rho I = \rho \end{aligned} \right) \quad \text{Equivalent superoperator channel representation}$$

Equivalent trajectory unraveling



The ideal expectation value is

$$\hat{Z}_{\text{ideal}} = \langle \hat{Z} \rangle = \text{Tr}(\hat{Z} \mathcal{U} \rho_0) = \text{Tr}(\hat{Z} \rho_0) = \rho_Z$$

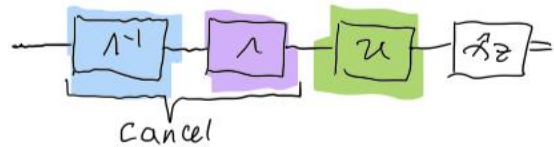
When the channel introduces an error however,

$$\begin{aligned} \mathbb{E}[\hat{Z}_X] &= \text{Tr}(\hat{Z} X \rho X) = \text{Tr}(X \hat{Z} X \rho) \\ &= \text{Tr}(-\hat{Z} \rho) \\ &= -\rho_Z \end{aligned}$$

Blackboard derivation

Noise Inverse

To undo the noise, we'd like to introduce the inverse noise

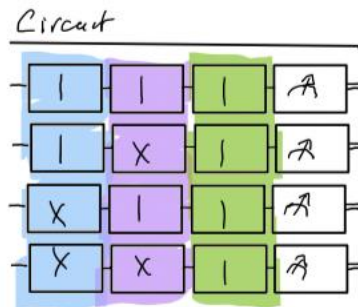


$$\Lambda^{-1}\Lambda = \Lambda\Lambda^{-1} = \mathbb{I}$$

Taking the ansatz $\Lambda^{-1}(p) = (1-r)\mathbb{I}\cdot\mathbb{I} + r(X\cdot X)$

we see 4 cases of unraveling

inverse	noise	no error	prob
I	I	✓	$(1-r)(1-p)$
I	X	✗	$(1-r)p$
X	I	✗	$r(1-p)$
X	X	✓	rp



ideally, we want to interfere trajectories so that the no-error ones will coherently add to unity probably, and the ones with an error will cancel.

$$\begin{aligned} \therefore \textcircled{A} \quad (1-r)(1-p) + r \cdot p &= 1 & \textcircled{B} \quad (1-r)p + r(1-p) &= 0 \\ 1 - r - p + 2rp &= 1 & p + r - 2rp &= 0 \\ r + p - 2rp &= 0 & \text{same condition} & \end{aligned}$$

$$\Rightarrow r(1-2p) = -p$$

$\therefore$

$$r = \frac{-p}{1-2p}$$

Recall p is a probability $0 \leq p \leq 1$,

$$p=0 \Rightarrow r=0$$

$$p=1 \Rightarrow r=1$$

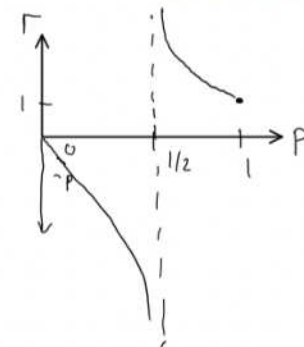
$$p=1/2 \Rightarrow r=\infty$$

$$p \ll 1 \Rightarrow r \approx -p$$

no noise, no need to do anything

deterministic bit-flip, requires deterministic bit flip usually for Λ^{-1}

singular value, since at $p=1/2$, we hit 0, scramble the state



Blackboard derivation

Note that we could equivalently have used the algebraic condition and solved for r

$$\Lambda(\Lambda^{-1}(\rho)) = \mathcal{I}(\rho) = \rho \quad \text{Solve for } r$$

$$= \Lambda((1-r)\rho + rX\rho X)$$

$$= \underbrace{(1-p)(1-r)\rho + prX\rho X}_{\text{no error}} + \underbrace{(1-p)rX\rho X + (1-r)pX\rho X}_{\text{error}}$$

$$= \underbrace{[(1-p)(1-r) + pr]\rho}_{\text{no error}} + \underbrace{[(1-p)r + (1-r)p]X\rho X}_{\text{error}}$$

Same conditions as above? solution $r = \frac{p}{1-2p}$

Blackboard derivation

How to implement? Quasi-Probability

$$\begin{aligned}\Lambda^{-1} &= (1-r)I_Z I + r X_Z X \\ &= \left[\frac{|1-r|}{|1-r|+|r|} \operatorname{sgn}(1-r) I_Z I + \frac{|r|}{|1-r|+|r|} \operatorname{sgn}(r) X_Z X \right] (|1-r|+|r|) \\ &= \gamma \left[S_I P_I I_Z I + S_X P_X X_Z X \right]\end{aligned}$$

with

$$\gamma = |1-r| + |r|$$

$$P_I = \frac{|1-r|}{\gamma} \quad S_I = \operatorname{sgn}(1-r)$$

$$P_X = \frac{|r|}{\gamma} \quad S_X = \operatorname{sgn}(r)$$

valid prob distribution

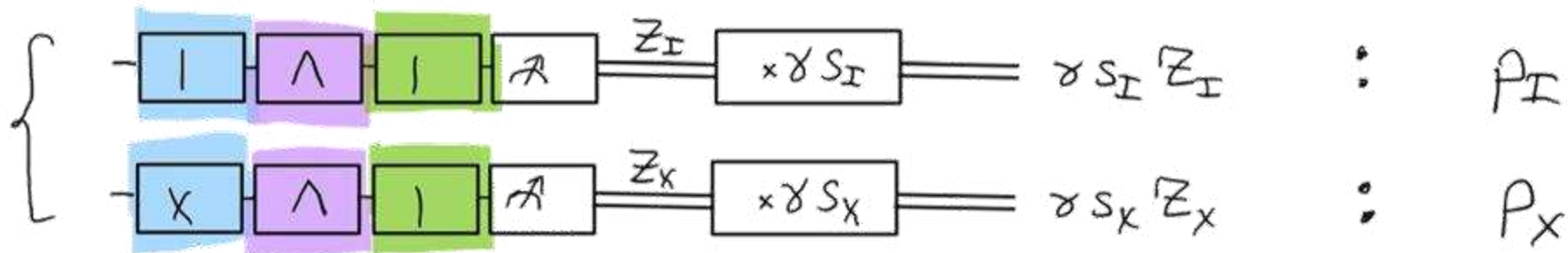
$$0 \leq P_I, P_X \leq 1 \quad \text{and} \quad |P_I| + |P_X| = 1$$

Blackboard

How to sample?

$$\begin{aligned}
 \langle Z \rangle &= \text{Tr}(Z \Lambda \Lambda^{-1} \rho_0) \\
 &= \text{Tr}(Z \tilde{I} [\gamma S_I \rho_I \rho_0^\dagger + \gamma S_X \rho_X \rho_0^\dagger]) \\
 &= \gamma S_I \rho_I \text{Tr}(Z \tilde{I} \rho_0) + \gamma S_X \rho_X \text{Tr}(Z \tilde{I} \rho_0) \\
 &= \gamma [S_I \rho_I \underbrace{\langle Z \rangle_I}_{\substack{\text{quantum} \\ \text{circuit eq. val. we can find}}} + S_X \rho_X \underbrace{\langle Z \rangle_X}_{\leftarrow}]
 \end{aligned}$$

Equivalent interpretation:



Blackboard

Estimator

$$E_{\text{mitg}} = \gamma S_I Z_I + \gamma S_X Z_X$$

$$\mathbb{E}[E_{\text{mitg}}] = \langle \hat{Z} \rangle_{\text{ideal}}$$

$$\mathbb{V}[E_{\text{mitg}}] = \mathbb{V}[\gamma S_I Z_I] + \mathbb{V}[\gamma S_X Z_X]$$

$$= \gamma^2 \mathbb{V}[Z_I] + \gamma^2 \mathbb{V}[Z_X]$$

$$= \gamma^2 (2 \sigma_{\text{ideal}}^2)$$

$$\sigma_{\text{ideal}}^2 = \mathbb{V}[Z_I] = 4q(1-q)$$

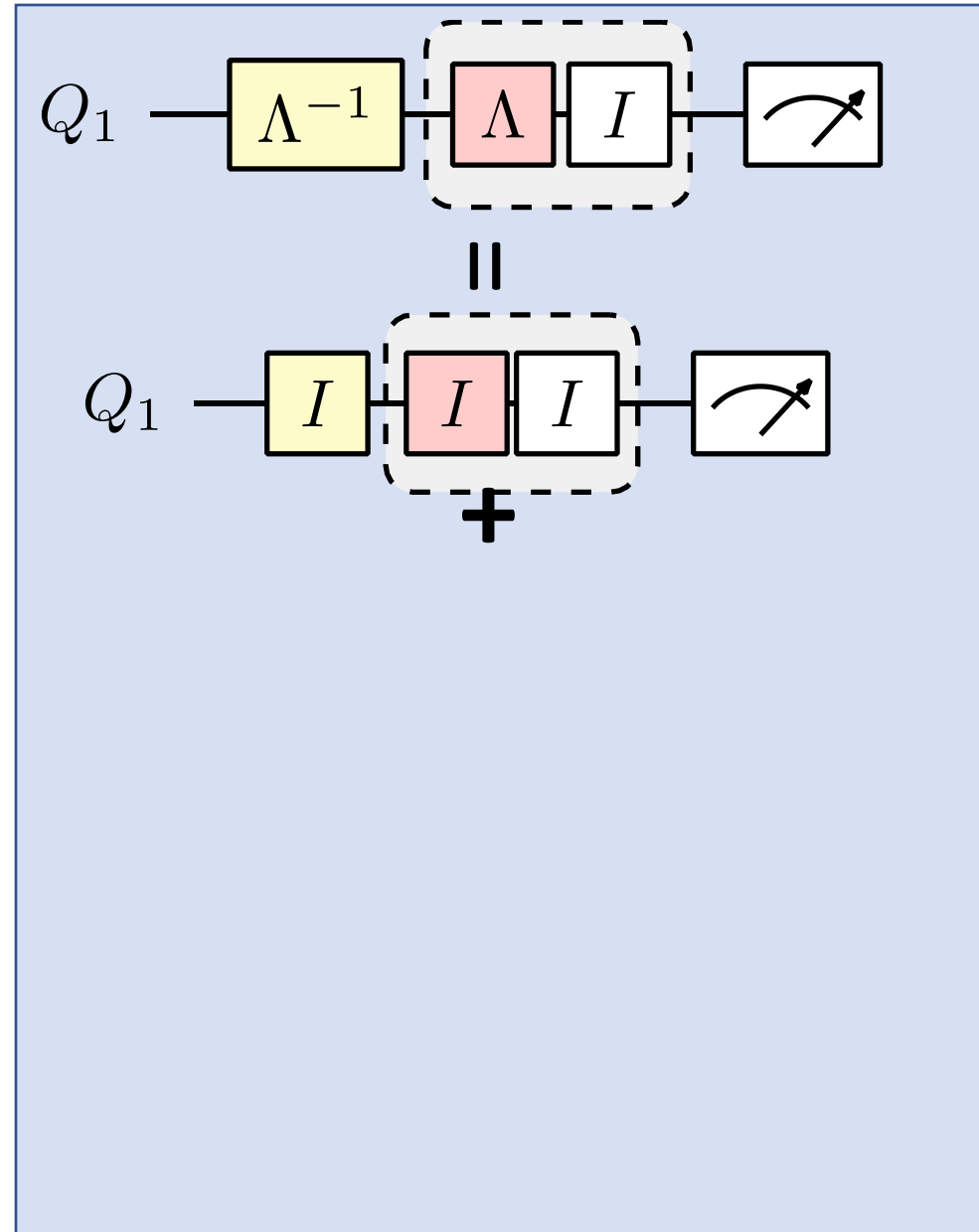
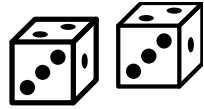
$$q = 1 - 2P_Z = \left\langle \frac{1 - \hat{Z}}{2} \right\rangle$$

Since the X just flips $Z \rightarrow -Z$ of P , it follows that the variance is the same, since symmetric

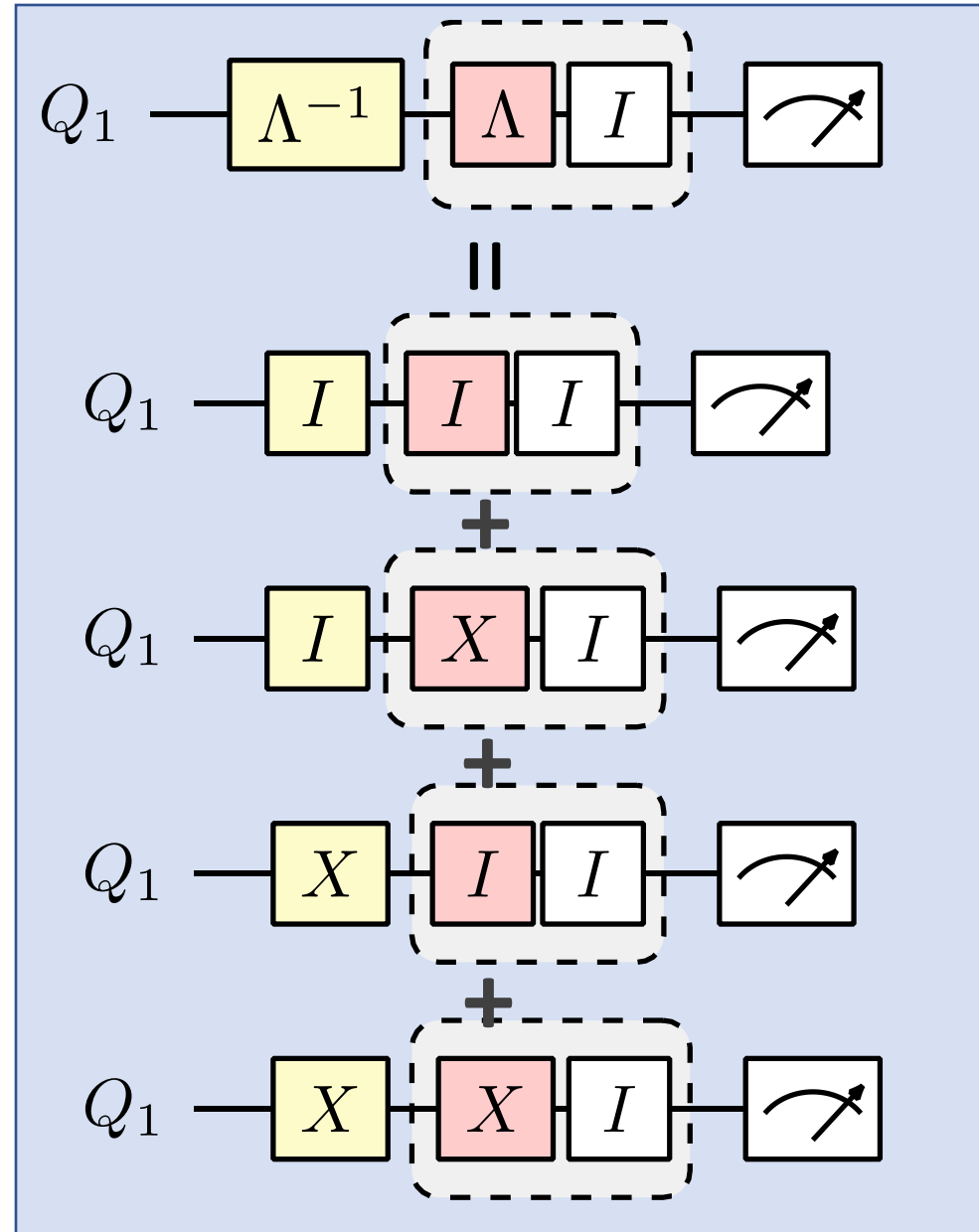
Raveling quantum trajectories to undo noise

No error

probability
 $(1-q)(1-p)$

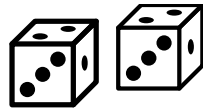


Raveling quantum trajectories to undo noise



No error

probability
(1- q) (1- p)



ERROR!

(1- q) p

ERROR!

q (1- p)

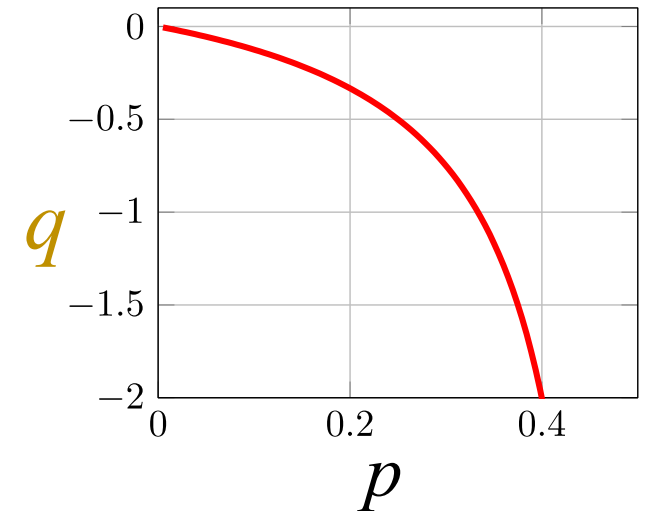
Error
CANCELED!

$q p$

Solution to noise free!

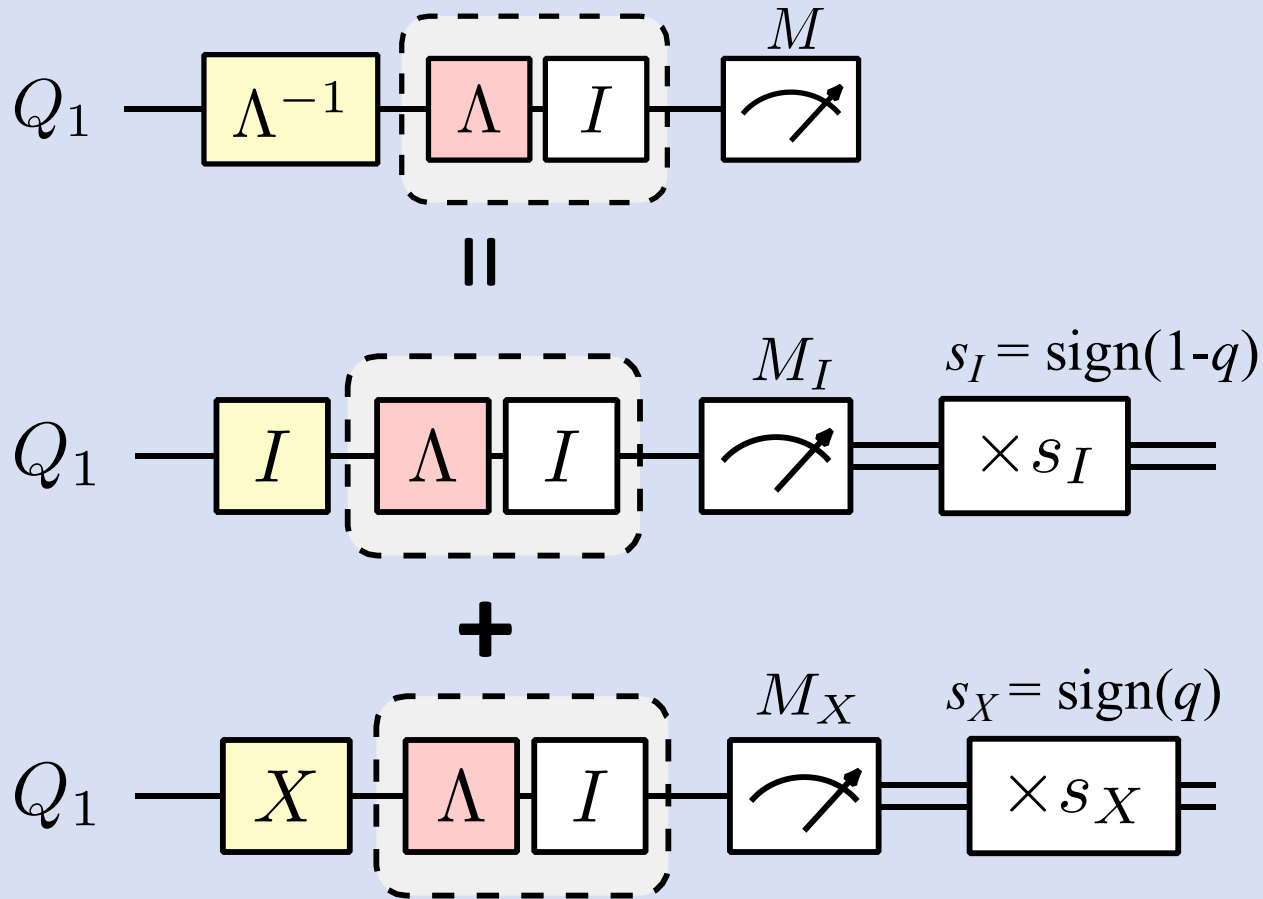
$$q = \frac{-p}{1 - 2p}$$

Sign & scale:
quasi-probability



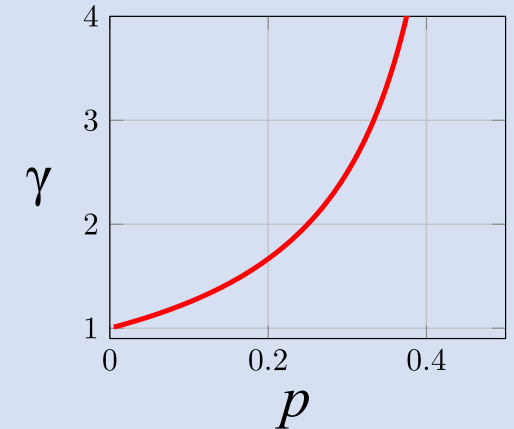
How to implement?

random circuits
(quantum trajectories)



sampling overhead

$$\gamma = |1-q| + |q|$$



mitigated expectation

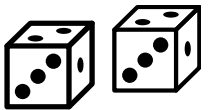
$$\langle M \rangle = \gamma (s_I P_I M_I + s_X P_X M_X)$$

Gain: Bias-free estimate!

Cost: Variance

probability

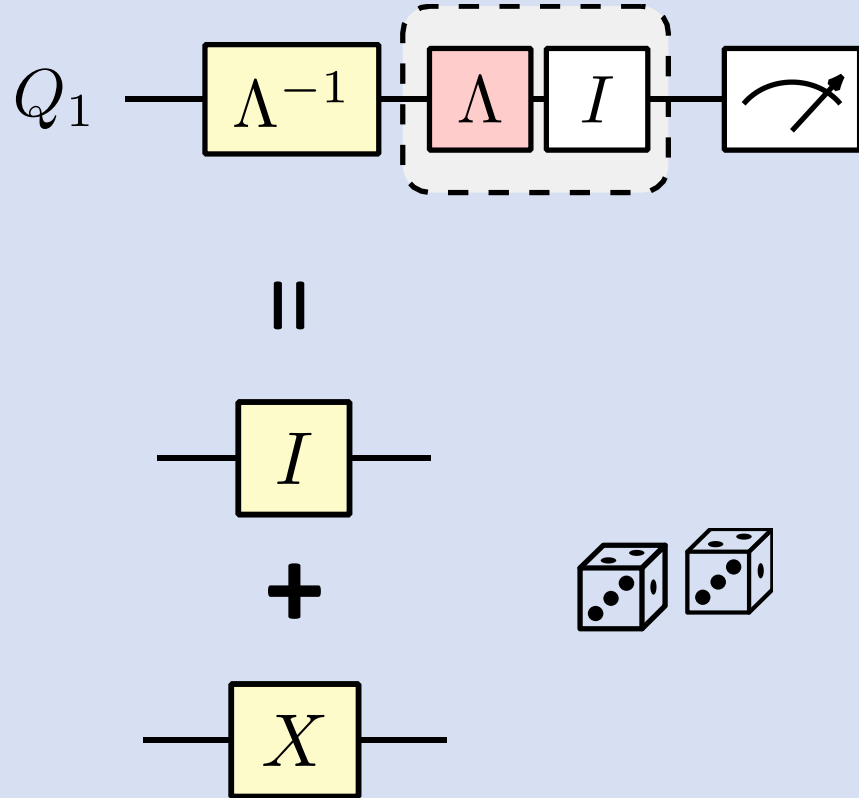
$$P_I = |1-q|/\gamma$$



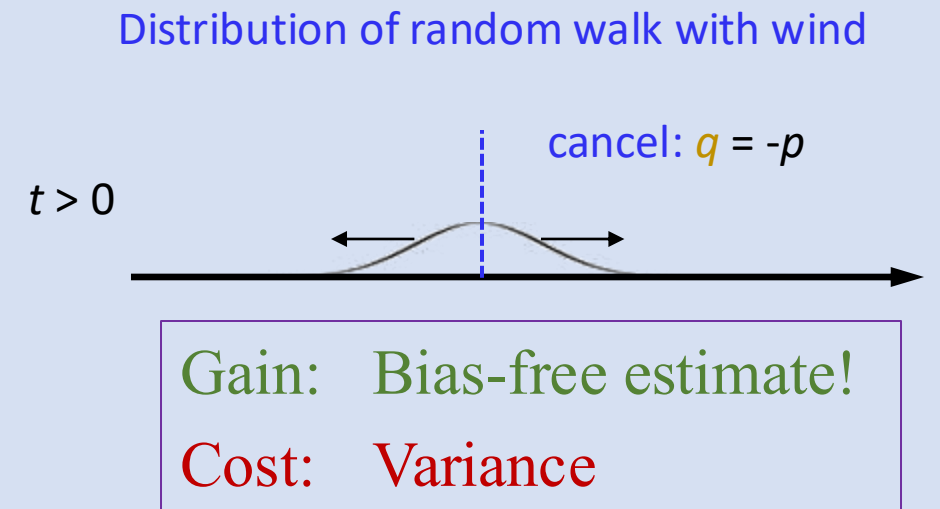
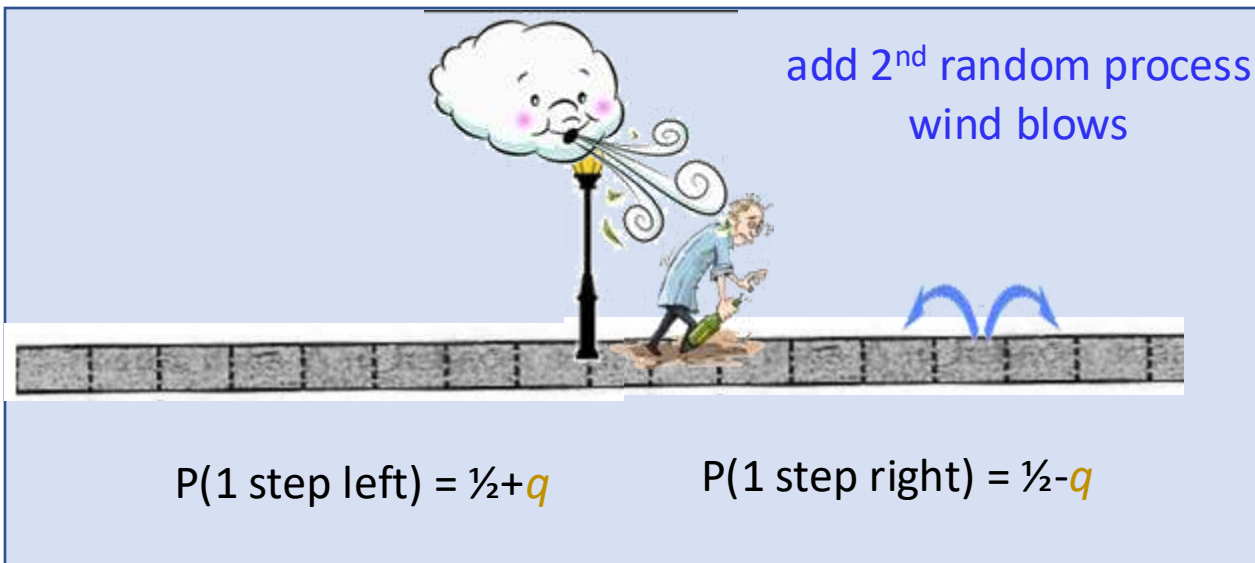
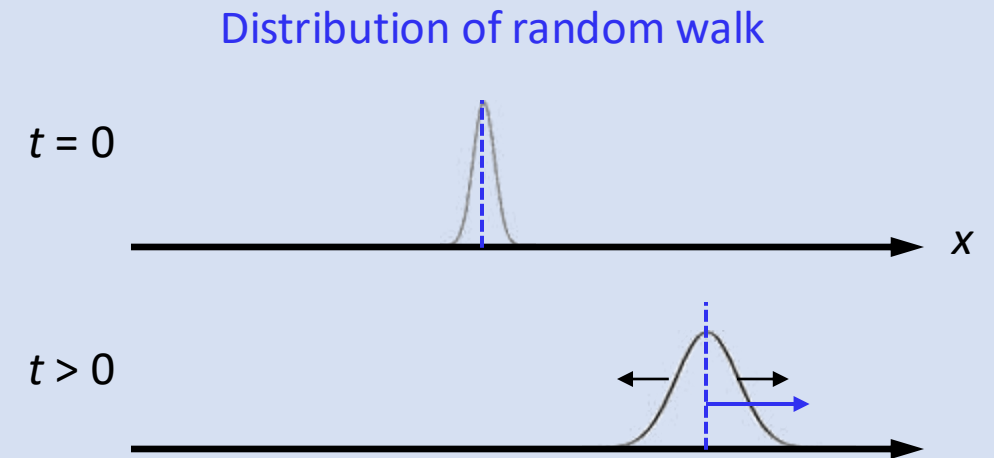
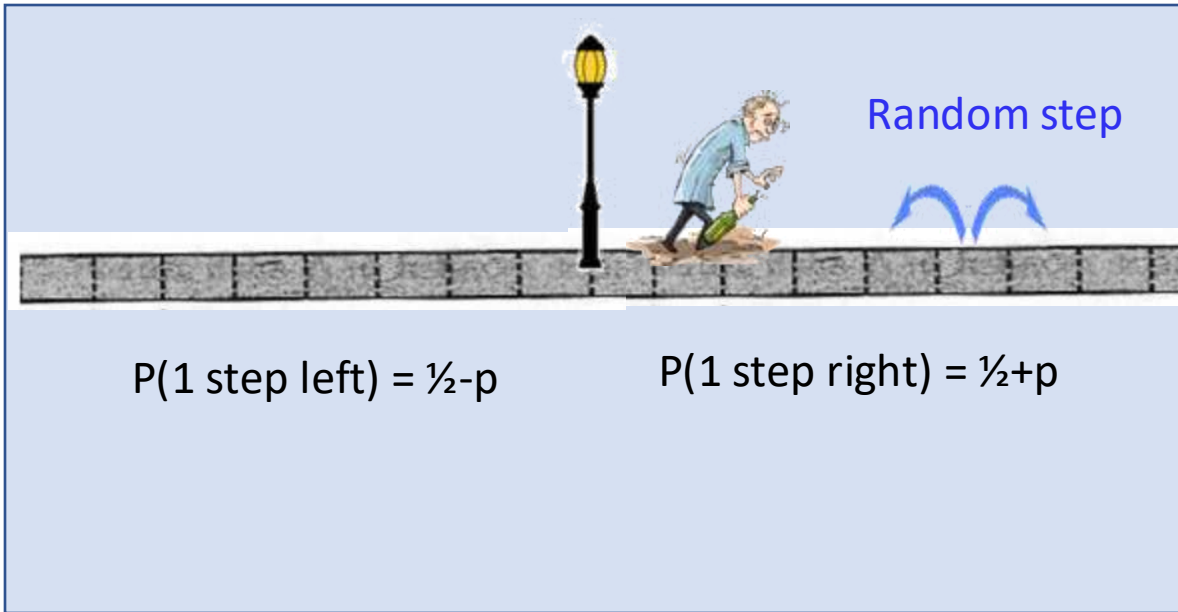
$$P_X = |q|/\gamma$$

$$P_I + P_X = 1$$

Canceling noise with noise



Canceling noise with noise: Drunkard's classical random walk analogy



Quantum Noise and Error Mitigation

Lecture 1

Big picture

Quantum computers status

Why error mitigation?

Noise in quantum computers
Overview of error mitigation

Mitigation fundamentals

Probabilistic error cancelation (PEC)

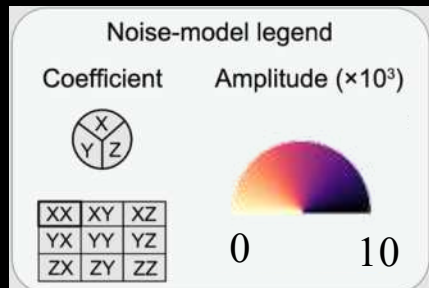
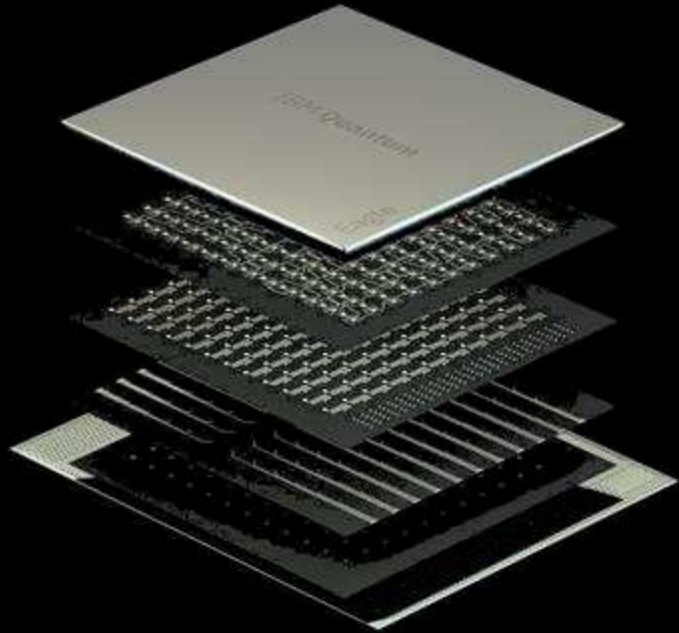
Introduction
One qubit example

Next lecture

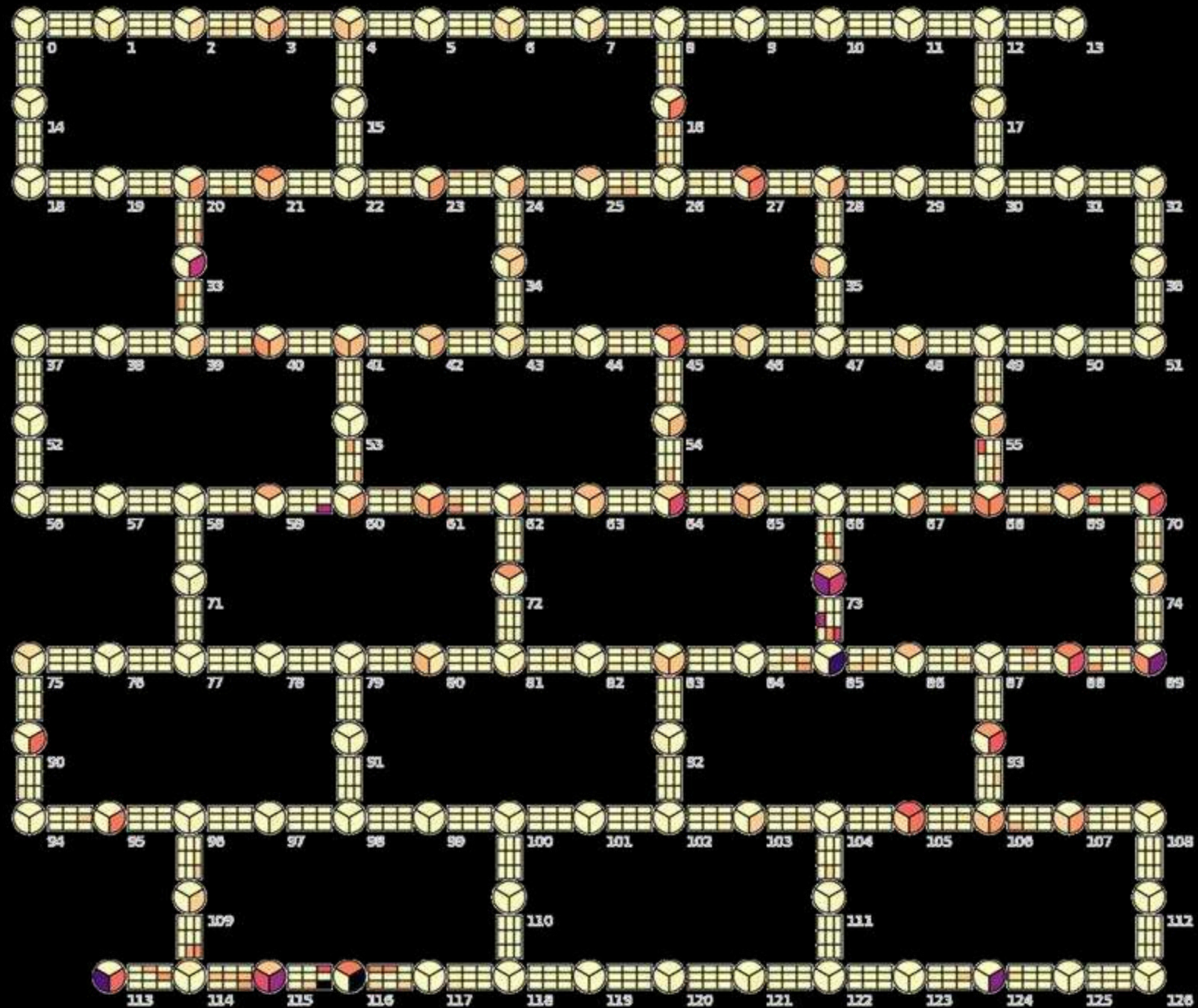
Learning noise
State-of-art mitigation experiments
Hardware
Outlook



Noise tomogram for 127Q Trotter layer

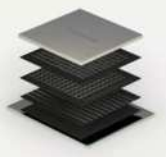


Same number of
learning circuits
as for 4Q



Scaling and error budget

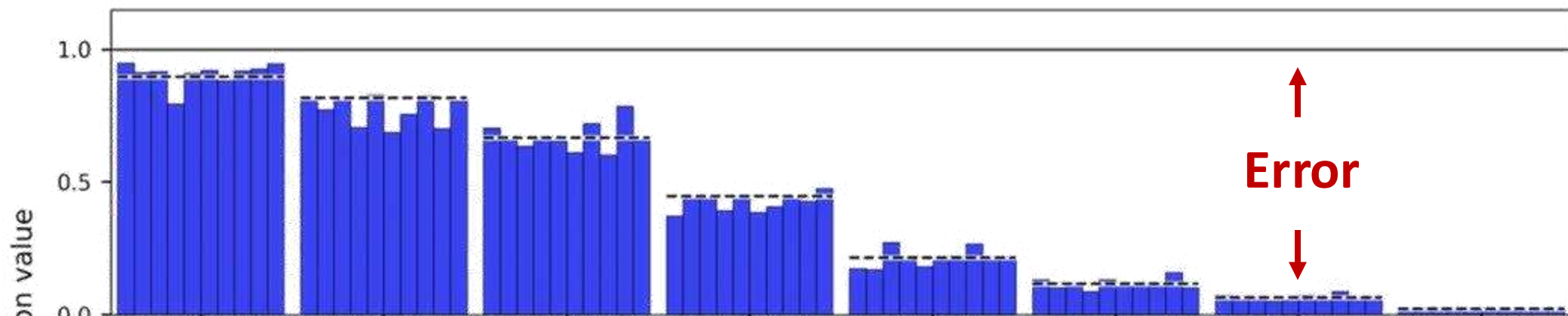




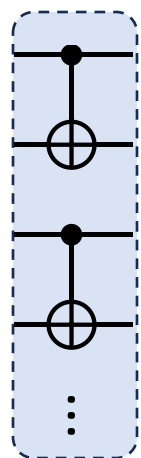
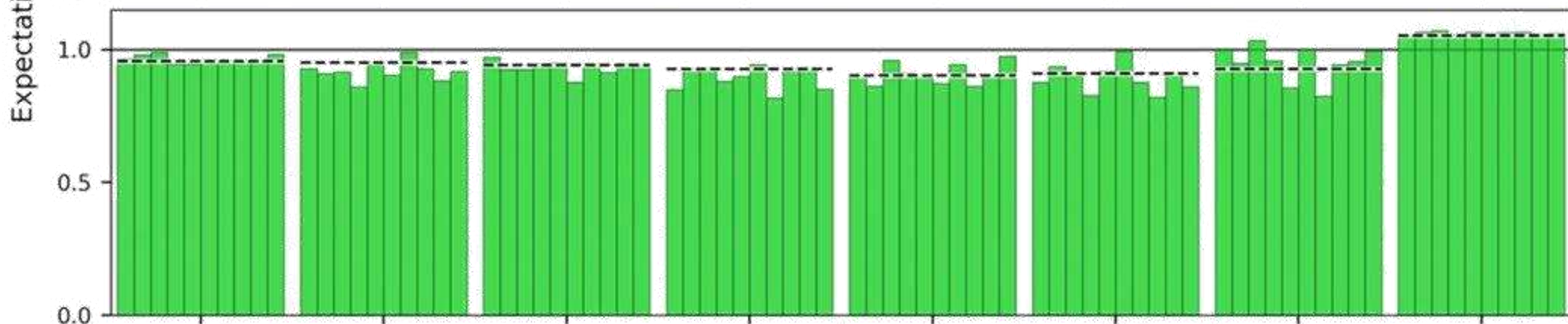
PEC on 50 qubit observables

Z stabilizers of increasing weight

Without PEC



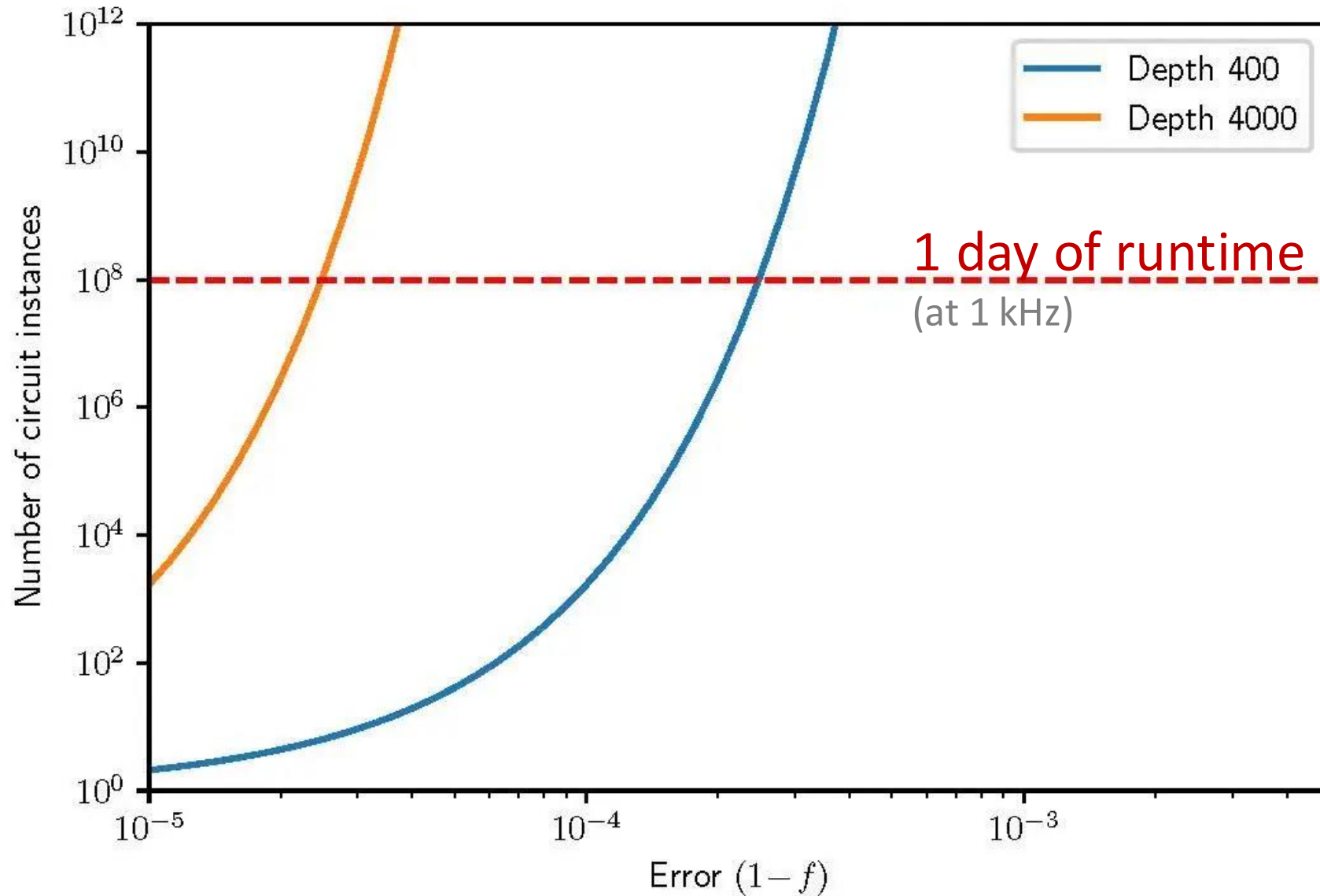
With PEC



50Q
2 layers of
cNOT gates

Path to 100+ qubits?

Estimating
PEC overhead
for Trotter
circuits
comprising
100 qubits



See also on speed: A. Wack, et al., Quality, speed, and scale: three key attributes to measure the performance of near-term quantum computers (2021).

Path to quantum computing

Noise-free estimators can be obtained from noisy quantum computers TODAY, at a runtime cost that is exponential in number of qubits n and circuit depth d

$$\text{Runtime} = \beta d (\bar{\gamma})^{n d} \text{ seconds}$$

d is the depth of the quantum circuit

β is a measure of the time per circuit layer operation (CLOPS) (increase by pushing **speed**)

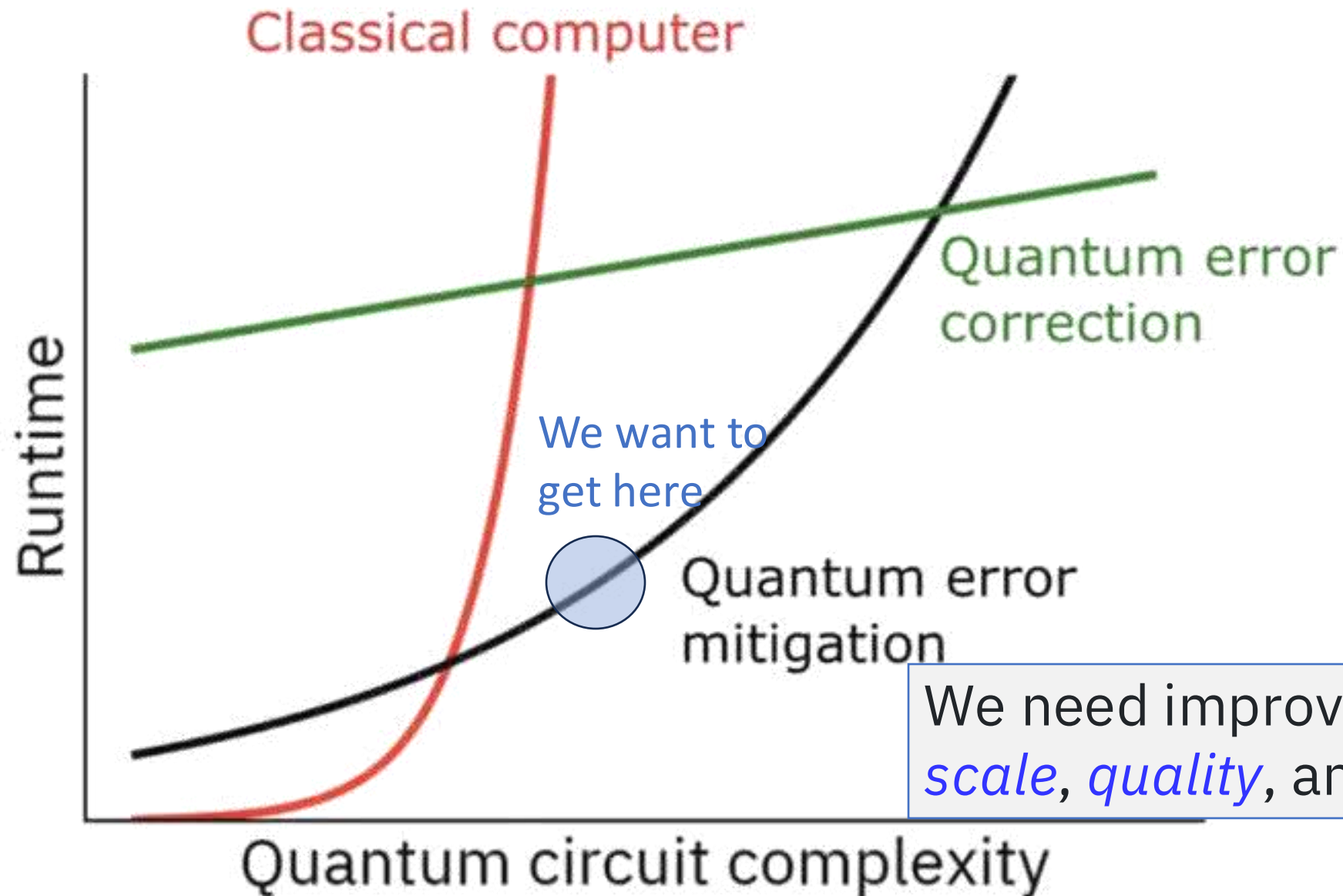
$\bar{\gamma}$ is a measure of the collective quantum noise (increasing **quality** brings it closer to 1)

n is the number of operational qubits (increase by pushing **scale**)

You can further reduce runtime using light cones and other strategies.

	Improvements	$\bar{\gamma}$
Hummingbird r2 (Brooklyn, 65Q)		1.038
Hummingbird r3 (Ithaca, 65Q)	2-3x coherence improvements over r2	1.024
Falcon r10 (Prague, 32Q)	State-of-the-art two-qubit gates, reduced crosstalk	1.012

Path to utility of quantum computers before error correction

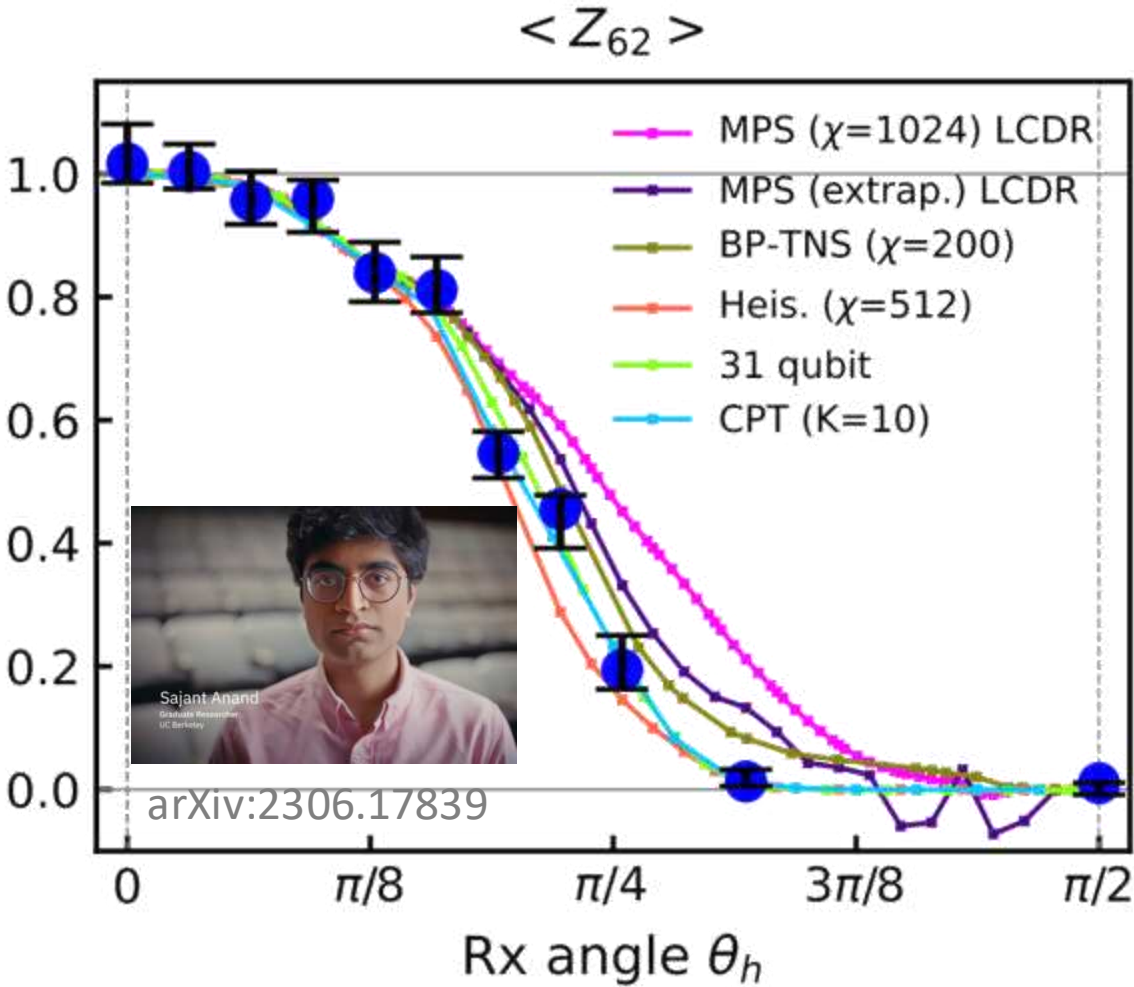
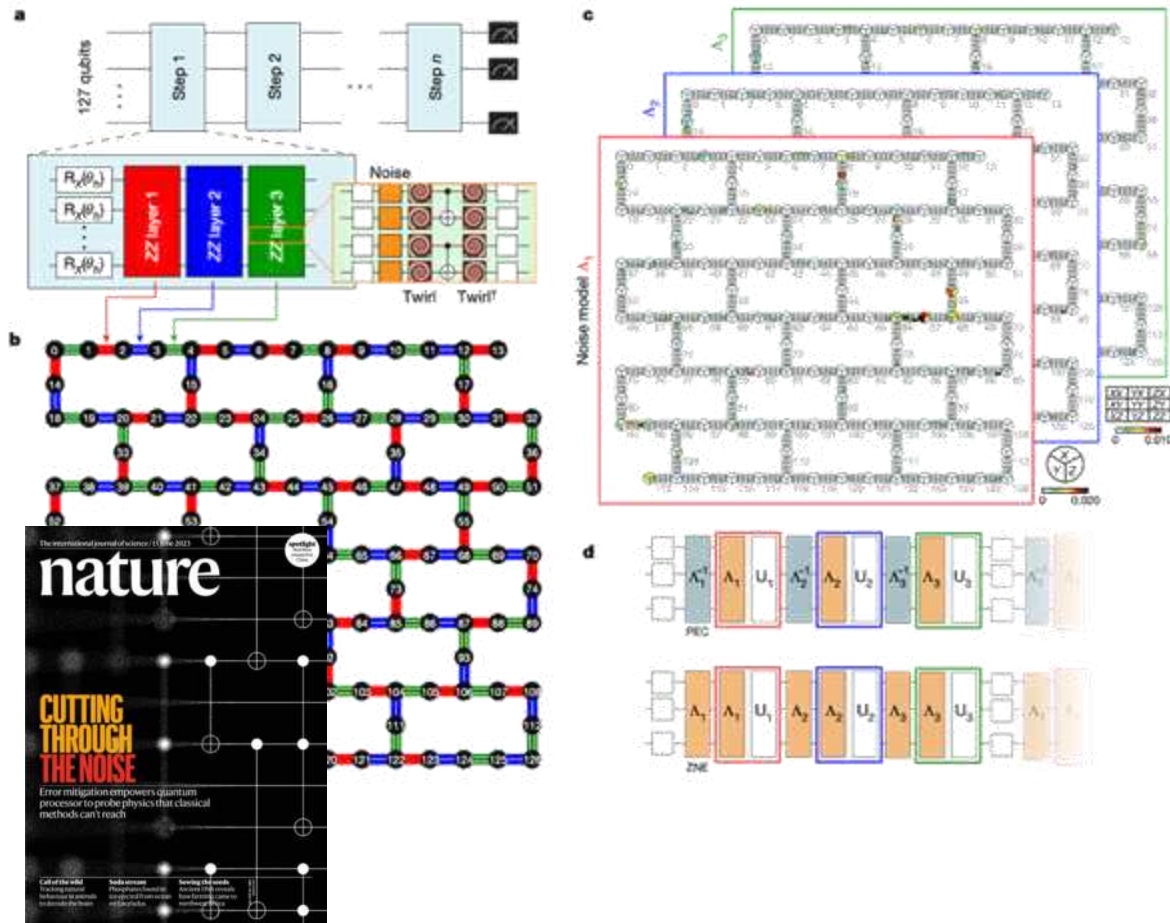


PEC + ZNE for 127 qubits

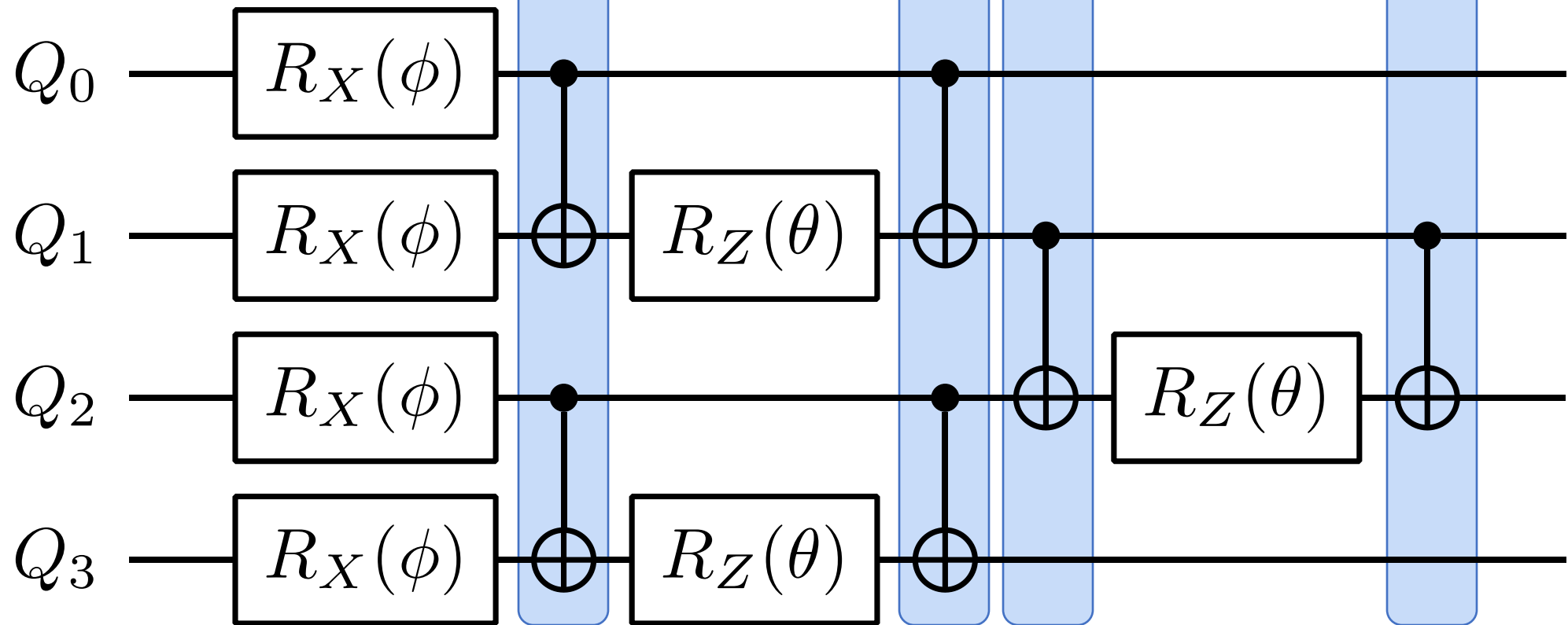
Article

Evidence for the utility of quantum computing before fault tolerance

Youngseok Kim^{1,6}, Andrew Eddins^{2,6}, Sajant Anand³, Ken Xuan Wei¹, Ewout van den Berg¹, Sami Rosenblatt¹, Hasan Nayfeh¹, Yantao Wu^{3,4}, Michael Zaletel^{3,5}, Kristan Temme¹ & Abhinav Kandala¹

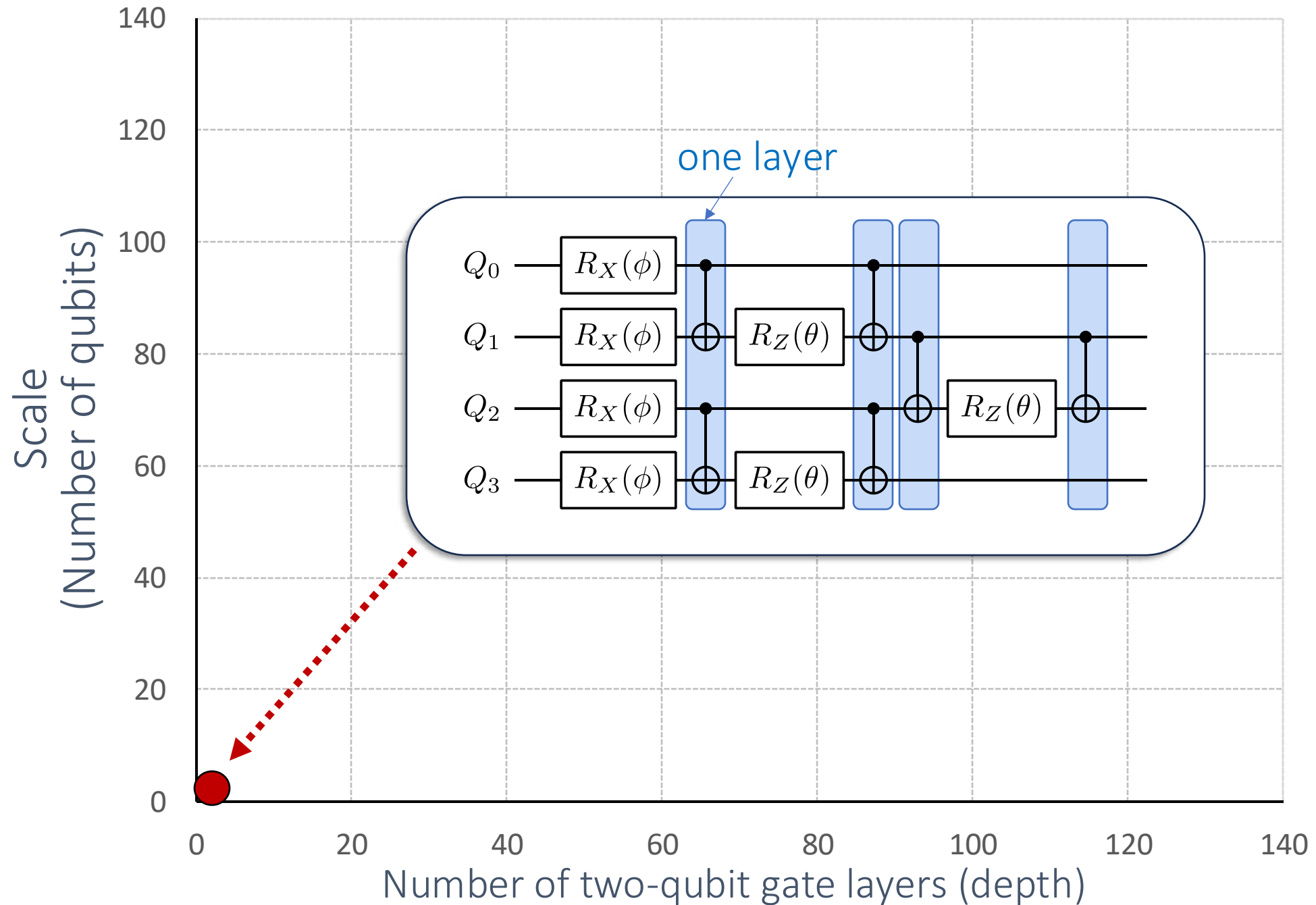


one gate layer



4 qubits x 4 gate layers

Circuit
width $\times$ depth



Some early utility-scale experiments



[0] Kim, Eddins, ..., Temme, Kandala.
Nature 618, 500–505 (2023)



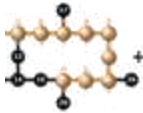
[1] Yu, Zhao, Wei.
arXiv: 2207.09994 (2022)



[2] Shtanko, Wang, Zhang, Harle, Seif,
Movassagh, Minev.
arXiv: 2307.07552 (2023)



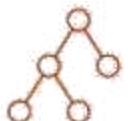
[3] Farrell, Illa, Ciavarella, Savage.
arXiv: 2308.04481 (2023)



[4] Bäumer, Tripathi, Seif,
Minev. APS Y45.4
arXiv: 2308.13065 (2023)



[5] Chen, Zhu, Verresen, Seif,
Baümer, ... Trebst, Kandala.
arXiv: 2309.02863 (2023) APS N51.13

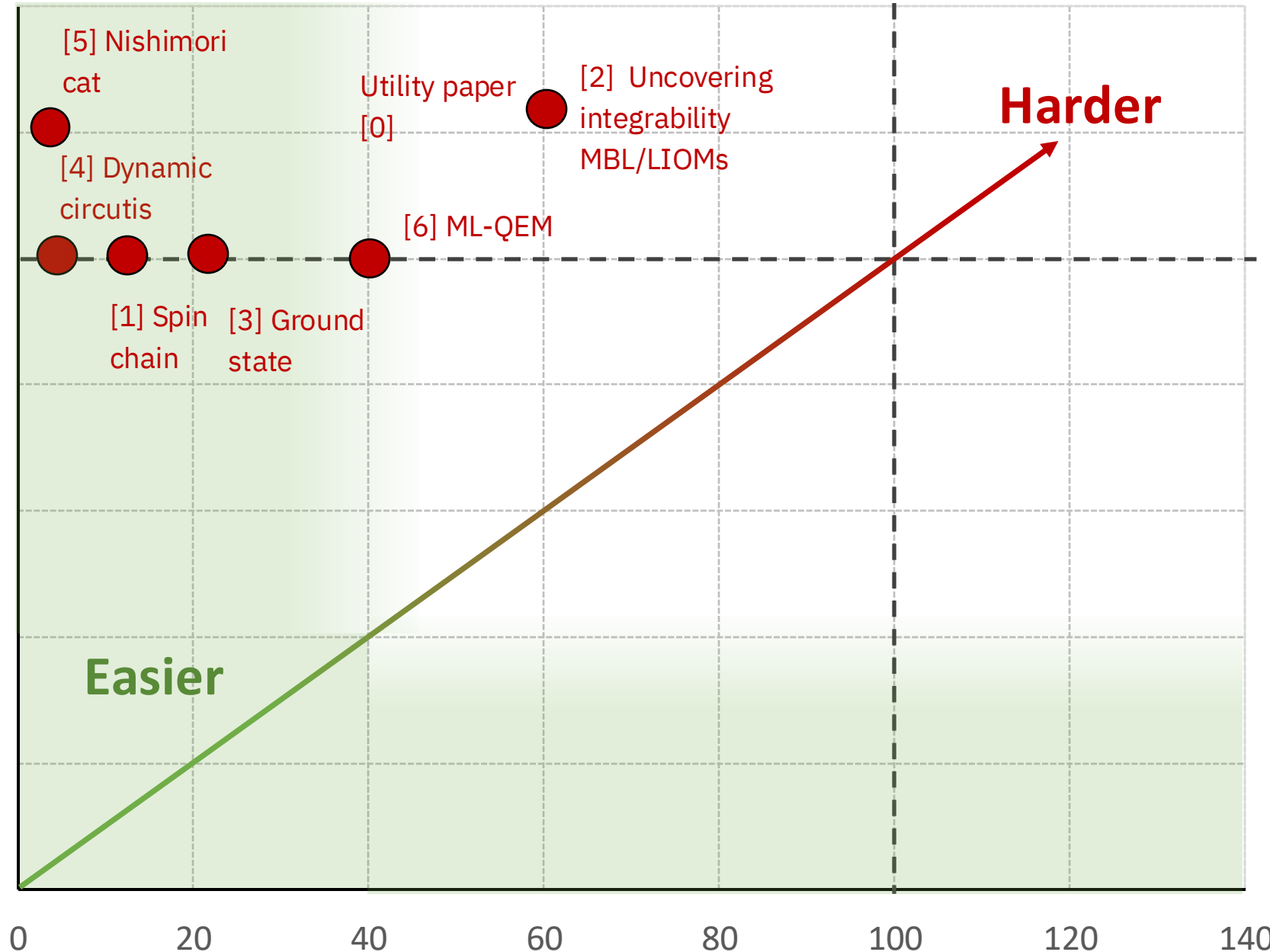


[6] Liao, Wang, Sitdikov, Salcedo, Seif,
Minev.
arXiv: 2308.13065 (2023)

...

Scale
(Number of qubits)

140
120
100
80
60
40
20
0



Number of two-qubit gate layers (depth)

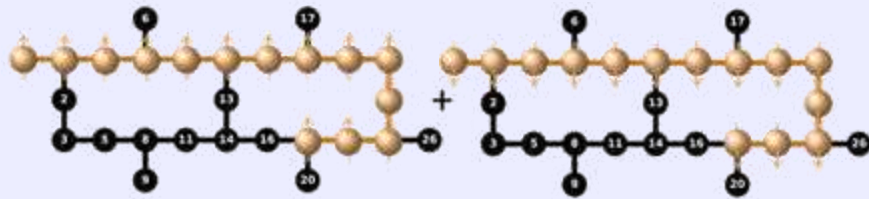
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IBM Quantum

* signal stops or decays more than 50% beyond this depth, capped at 140

* note: quantum advantage with shallow circuits, Bravyi, Gosset, Konig, and related

Check out related utility-scale work

Efficient Long-Range Entanglement using Dynamic Circuits (invited)

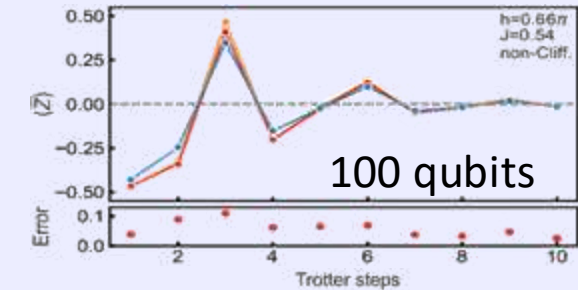


Gate teleportation over 100+ qubits

arXiv:2308.13065 (2023)

Bäumer, Tripathi, Wang, Rall, Chen, Majumder, Seif, Mineev

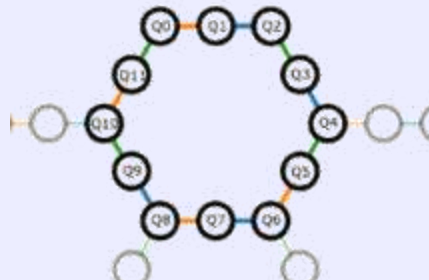
Machine Learning for Practical Quantum Error Mitigation (ML-QEM)



arXiv:2308.13065 (2023)

Liao, Wang, Sitdikov, Salcedo, Seif, Mineev

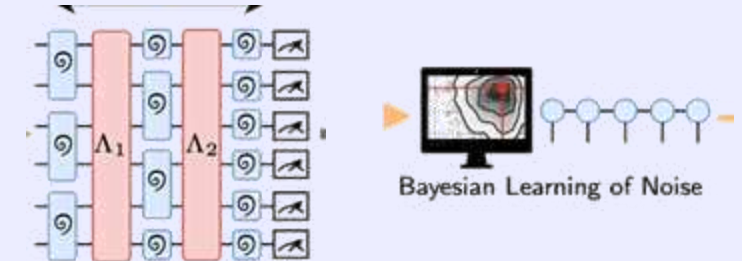
Learning about Quantum Noise at Scale



Seif, Liao, Majumder, Chen, Wang, Bäumer, Malekakhlagh,
arXiv:2408.03376 (2024)

Javadi-Abhari, Jiang, Mineev

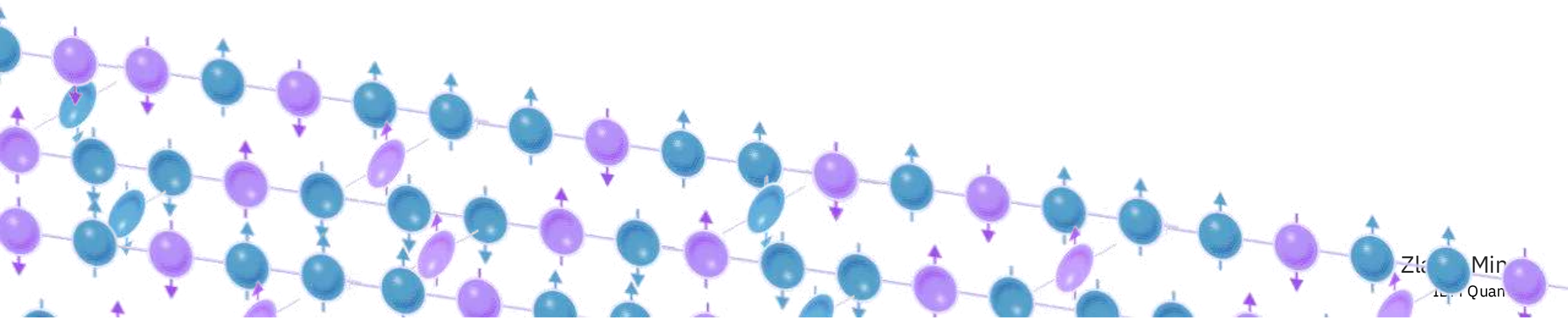
Demonstration of Robust and Efficient Quantum Property Learning with Shallow Shadows



arXiv:2402.17911 (2024)

Hu, Gu, Majumder, Ren, Zhang, Wang, Mineev, You, Seif, Yelin

Preview



Can quantum computers be useful for
many-body physics in the next 3-5 years?

Uncovering the dynamics of many-body systems

Many-body quantum systems and their dynamics

- fundamental and technological
- but generically difficult to simulate and understand

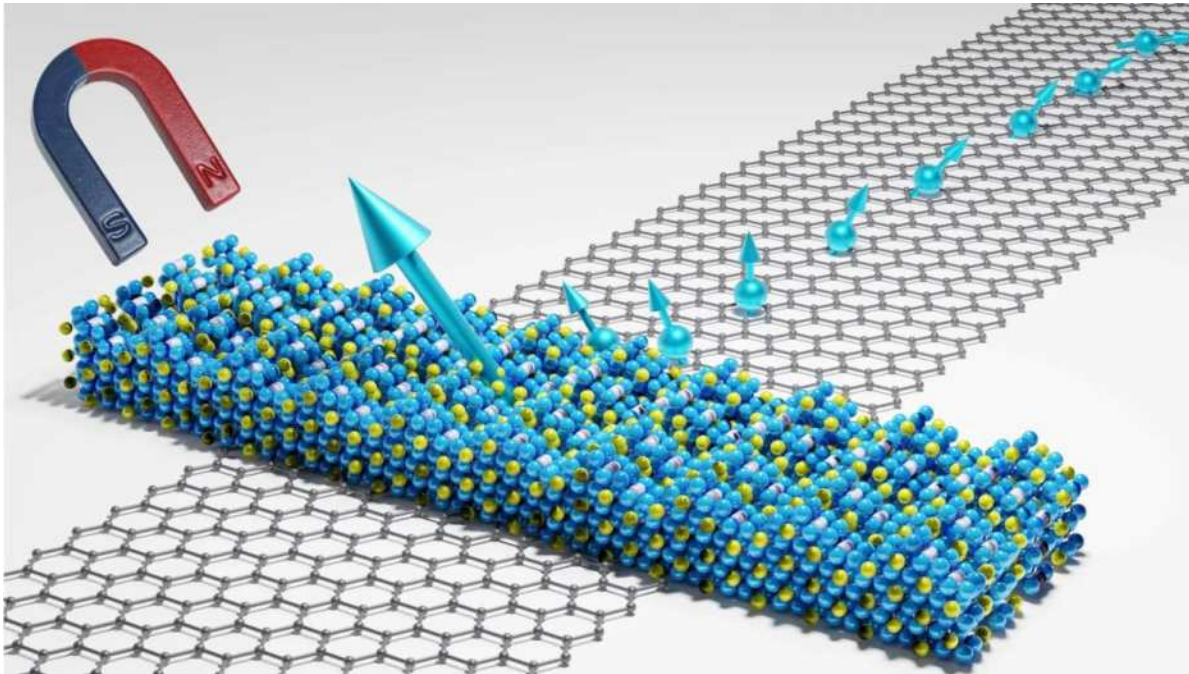


Image: [Chalmers](#)

Symmetries, conservation laws, and integrability

- can unravel intricacies of these complex systems
- but generically difficult to discover

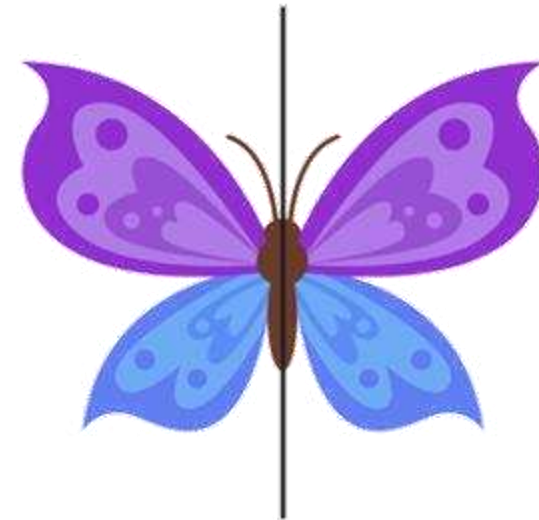


Image: [SuperSimple](#)

Classical Physics: Integrals of Motion

Before



Integrals of motion:
Total Linear Momentum
Total Kinetic Energy

After



Classical Physics: Integrals of Motion

Before



If you can find enough
Integrals of Motion (IOM),
you fully learn the system.

Integrals of motion:
Total Linear Momentum
Total Kinetic Energy



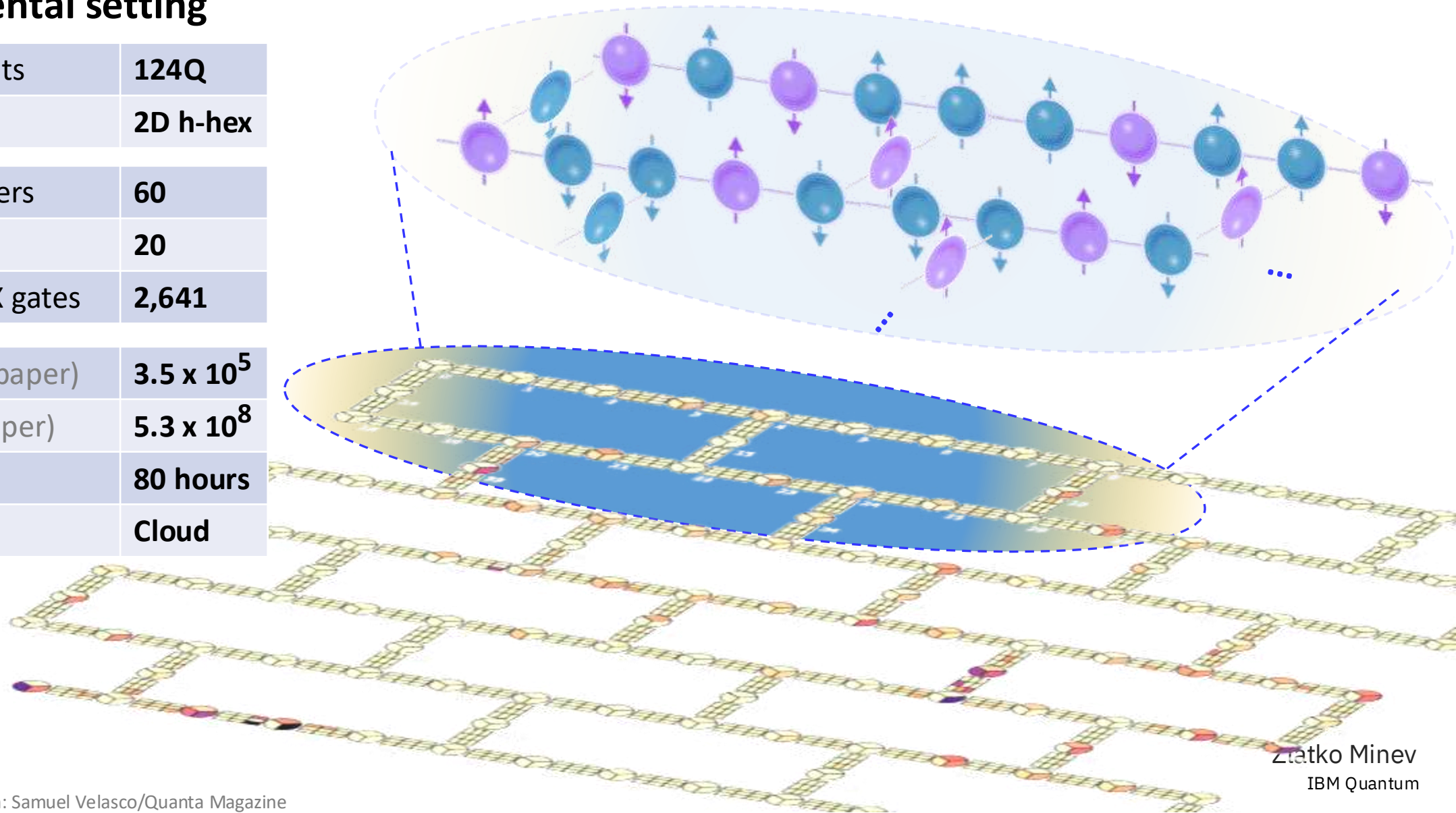
After



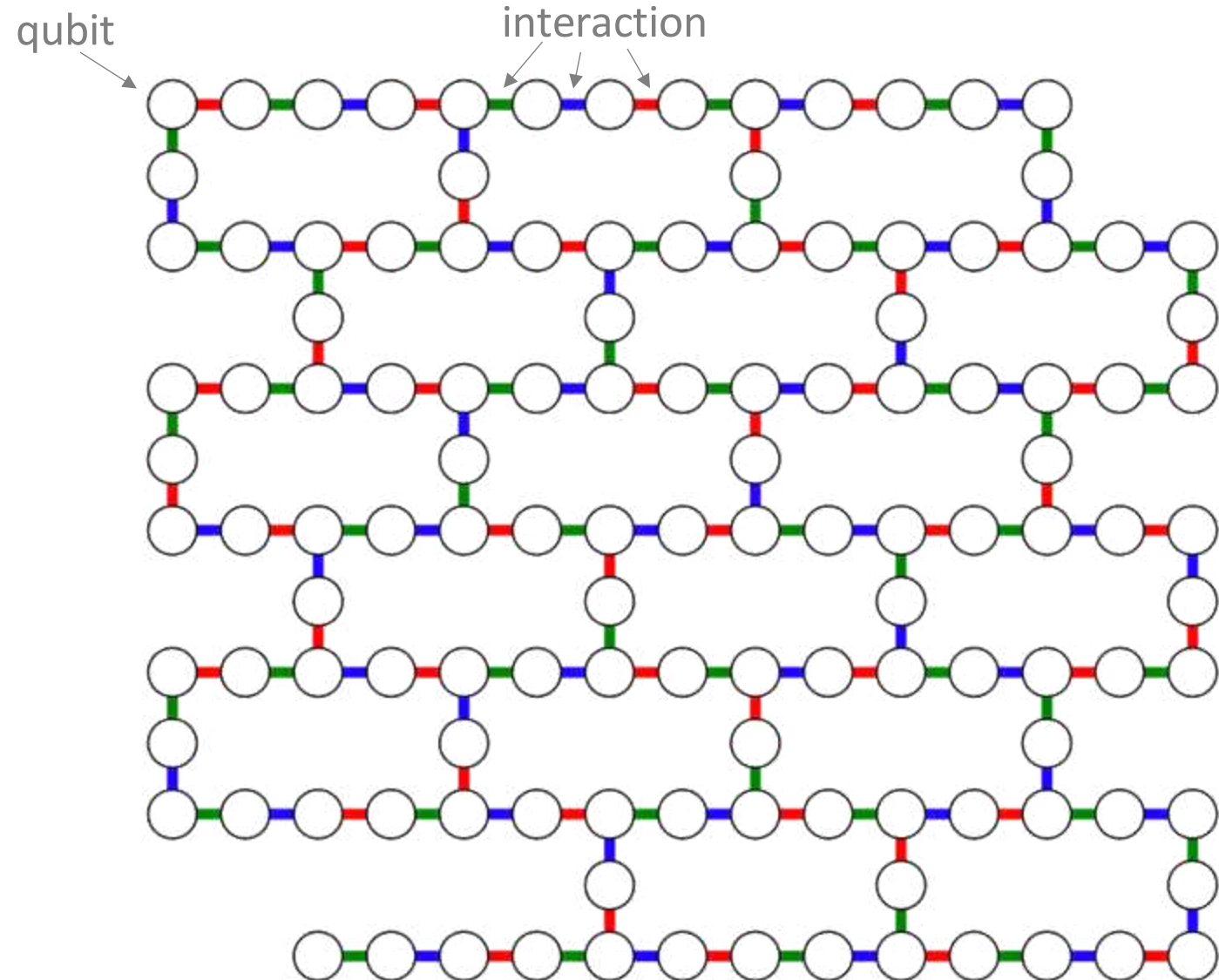
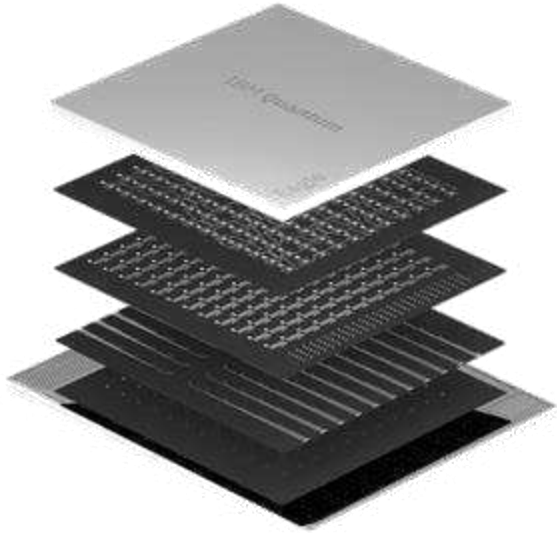
QSim on 100Q+: 2D interacting many-body Floquet system

Experimental setting

Number of qubits	124Q
Connectivity	2D h-hex
Depth in 2Q layers	60
Floquet steps	20
Total num. of cX gates	2,641
Circuits (entire paper)	3.5×10^5
Shots (entire paper)	5.3×10^8
QPU runtime	80 hours
Environment	Cloud

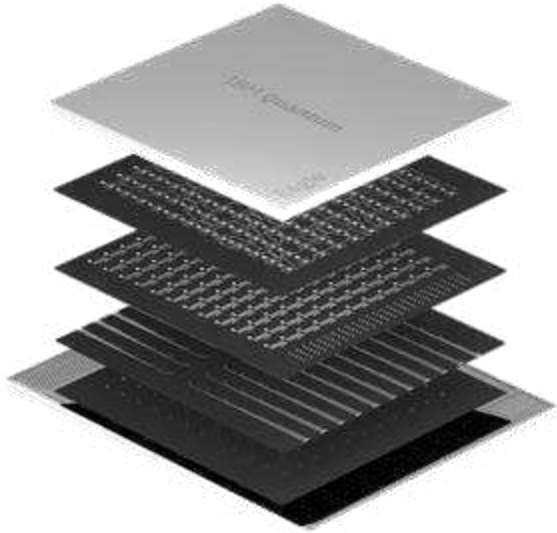


Interaction map and device layers



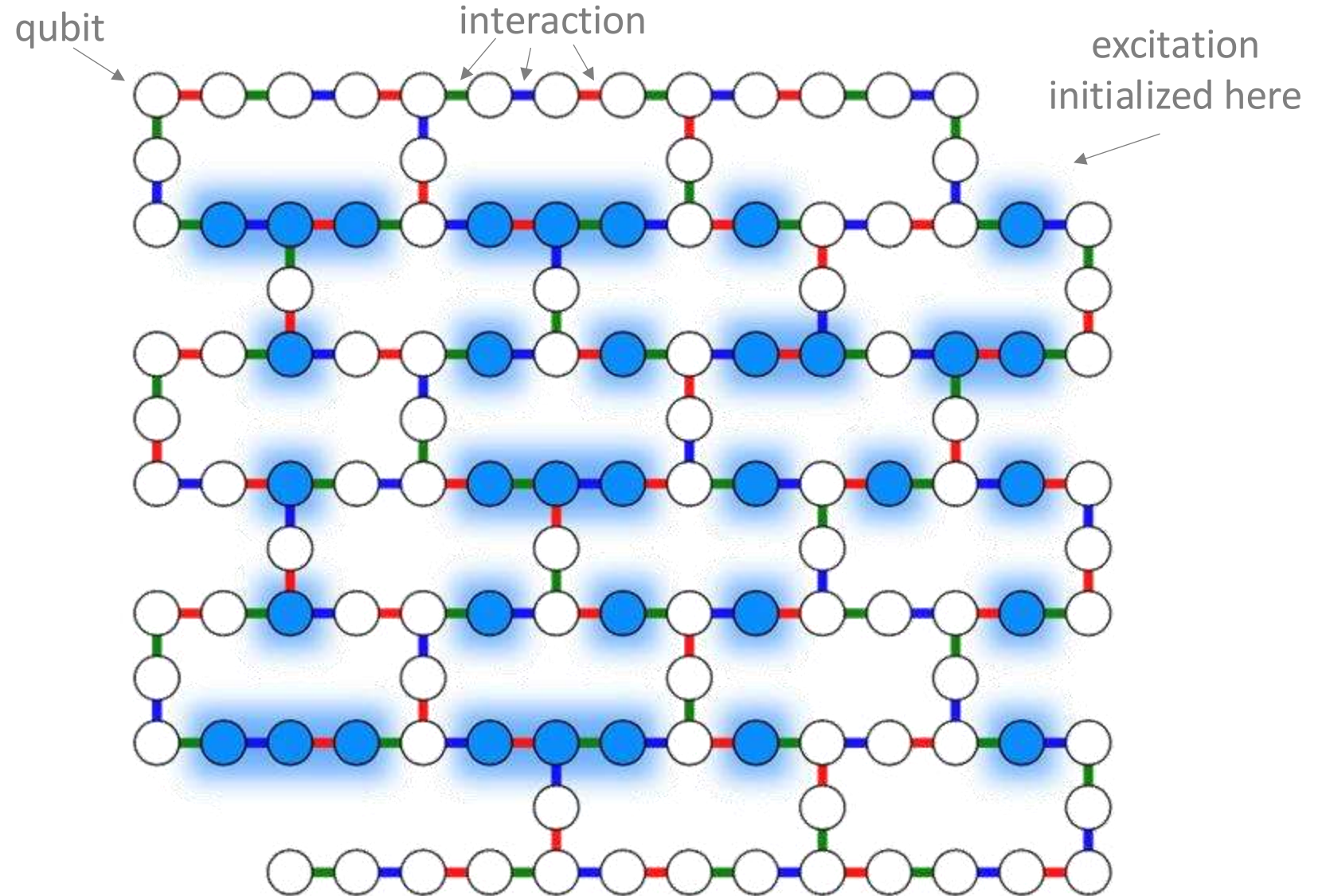
Number of qubits	124Q
Connectivity	2D h-hex
Depth in cX gates	60
Floquet steps	20
Total number of cXs	2,641

Initialize lattice in fun states



Number of qubits	124Q
Connectivity	2D h-hex
Depth in cX gates	60
Floquet steps	20
Total number of cXs	2,641

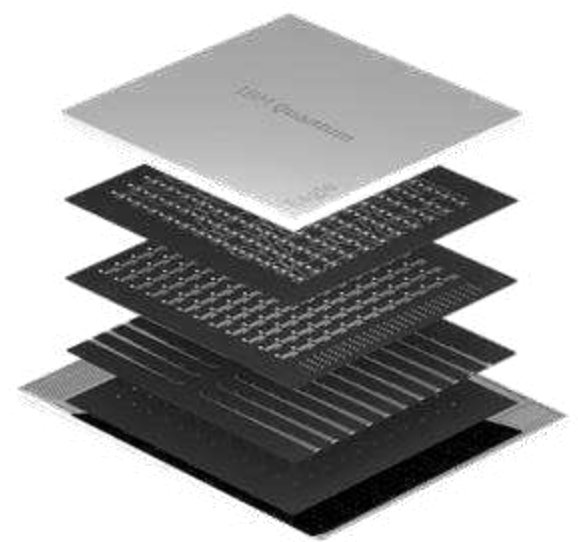
arXiv:2307.07552



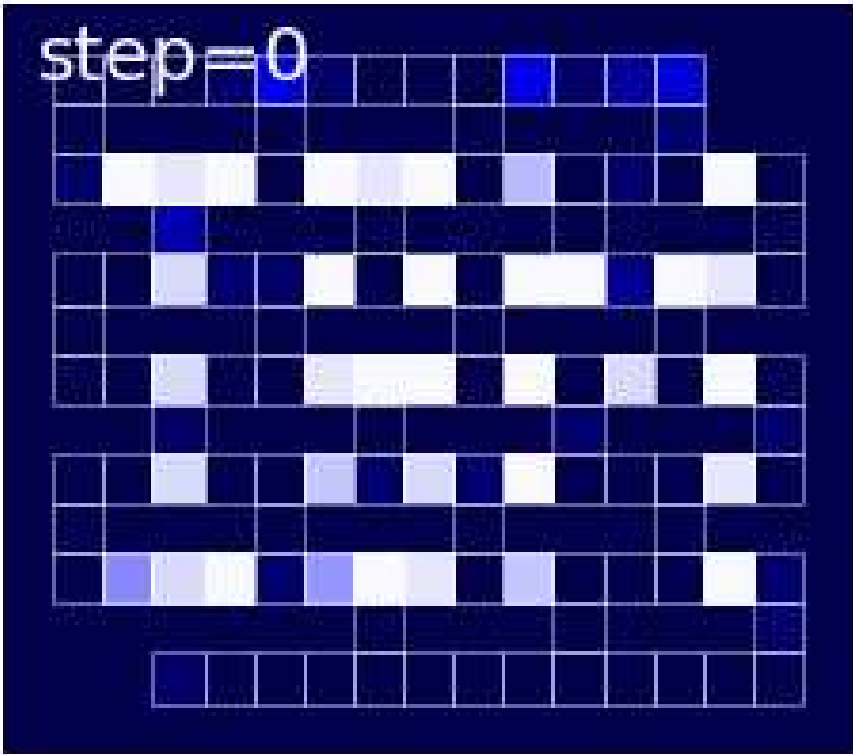
Color represent spin polarization

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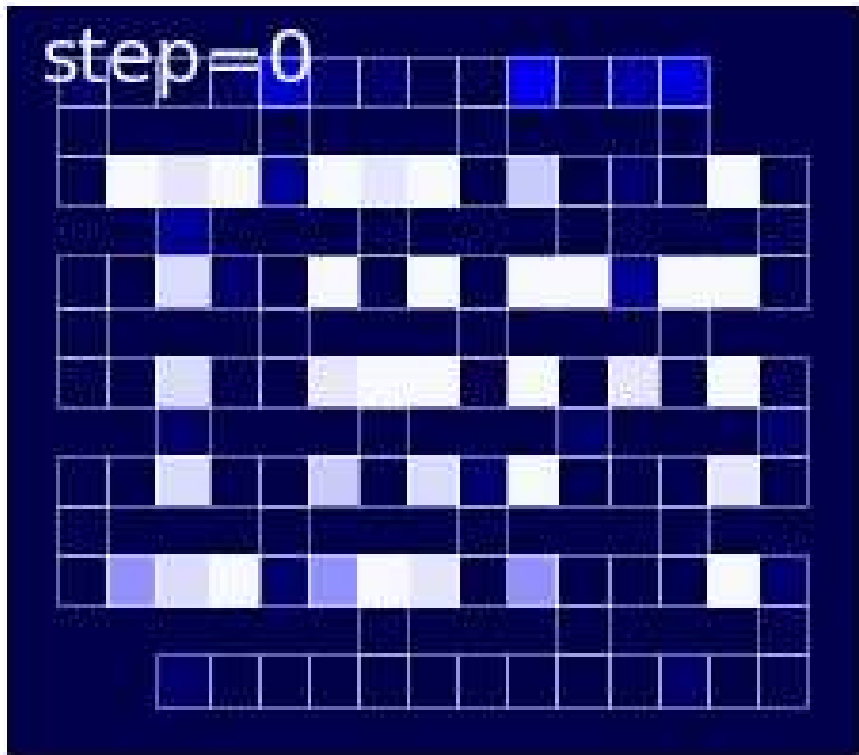
Quantum dynamics in different regimes



Number of qubits	124Q
Connectivity	2D h-hex
Depth in cX gates	60
Floquet steps	20
Total number of cXs	2,641



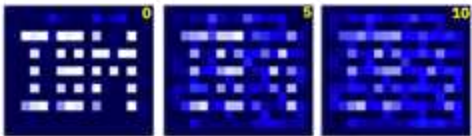
Thermalizing regime



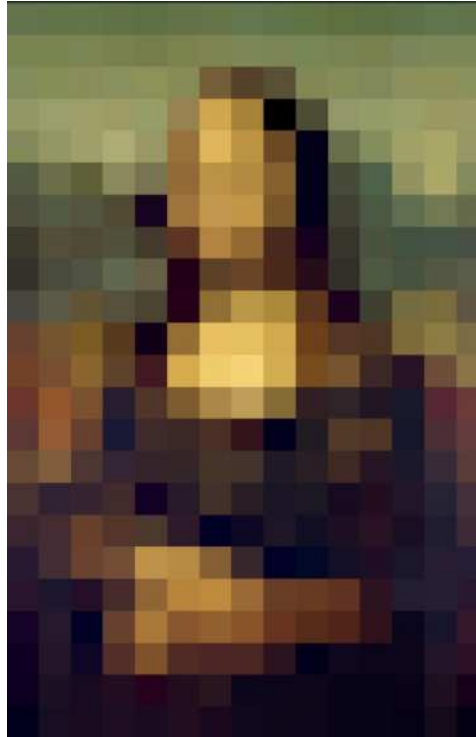
Prethermal regime

Color represent spin polarization

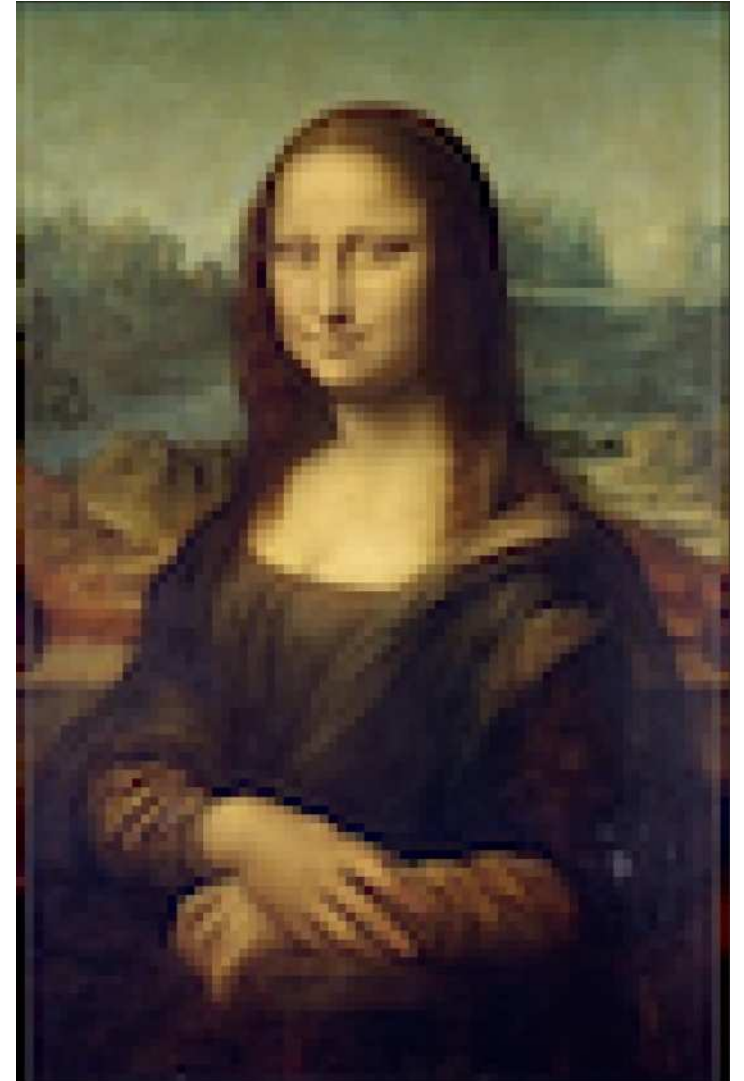
A more detailed portrait



Spin
imbalance

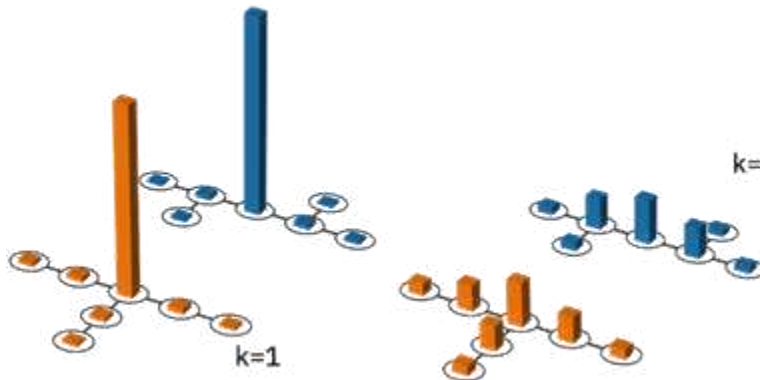
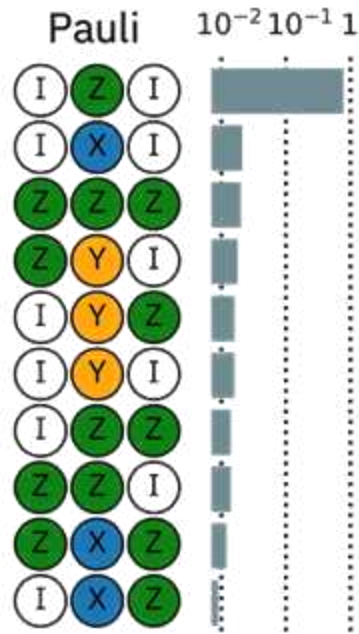
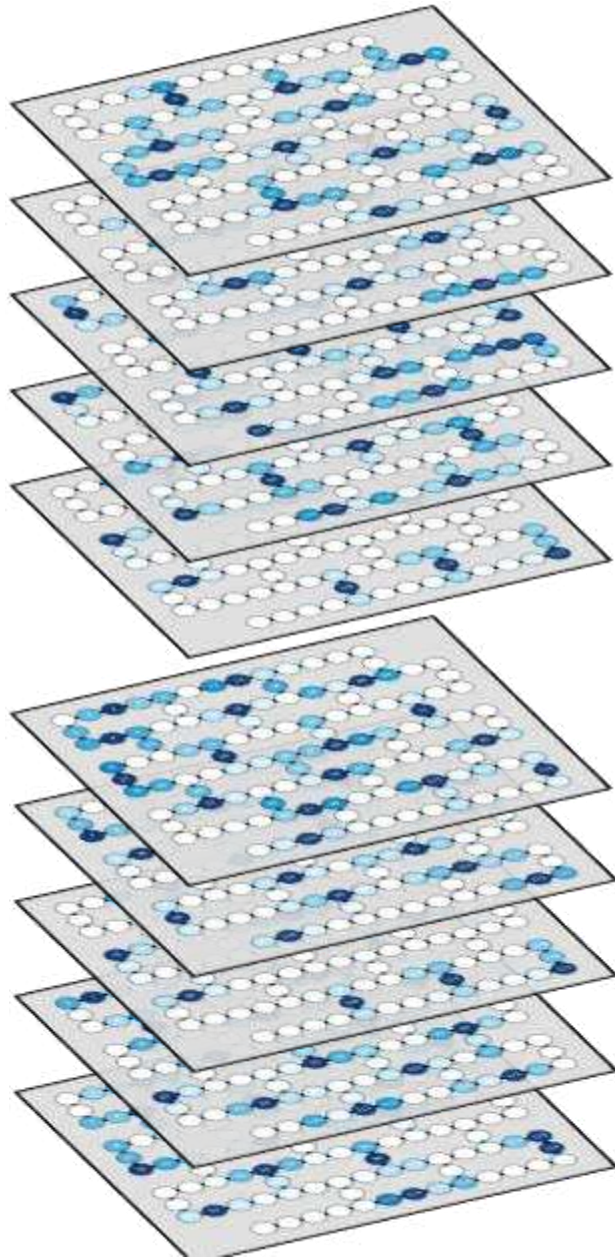


One-particle density
matrix (OPDM)



Local integrals of motion
(LIOMs)

Operational LIOM description on observation timescales



The important thing is not to stop questioning. Curiosity has its own reason for existence.

One cannot help but be in awe when they contemplate the mysteries of eternity, of life, of the marvelous structure of reality.

It is enough if one tries merely to comprehend a little of this mystery each day.

Albert Einstein

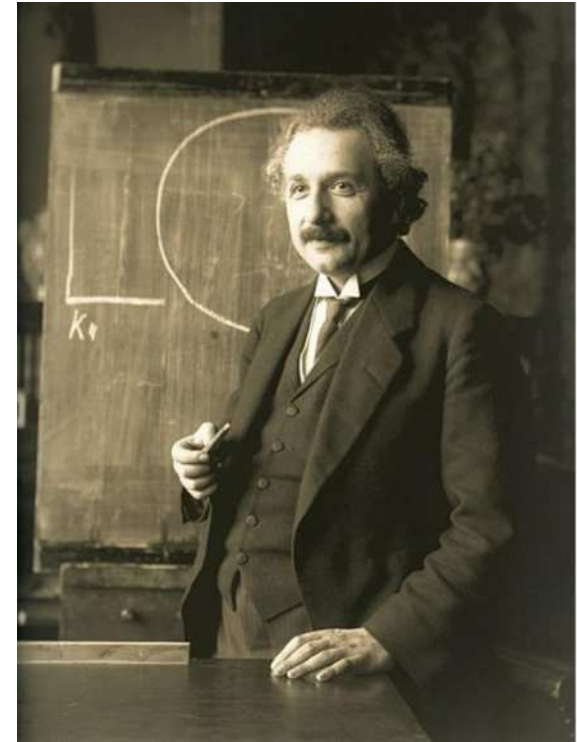


Photo: F. Schmutzer



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IBM Quantum

Quantum Working Groups

From utility to advantage

IBM Quantum working groups bring together the best scientists in our field to accelerate our path to achieving Quantum Advantage by 2025, across domain areas:

Optimization

Quantum Optimization: Potential, Challenges, and the Path Forward

arXiv:2312.02279



Materials & HPC

Quantum-centric Supercomputing for Materials Science: A Perspective on Challenges and Future Directions

arXiv:2312.09733



High-Energy Physics

Quantum Computing for High-Energy Physics: State of the Art and Challenges. Summary of the QC4HEP Working Group

arXiv:2307.03236



Healthcare & Life Sciences

Towards quantum-enabled cell-centric therapeutics.

arXiv:2307.05734



Sustainability

Collaborative projects in the fields of Materials and Energy leveraging quantum computers.

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